

# HARMONY TUTOR

## The intelligent tonal-harmony checker and multi-voice writing

### USER MANUAL

Version 1.19.0 — November 2026

macOS · Windows

# What's new in version 1.19

For those who already know Harmony Tutor: what is new, in short, and where to read the details. With the next version this page makes way for its own news; the explanations stay in the chapters.

**Teaching and studying**

- Notes on the piece: a panel for the teacher's analysis and for the reminders of whoever studies the piece, saved in the file; the bars mentioned in the text become links. → 9.14
- Prepare for your AI: the piece and Harmony Tutor's analysis, with a ready-made request, to paste into the chat of your own artificial intelligence. → 9.14
- Method texts: the teacher rewrites the title, explanation and suggestion of the findings and hands them to the students in a file. → 15.4

**Import**

- File › Import MuseScore...: .mscz and .mscx files, converted by the MuseScore installed on the computer. → 13.1

**Writing**

- Choir on a single staff: with Cmd the note stays in the chosen voice at any pitch, and several notes on the same beat go to the free voices. → 3.2
- A chord written from its symbol follows the key of the bar and does not double the leading note. → 4.14

**School licence**

- The code is checked at every start; «Manage License…» shows the school and the date it is valid until. → 2.5

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# 1. Introduction

## 1.1 What is Harmony Tutor

Harmony Tutor is a SATB (Soprano, Alto, Tenor, Bass) music editor with automatic, real-time harmonic analysis. It is designed for harmony students and teachers, musicians and composers who want to check and improve their four-part writing exercises.

The application analyses each chord the moment it is entered, identifies voice-leading errors, recognises cadences and modulations, and provides a didactic explanation for every flag. It is currently the only SATB editor with real-time harmonic analysis available.

## 1.2 Who it is for

- **Conservatory and music-school students:** immediate checking of exercises, understanding of mistakes, self-study.
- **Harmony teachers:** a fast correction tool, teaching support, time saved on recurring errors.
- **Self-taught musicians:** learning guided by practice and feedback.
- **Composers and arrangers:** voice-leading checking and harmonic analysis in a classical context.

## 1.3 Core principles

- **Spelling-First:** analysis is based on note names (spelling), not only on MIDI numbers. This guarantees correct distinction of enharmonic intervals and reliable recognition of augmented-sixth chords (It6, Fr6, Ger6).
- **Educational correction:** every violation shows a rule code, an expandable description and a practical tip for fixing it.
- **International academic standard:** nomenclature follows the academic standard, supporting both Roman numerals (functional figuring) and modern chord symbols (Cmaj7, Dm, G7/F, etc.).

## 1.4 System requirements

| Operating system | Minimum requirements |
|---|---|
| macOS | macOS 12 (Monterey) or later, Intel and Apple Silicon processors |
| Windows | Windows 10 (64-bit) or later |

Working does not need an internet connection: it is needed only to activate the licence, to check it from time to time (see 2.5) and to receive updates. Disk space: about 1 GB. Recommended resolution: 1280×800 pixels or higher; on a narrower screen the command bar scrolls sideways.

# 2. Installation and first launch

## 2.1 macOS

1. Download the Harmony Tutor-mac.dmg file from the website.
2. Open the DMG and drag “Harmony Tutor” into the Applications folder.

## 2.2 Windows

1. Download Harmony Tutor.win.exe from the website and run the installer.
2. If the “Windows protected your PC” (SmartScreen) warning appears, click “More info” and then “Run anyway”.
3. Follow the wizard. The app will be available in the Start menu.

## 2.3 First launch

On opening, Harmony Tutor presents an empty project in C major, 4/4 time, with 4 measures. The interface is in dark mode.

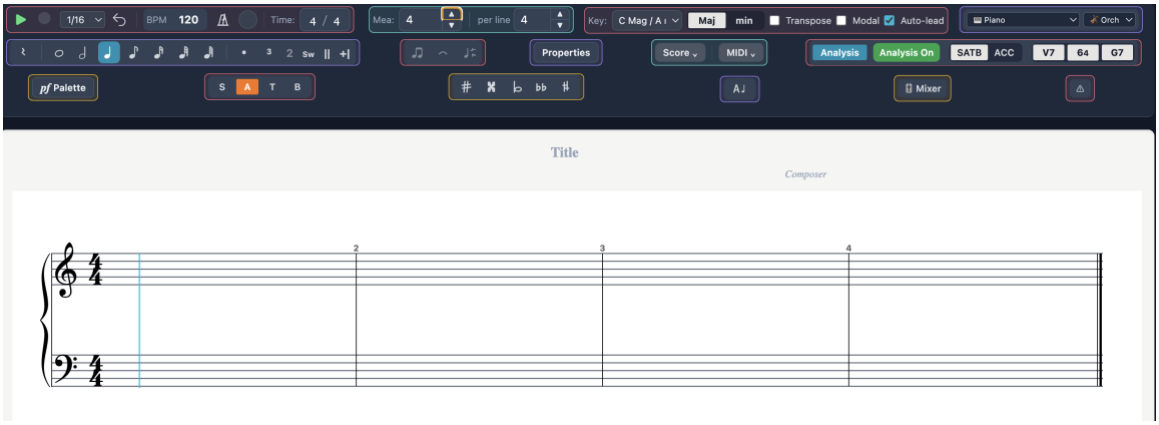



*Figure 1 — First launch: empty project in C major, 4/4, 4 measures*

## 2.4 New project (initial tracks and templates)

Creating a new project (File ▸ New, or the New action in the side menu) opens the New project dialog, where you set up the starting tracks before the project is created. You can always add or remove tracks later from the Mixer.

- SATB staff — the “SATB (4-voice choir)” checkbox shows or hides the four-part vocal staff. Uncheck it for an accompaniment- or percussion-only project.
- Accompaniment tracks — pick an instrument from the drop-down and click “+ Instrument” to add a pitched track; “+ Drums” adds a percussion track. The chosen tracks are listed below; ✕ removes one.
- Templates — “Save current configuration…” stores the current set of tracks under a name; click a saved template to load it into the selection, to delete it. Templates are kept locally and are offered for every new project.
- Create project / Cancel — confirm the configuration to create the project, or close the dialog without creating it.

## 2.5 School licence

If your school has adopted Harmony Tutor you do not receive a personal key: you receive a code beginning with HTS- that belongs to you. Paste it into the same window used for a normal licence, either at first launch or at any time from the application menu.
One code covers TWO computers. That is deliberate: the one at home and the one in the classroom, without having to ask anyone. Reinstalling the program on the same computer does not use up the second slot.
The code is valid until the date your school has set, and the expiry applies with or without a connection. The program checks the code at every start, when there is a connection, but it does not need the network in order to work: it keeps going for weeks without one, and asks for a connection only when too long has passed since the last check.
If the school revokes the code, or it has expired, a window says so at start and gives the reason: the open work can be finished and saved, and next time the program asks for a valid code. If instead the school moves the expiry date — for a student repeating the year, for example — the program picks up the new date by itself.
To see the active licence: View menu › «Manage License…» (on the Mac also in the Harmony Tutor menu). The window shows the school, the code, the date it is valid until and the last check.
If you change computer, deactivate the licence on the old one before activating it on the new: the slot is freed. If the old computer is gone, ask your school for a new code.
**For the school's technician.** The program uses the network only over HTTPS (port 443), towards these addresses: license.harmonytutor.it to activate and check the licence; github.com and

*.githubusercontent.com for updates. If the school network has a filter, they must be allowed through. The computers do not need to be connected to each other.

# 3. The interface

## 3.1 Overview

- **Toolbar (top):** controls for voice, durations, accidentals, key, meter, playback and instrument.
- **Central editor:** the staff with the SATB voices, harmonic labels (Roman numerals) below the staff and chord symbols above (toggleable).
- **Analysis panel (right):** list of violations, severity filters, rule search, explanation of harmonic labels.

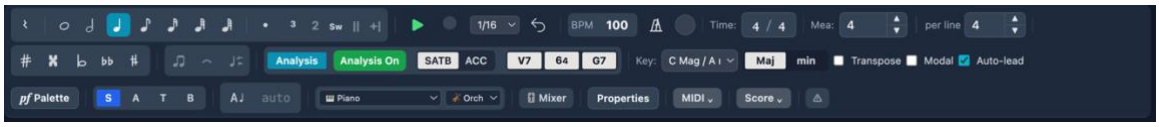



*Figure 2 — The main toolbar*

## 3.2 Display modes

The editor supports several layouts, selectable from the Preferences panel (Editor tab) or the side menu. Switching layout is instantaneous: the notes are unchanged.

| Mode | Description | Staves |
|---|---|---|
| Grand Staff (default) | Treble clef + bass clef | 2 |
| Treble clef | A single staff | 1 |
| Old SATB clefs | Four staves with old clefs (S, A, T, B) | 4 |
| Close score | Grand Staff with reduced spacing | 2 |
| Open score | Separate staves with wide spacing | 4 |

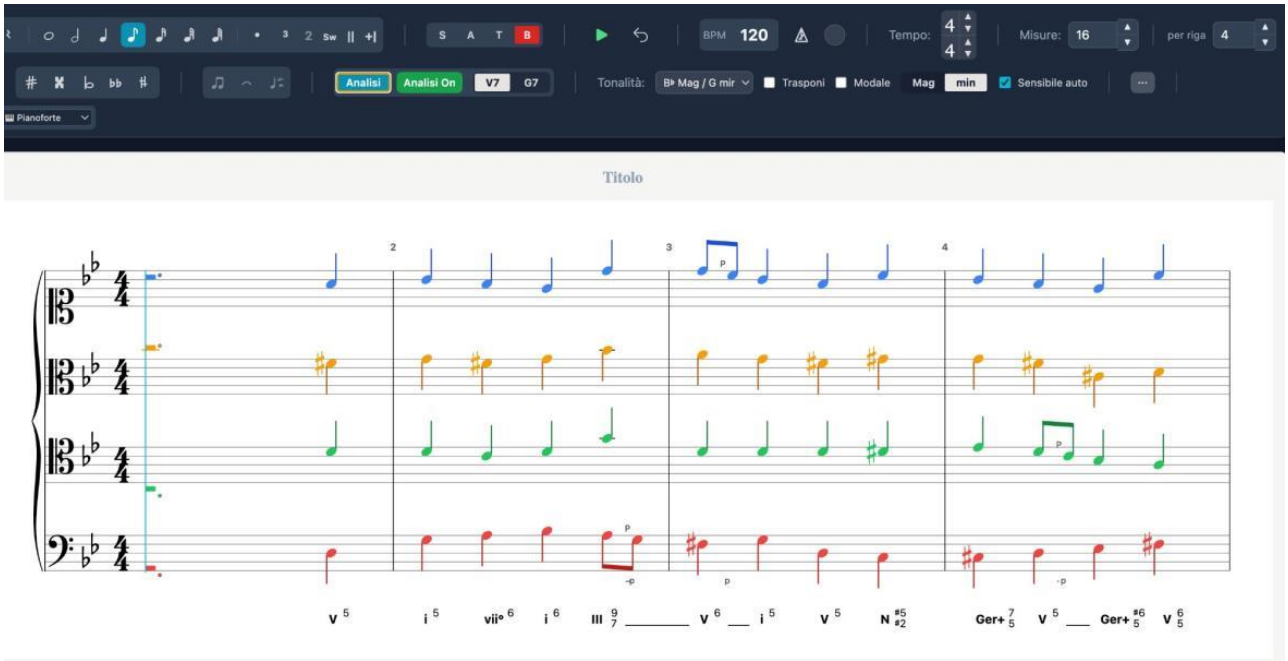



*Figure 3 — SATB layout with old clefs, coloured voices, augmented sixths and Neapolitan*

### The choir on a single staff: which voice you write in

In the «Treble clef» layout the four voices share one staff. A plain click chooses the voice from the height you click at: above the staff the soprano; from the top line to the middle line (the B) the alto; from the middle line to the bottom line the tenor; below the staff the bass.

Writing with Cmd (Ctrl on Windows) the note stays in the chosen voice at any pitch: a soprano can go down inside the staff without becoming the alto. The ghost note takes the colour of the voice it will write in.
If the chosen voice already has a note at that point, the new one goes to a free voice and replaces nothing: above the staff the soprano and below it the bass, if they are free; inside the staff the first free voice from the top that keeps the order of the notes already written. So a chord can be written note by note on the same beat. Only when every voice is taken does the note replace the chosen voice's one.

## 3.3 Side menu

The "..." button in the toolbar opens the side menu, which gathers the layout, format and MIDI controls.

- **Layout:** Grand Staff (2 staves), Treble clef (1 staff), Old SATB clefs (4 staves), Close score.
- **Format:** Page (portrait) or Landscape. Affects the on-screen layout and the PDF export.
- **MIDI:** enable/disable MIDI output and select the destination device (Internal Audio, IAC Driver Bus, Logic Pro Virtual In or other connected devices).
- **Step Input:** when active, notes are entered one at a time from the connected MIDI keyboard; the position advances automatically to the next beat.

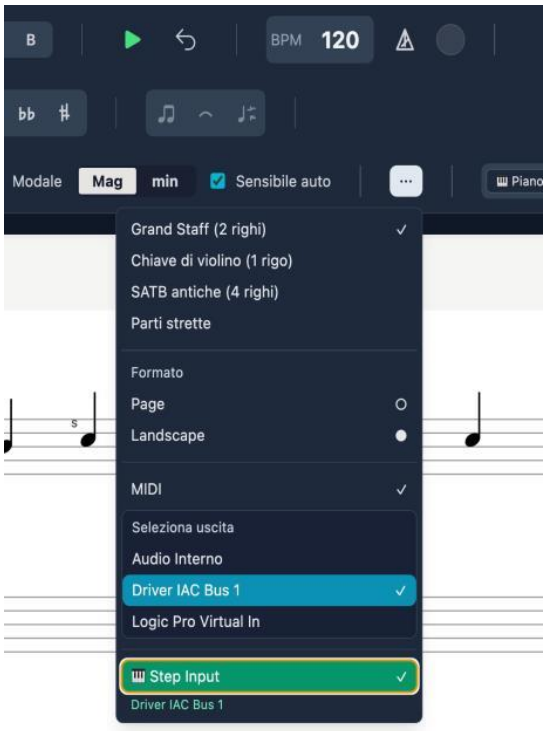



*Figure 4 — Side menu: layout, format, MIDI and Step Input*

## 3.4 Voice colours

By enabling "Voice colours (BTAS)" in Preferences → Editor, each voice is shown in a distinct colour: Soprano in blue, Alto in orange, Tenor in green, Bass in red.

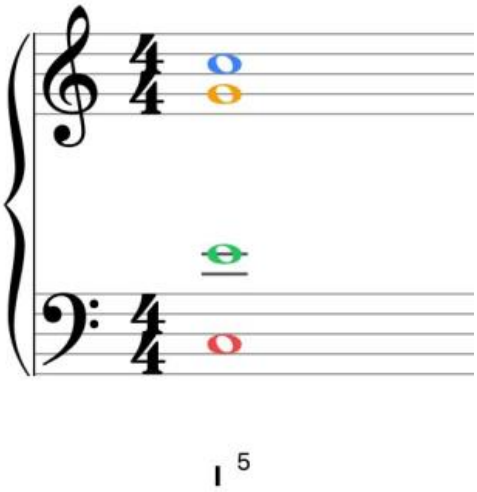


*Figure 5 — Coloured voices: S (blue), A (orange), T (green), B (red)*

## 3.5 Voice-leading indicators

On the staff, coloured dashed lines connect the notes between one chord and the next:

- **Green lines:** permitted or tolerated motion, or recognised cadences (PAC, IAC, HC, plagal). They appear only where the engine detects a meaningful pattern.
- **Orange lines:** a warning (situation to check).
- **Red lines:** a serious error (parallel fifths, parallel octaves, voice crossing, etc.).

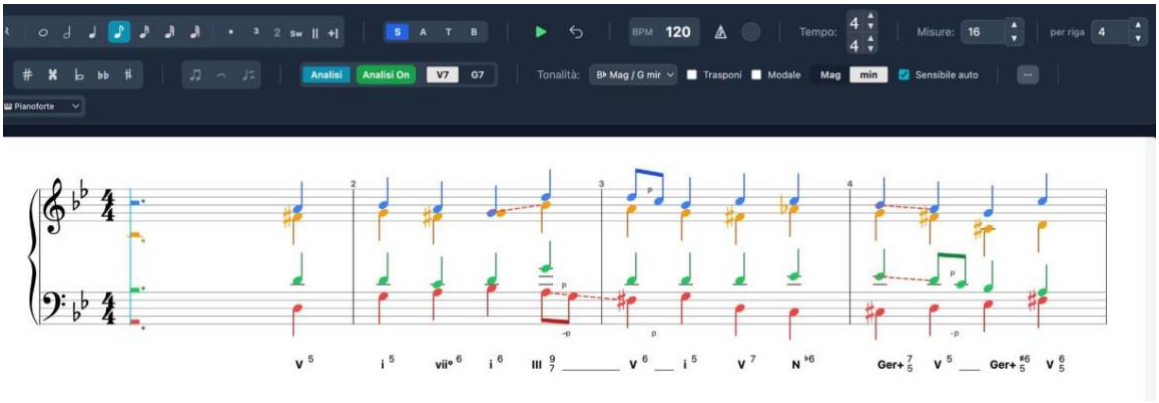



*Figure 6 — Voice-leading with green, orange and red lines, Ger+ and Neapolitan*

## 3.6 Chord symbols

In addition to Roman numerals, modern chord symbols (C, Am, D7/F#, Dm7/F, G7, etc.) can be shown above the staff. Toggle them from the toolbar selector or from Preferences → Analysis → “Show chord symbols”.

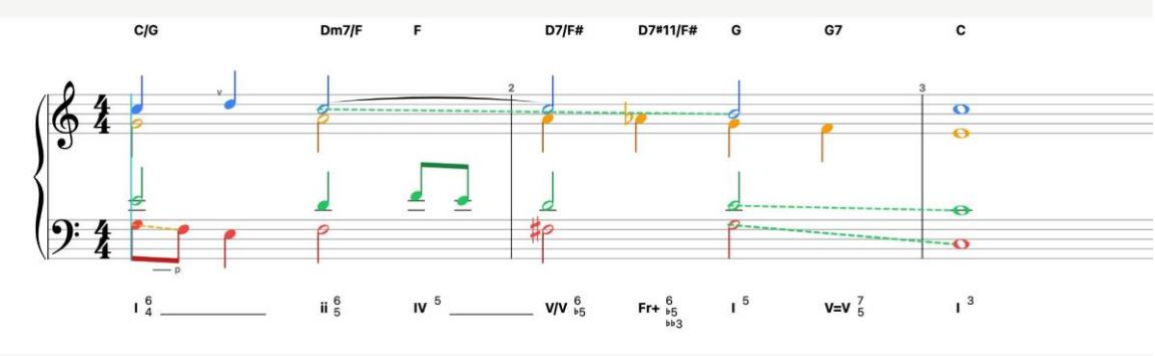



*Figure 7 — Dual display: chord symbols above + Roman numerals below, with a French augmented sixth (Fr+)*

## 3.7 Listening to single voices

To hear a single voice in isolation, double-click its voice button (S, A, T or B). Can be enabled on several voices. Double-click again to disable.

Alternatively, more appropriately, by going to the mixer panel accessible from the toolbar, you can solo, mute, or adjust the volume of any track.

More precisely: a double-click on a toolbar S/A/T/B button toggles SOLO for that voice (you can solo several at once). In the Mixer, every voice and ACC-track strip has its own M (mute) and S (solo) buttons, with unified semantics across voices and tracks: as soon as anything is soloed — a voice or a track — only the soloed channels sound, while mute always silences. This is the most flexible way to isolate or compare lines while writing.

## 3.8 Vocal ranges

| Voice | Comfortable range | Extended range |
|---|---|---|
| Soprano | C4 – F5 | B3 – A5 |
| Alto | F3 – C5 | E3 – D5 |
| Tenor | A2 – E4 | A2 – A4 |
| Bass | E2 – B3 | D2 – C4 |

## 3.9 Incomplete-measure warnings (⚠ toggle)

Measures that do not reach the duration required by the meter are flagged with a red rectangle. The warning can be hidden:

- Click directly on the red overlay on the measure.
- **⚠ button in the toolbar:** red = warnings visible, grey = warnings hidden.

> ***Note:*** *The setting is not persistent: it resets to “visible” each time the file is opened.*

### What the box tells you

The box does not merely say that the measure does not add up: it says what does not add up. Hovering over it, or reading the label, you find "an eighth missing on beat 3", or "a quarter too many", or "two notes overlapping by an eighth on beat 2".
Inside the box the same thing appears in two characters, because on a narrow screen the label can be cut off: −0.5 means half a beat is missing, +1.5 that there is a beat and a half too much, ↑↓ that two notes of the same part are sounding together without being written that way.
That last warning is the most useful of the three, because the fault it reports cannot be seen on the page: every note looks to be in its place, but one lasts longer than the space allows and overlaps the next. It is the usual reason why a measure seems to refuse new notes.

## 3.10 The command bar is yours to arrange

Everyone composes the toolbar their own way. The customise command turns groups of buttons on and off: if you never use accompaniment tracks you can put them away, and what remains spreads out to fill the space.
Every group of buttons has a position of its own: you pick it up, put it where you want it and there it stays. Groups do not wrap by themselves when the window narrows: you can keep the things you actually use within reach and push the rest out of the way.
Every group sits in a coloured FRAME, and the frame moves as one block with everything inside it. To keep together two groups you use together, drop a group — or a whole frame, taken by its ✣ handle — onto another frame: they merge into one. To separate it, drag the group out: it takes a new frame with it. No frame is ever left empty, and a click on the dot changes its colour.
While you drag, the frame already shows where it will land, and a magnet aligns it with its neighbours; holding Opt (Alt on Windows) silences the magnet and the frame is placed to the pixel. What you drop stays where you put it: if it covers other frames, they move aside, only as far as needed. The arrangement you have built can be saved to a file and loaded back: useful to carry it to another computer, or to keep two arrangements for two ways of working. Anyone who has never touched anything starts from a ready-made arrangement, not from a bar to be built.

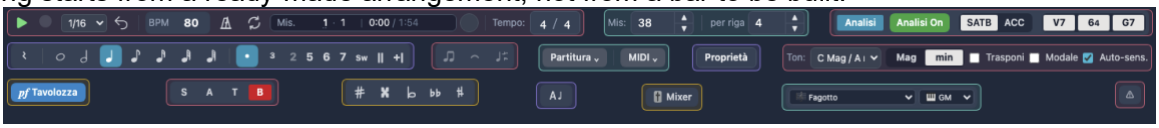



Figure — The arranged toolbar: button groups in their coloured frames, each one where it was put.

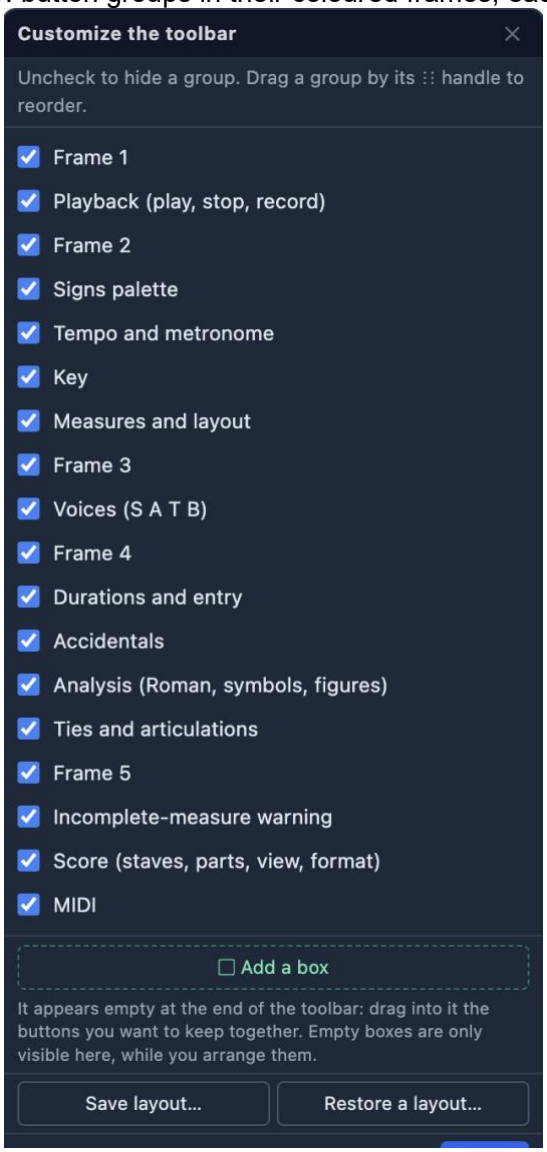



Figure — The «Customise the toolbar» panel: here you hide a group by unticking it, and save or load an arrangement.

## 3.11 Toolbar and command bar

These are two different, independent strips, and each is switched on and off on its own from the View menu. The toolbar is the fixed one at the top, with its ⌥T shortcut; the floating command bar is the one you drag to wherever you are writing.

# 4. Entering notes

## 4.1 Voice and duration selection

| Action | Shortcut | Effect |
|---|---|---|
| Cycle voice | V | B → T → A → S → B |
| Whole note | 1 | 4 beats |
| Half note | 2 | 2 beats |
| Quarter note | 3 | 1 beat |
| Eighth note | 4 | 1/2 beat |
| Sixteenth note | 5 | 1/4 beat |
| Thirty-second note | 6 | 1/8 beat |
| Sixty-fourth note | 7 | 1/16 beat |

When you write several notes at the same point, the program can work out on its own which voice each belongs to, by looking at how they sit on the staff: the highest goes to the upper voice of that staff, the lowest to the bottom one. As long as the arrangement is unclear, the voice chosen in the toolbar applies.

### A tied value: Shift + duration

To lengthen a note by a value that does not fit one note shape — a half note tied to an eighth, for example — select the note and press Shift together with the key of the duration to add (1 to 7, as in the table). Right after the note a new one appears at the same pitch, with that value, already tied.
The added note writes over whatever it finds in the same voice; if it crosses the barline it continues into the next bar, still tied, and the uncovered part of a half-covered note becomes a rest. With Shift + Option (Alt on Windows) + duration, instead, the following notes move forward to make room.
With a chord selected the value is added to every voice. If the added note would go past the end of the piece, or the shift would put a tuplet across a barline, nothing happens and a message says why. Cmd+Z undoes the whole gesture.

## 4.2 Accidentals

| Shortcut | Accidental | Effect |
|---|---|---|
| # | Sharp (♯) | +1 semitone |
| b | Flat (♭) | −1 semitone |
| n | Natural (♮) | Cancel |
| ## (double) | Double sharp | +2 semitones |
| bb (double) | Double flat | −2 semitones |

## 4.3 Modifiers

| Key | Effect |
|---|---|
| . | Augmentation dot (duration ×1.5) |
| R | Writes a rest at the cursor and moves on — see 4.4a |
| Control + R | General rest: the same rest in every part — see 4.4a |
| L | Toggle the tie on the current note |
| **Option/Alt + F** | Fermata — see 4.3a |

### 4.3a Fermata

Applies a suspension of metronomic time to a note or rest, lengthening its actual duration in playback (default 2×). The playhead visually freezes and the following events shift accordingly.

1. Select the note or rest.
2. Press Option+F (Mac) or Alt+F (Windows).
3. The fermata appears above the note on the staff.

> ***Note:*** *To remove the fermata: select the note and press Option/Alt+F again.*

## 4.4 Ghost note and entry

Entering notes with the mouse requires the **Ctrl** (Windows) or **Cmd** (Mac) modifier key, to prevent accidental clicks.

1. **Select the voice** you want (S, A, T or B) from the toolbar or with the V key.
2. **Position yourself on the correct staff** (treble clef for Soprano/Alto, bass clef for Tenor/Bass). If voice and staff are inconsistent, the system blocks entry and shows no note.
3. **Hold Ctrl/Cmd:** the semi-transparent “ghost note” appears as a guide.
4. **Click** on the staff to insert the note definitively.

Without holding Ctrl/Cmd, the mouse acts only as a pointer and selection tool.

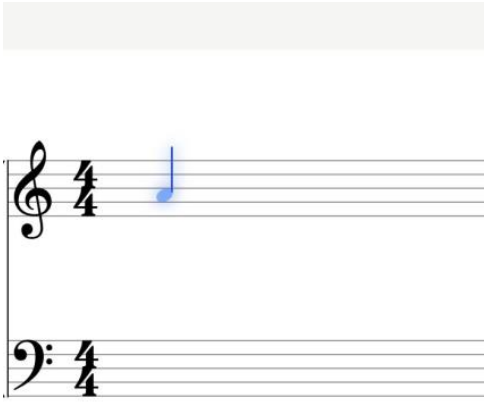


*Figure 8 — Ghost note: preview before insertion*

### 4.4a Writing rests

Rests are written from the keyboard, just like notes: R writes a rest at the cursor — the vertical line — in the active voice and with the duration chosen in the toolbar, dots and tuplets included. Then the cursor moves on, exactly as after a note: press R three times and you get three rests in a row.
Control + R writes a GENERAL REST: the same rest in every part at the same point, and the cursor moves on only once. The key is Control on the Mac too, not Cmd. On an ACC track it covers the four voices if the track is a voiced grand staff; on the other tracks, which have a single line, it does what R does.
If there was already a note at that point, the rest takes its place.
When a piece has just been opened the cursor is not anywhere yet, and R does nothing: that is deliberate, so a key pressed by mistake cannot change the piece at a point you were not looking at. One click on the staff places the cursor.
The rest button in the toolbar is still there: it is for placing a rest with the mouse, wherever you like, with Cmd/Ctrl + click as for notes.

### 4.4b With your finger: touch screens and Windows tablets

On a computer with a touch screen — a convertible laptop or a Windows tablet — Harmony Tutor can also be used with a finger or a stylus. Keep a keyboard at hand, a Bluetooth one will do: the on-screen keyboard covers half the display, and shortcuts are still the quickest way to write.
Every button responds to a tap as it does to a click. On the score, what matters is where your finger starts:

1. To write a note, hold Ctrl (Cmd on the Mac) and put your finger on the staff: the ghost note appears. Without lifting, slide your finger and the ghost moves up or down one step at a time. Above your finger the pitch you are about to write appears in large type — for example «G4 · S» — because your fingertip covers the very spot where the note lands: you look at the label, not at your finger. Lift your finger and the note is written.
2. Move your finger off the staff before lifting it and nothing is written: a wrong touch cancels itself.
3. A finger that starts on a note moves it: as you slide, the note goes up or down by step, with the same label. A quick tap selects it instead, and Shift + tap adds more notes to the selection.
4. A finger that starts on an empty spot scrolls the page, as in any other program.

The signs in the palette can also be dragged onto the score with your finger; or tap the note and then the sign. A hairpin placed this way covers the chosen note.
To the screen, a stylus with a rubber tip is a finger: everything said here applies. The selection rectangle, which you draw by dragging with the mouse, is not available with a finger — that gesture scrolls the page: to select several notes use Shift + tap.

## 4.5 Selecting and editing

| Action | Shortcut |
|---|---|
| Select note | Click |
| Multi-selection | Shift + Click |
| Delete | Backspace / Delete |
| Transpose ±1 semitone | ↑ / ↓ |
| Transpose ±1 octave | Alt + ↑ / ↓ |
| Undo / Redo | Cmd/Ctrl + Z / Shift+Z |

Deleting a note leaves a rest in its place: it keeps everything that follows from sliding backwards. What belonged to the note, though, goes with it — accidentals, fermatas, articulations, ornaments, stems and beams set by hand, and any ties or slurs involving it. Cmd/Ctrl+Z brings it all back.
Cmd/Ctrl+Z undoes the whole GESTURE, whatever it touched: the notes of the choir, those of the tracks and the signs on the score (dynamics, texts, slurs, pedals, changes). A command that changes several things at once — deleting a measure, applying a figure — is undone in one go, and a drag is undone back to where the sign was before you grabbed it. Opening a file or creating a new one clears the history: Cmd/Ctrl+Z does not bring back the previous piece.

## 4.6 Manual ornament marking

With the T panel open, the “Ornament marking” section lets you force the ornament type. Shortcuts use the Option (Mac) or Alt (Windows) key:

| Shortcut | Type |
|---|---|
| Option+H | Structural (chord tone) |
| Option+P | Passing tone |
| Option+V | Neighbor tone |
| Option+A | Appoggiatura |
| Option+N | Anticipation |
| Option+S | Escape tone |
| Option+R | Suspension / Retardation |
| Option+O | Ornamental (figuration) |
| Option+C | Changing tone |

> ***Note:*** *On Mac the Control key is reserved for toggling note colours in the voices.*

What it applies to. The marking concerns the notes of the CHOIR: it tells the analysis whether a note belongs to the harmony or decorates it. On the notes of an accompaniment track the marking has no

effect on the choir's analysis, which reads the four voices: to make a track note count towards the choir's harmony, use the collapse into a single chord (9.9).

## 4.7 Voice reassignment

Notes that have already been written can be moved from one voice to another by reusing the toolbar S/A/T/B buttons, without rewriting them. The feature works both in SATB editing and on "Grand staff with voices" ACC tracks.

1. Select the notes (rectangle selection is supported).
2. Click the destination voice (S, A, T or B) in the toolbar: the notes move to that voice.
3. The clef is recomputed from the voice: for example S → Tenor/Bass moves correctly to the lower staff.

Behaviour:

- **Collisions → chord:** if a note already exists at that position in the destination voice, the two become a chord (nothing is lost).
- **Automatic rests:** the source voice fills in rests where the moved notes used to be.
- **Undoable:** the operation is reverted with Cmd/Ctrl+Z.

> ***Note: the S/A/T/B buttons convert notes only when you made the selection yourself, by clicking a note or dragging a marquee. The note you have just written stays selected for convenience, but it is not converted: pressing another voice simply sets up the next note.***

## 4.8 Moving notes with the mouse

A note can be moved vertically by dragging it: press on the notehead and move up or down. The preview follows the mouse one scale degree at a time — line, space, line — and the pitch is written on release. If the note belongs to the current selection the whole selection moves, so a chord can be transposed in a single gesture; otherwise only that note moves. It works on accompaniment tracks too.

The move is diatonic: you change line or space and the spelling comes from the key signature (in F major, B comes out as B flat), exactly as when clicking to enter a note. The ↑↓ arrows remain chromatic, semitone by semitone: they are two different gestures. If any note of the group would fall outside the piano range, the move is not applied.

### Moving a note in time

Dragging a note sideways, instead, moves it in time. The first movement decides the direction: start horizontally and the note changes position, start vertically and it changes pitch — one gesture never mixes the two.
The step is the value chosen in the toolbar: with sixteenths the note moves by as many sixteenths as you like, never by a thirty-second. The place it leaves becomes a rest. Holding Option (Alt on Windows) on release leaves the note where it was and moves a copy.
The note always stays in its own part. If it lands on another note of the same part, the one that starts earlier is shortened: dropped across the start of the next note, it shortens itself; dropped on the start of another note, it replaces it; dropped in the middle of a long note, that note is shortened up to there.
The barline shortens nothing: a note that crosses it keeps its value, splits and continues into the next bar with a tie. A tied note moves with its whole chain and is rewritten as a single value where it fits, and neighbouring rests merge into whole figures. Several selected notes move as a block. Notes in tuplets (triplets and the like) cannot be moved this way.

## 4.9 Enharmonic respelling (J)

The J key respells the selected notes with their enharmonic equivalent: D sharp becomes E flat and vice versa. The sound never changes; letter, accidental and staff position do. It is mostly needed on imported

files, where a wrongly spelled note distorts the chord symbol, the figured bass and the scale degree (see 9.8, Inconsistent spelling).

## 4.10 Number of parts (4, 3 or 2)

The choir is not necessarily in four voices. Under Score ▸ Parts you can choose 4 parts (S A T B), 3 parts (S A B) or 2 parts (S B). Disabled voices disappear from the toolbar and from the V key cycle, and can no longer be written.

The convention is that the outer voices always stay and the inner ones are removed: the lowest voice remains the Bass, so the rules that depend on it — inversions, doublings, motion of the outer parts — keep their meaning. Reducing the parts never deletes anything: if the voices to be switched off contain music, the change is refused with a warning. The number of parts belongs to the project and is saved in the file.

## 4.11 Where a note may start

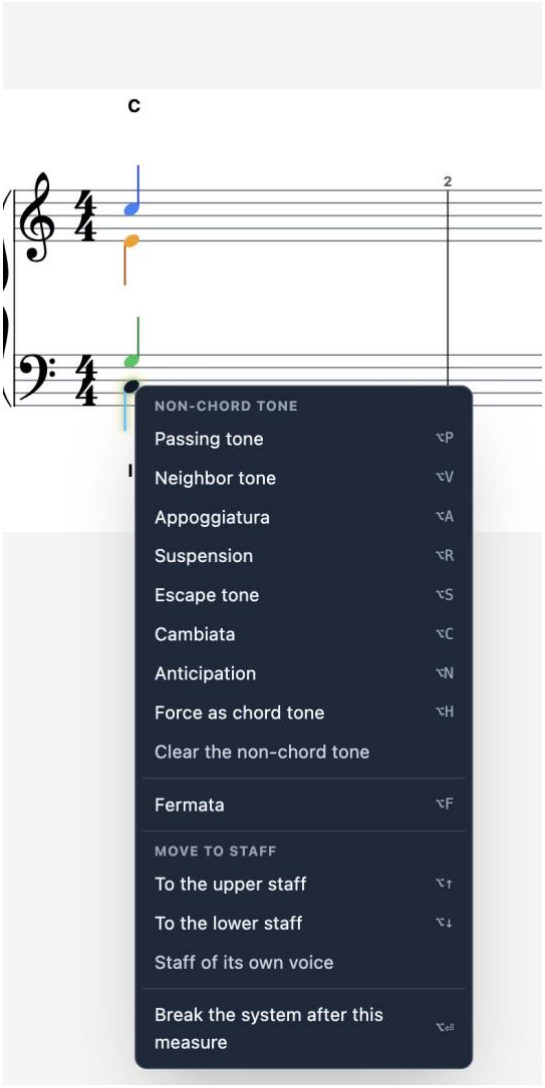
Where a note lands does not depend on the duration you are about to write, but on what is already written. A measure offers three kinds of place: the beats of the metre (an empty measure is written from its beats), the END of every existing event — that is where what follows begins — and the START of every event, which is how you stack a chord on top of a note. The place nearest your click wins.
Rests count exactly like notes: to attack on an offbeat, write the rest that precedes it — which is also the correct notation. If you have written an eighth note, the next place is an eighth away, and a quarter note can start there.
Tuplets are a case of their own, because their attacks do not fall on the ordinary grid: a triplet of quarter notes does not sit on the beats. Inside a tuplet already begun the ordinary places do not apply; when only the last note is missing, the group can still be abandoned.
Hold SHIFT to fall back to the free grid, for attacks that do not grow out of what is already there.
How strongly a note is drawn towards existing attacks is set in Preferences → Editor (“magnet strength on attacks”): stacking a chord and stepping just off one are the same gesture read from opposite sides, and the right amount depends on how you work.

## 4.12 Voice-crossing warning

When the note you have just written ends up below (or above) its neighbouring voice, a warning appears. It is not modal: you keep writing, and the warning waits.
It offers three answers: swap the two voices with one click, accept it as it is, or turn these warnings off — anyone writing counterpoint with deliberate crossings does that straight away (Preferences → Editor turns them back on).
It exists because the analysis reports crossings anyway (rule R-04), but as a matter of PART WRITING, and only when you open the panel — possibly days later. When the real cause is a note entered in the wrong voice — which happens above all with WHOLE NOTES, the one figure with no stem to show whose it is — that report arrives in the right place at the wrong time, dressed up as a harmony problem.

## 4.13 The right-click menu on a note

Right-clicking a note opens a menu with everything that concerns that note: the non-chord tones (passing note, neighbour note, appoggiatura, suspension, escape note, cambiata, anticipation), “force structural” to declare it part of the chord, and “clear ornament”.
Then the fermata, clearing articulations, moving the note to another staff (upper, lower, or the staff of its own voice) and “break the line after this measure”, which fixes where the system ends.
At the bottom of the menu, «Select the whole voice» selects every note of that note's voice (the menu says which, for example «Select the whole soprano voice»); «Add the voice to the selection» adds them to the notes already selected, for example to take soprano and bass together.




The right-click menu on a note: non-chord tones, fermata, move to staff and system break.

## 4.14 Writing a chord from its symbol

Besides placing notes one at a time, a chord can be written by naming it: type Cmaj7 and the four voices appear, already laid out.

Turn the command on with the **A♩** button in the toolbar, which turns cyan. A small box opens above the cursor; type the symbol there. Leave with Esc, or by pressing Return on an empty box.

The chord lands **where the cursor is** — not where you last clicked — and takes the duration armed in the toolbar. Confirm with the **right arrow** and the chord goes in while the cursor moves on by itself, so a whole progression can be written in one run. The left arrow steps back one chord.

The four voices are not a stack of pitches: they are built by the same chorale engine described in chapter 14, and they **connect to the previous chord**, looking for the best voice leading rather than repeating one fixed spacing.

The engine knows the key in force in the bar, also after a key-signature change and in minor, so it does not double the leading note.

Standard chord notation applies. A bass after the slash gives the inversion — Dm7/F puts F in the bass — and extended chords, which have more than four pitches, are reduced to four by omitting the fifth, as is customary. If the piece is written in three or two parts, the chord adapts.

Two conveniences for writing in sequence:

5. Cmd+Z and Cmd+Shift+Z act **on the score without leaving the box**: undoing the last chord does not mean closing the command and starting again.
6. On Italian keyboards the sharp appears where you expect it: the character à is turned into #.

If the active staff is an accompaniment track rather than the choir, the same symbol goes in there with the figure chosen in the palette (§ 11.3) — block chords, an arpeggio, an Alberti bass… — instead of the four chorale voices. With a figure, the chord lasts until the end of the measure.

## 4.15 More than one note in the same part

A voice sings one note at a time. An instrument does not: a violin plays double stops, a guitar strikes five or six sounds together, and often holds a bass while the melody moves above it. Harmony Tutor writes this too, with a single gesture.

### The gesture: Opt+Cmd

Cmd+click writes a note: if something of that part is already where you click, it replaces it. That is the gesture for correcting.
Opt+Cmd+click ADDS instead: the new note joins what is already there, removing nothing. It is the only gesture to remember, and it works on every choir staff. On accompaniment tracks the line you write in is chosen beforehand, with a click: see «On tracks» below.

### Chord or second line: the note value decides

What the added note becomes is not something you are asked: it follows from the value you have armed on the toolbar.
If it is the SAME value as the note already there, the two join into a chord: one stem, two heads. This is how the five sounds of a guitar chord are written — one note at a time, on the same onset.
If it is a DIFFERENT value, the new note takes a line of its own inside the same part, with the stem turned the other way. This is the sustained double stop: a half note held while eighths run above it. Two notes of different lengths cannot share one line without overlapping, and that is why the program separates them by itself.

### How the second line behaves

The second line exists only where it has something to say: no rests are written for it, neither where it is silent nor around its notes — this is how an engraver writes a secondary voice. For the same reason the measure check does not judge it: a bar containing a single double stop does not come out three quarters empty.
In the analysis the second line produces no voice-leading errors: parallels, leaps and resolutions speak of one line per part, and a double stop is not an extra part. Nor does it enter the chord reading, unless you say so: select one of its notes and press Opt+H to declare it harmonic, and from then on it counts in the chord symbol and the figured bass.
This is the opposite of what happens in the choir, and it is worth keeping in mind: there you exclude what is ornamental and everything else counts; here you declare what counts, and the rest stays out. The reason is that a line without rests gives the program no metrical context on which to base the judgement "this is a passing note".

### On tracks: the line is chosen with a click

A track staff — a violin, a flute, a guitar — can have two lines: the main one and a second one, with its stems turned down. As with soprano and alto sharing a choir staff, you choose the line to write in first: a plain click, with no keys, above the middle line of the staff chooses the main line; below the middle line it chooses the second. The choice stays until you change it.
Cmd+click (Ctrl+click on Windows) then writes in the chosen line, at any pitch: the second voice can also sit above the middle line. Opt+Cmd+click (Alt+Ctrl+click on Windows) adds another note on the same onset to the chosen line, making a chord. The preview note already shows the stem of the line it will go to; with voice colours on, the second line has a colour of its own — in treble clef blue and amber, like soprano and alto.
A MusicXML file with two voices on a single staff, for example written in MuseScore, comes in the same way: the first voice becomes the main line, the others the second line. On export the second line comes out as a voice of its own.

### Lengthening a note

When you change a note value, the notes that fall inside its new duration are removed: the long note wins, as in every notation program. Without this rule they would stay written underneath it, invisible as a problem but audible all the same. Cmd+Z undoes it, like any other change.
If instead you change the value of a note belonging to a chord, every note of that chord changes with it. Two heads on one stem necessarily have the same duration: that is not a convenience, it is the definition.

# 5. Measure-edit panel (T)

Pressing T with the playhead on a measure opens the side panel with the editing tools.

## 5.1 Modulation / Tonicisation

Sets a key change from the current measure. Select the new tonic and mode (Maj/min), then “Apply context”.

## 5.2 Measure

- Delete measure: removes the current measure and shifts everything after it back — the choir, the accompaniment tracks and the signs.
- **Repeat:** inserts repeat barlines (open, close, double).
- **Meter change:** applies a meter change from the current measure.

Adding or deleting a measure shifts the WHOLE piece: the notes of the choir and of the tracks, dynamics, texts, slurs, pedals, changes of metre, key and metronome, repeats, voltas, system breaks and chords fixed by hand. When deleting, whatever lay entirely inside the measure goes; a slur, a hairpin or a pedal crossing it is shortened; a change of metre or key written in that measure moves to the next one. A single Cmd/Ctrl+Z puts everything back.

## 5.3 Harmony override

Manually forces the harmonic label with three fields: Roman (e.g. I, V/vi, Ger+), Figure (e.g. 6/5) and Chord symbol (e.g. D7/F#).

## 5.4 Move to staff

Moves a note between the treble and bass staff with Option+arrow up/down (Mac) or Alt+arrow (Windows). Allows mixed open/close-score writing on the same system. “Restore default staff” returns the note to the staff assigned by its voice.

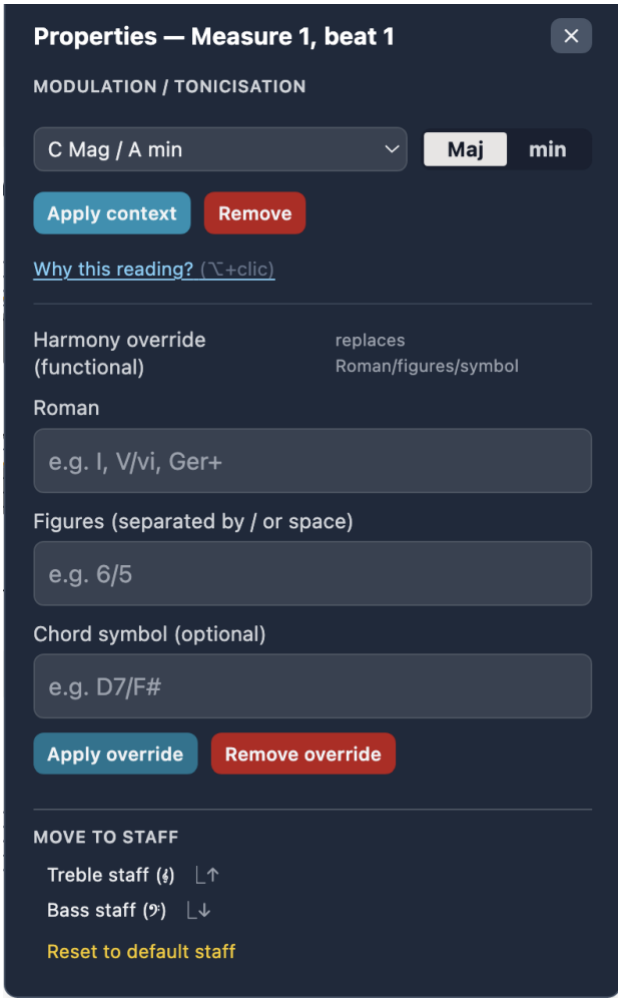



*Figure 9 — The T panel: modulation, override, repeat, delete measure*

# 6. Key and meter

## 6.1 Key selection

All 24 major and minor keys. Global change from the toolbar or per single measure from the T panel.

## 6.2 Minor mode

- **Natural (aeolian):** natural minor scale, with no raised leading tone.
- **Harmonic:** raised 7th degree to create the leading tone (V–i).

The “Auto leading tone” flag in the toolbar automatically inserts the leading-tone accidental in minor mode: the 7th degree is written already altered, without doing it on every note.

## 6.3 Modulations

- **Automatic recognition:** cadential patterns tested against all candidate tonics.
- **Manual marking:** via the T panel.
- **Look-ahead tonicisations:** the engine looks up to 2 measures ahead.

7. Modulation confirmed by a cadence: the new key starts from the predominant of the confirming cadence, not from the first chord that could already be read in the new key; after a double bar the example starts again in the home key.
8. ♭VII followed by IV, in major, is shown as «IV/IV», with ♭VII among the alternative readings.

## 6.4 Meter and BPM

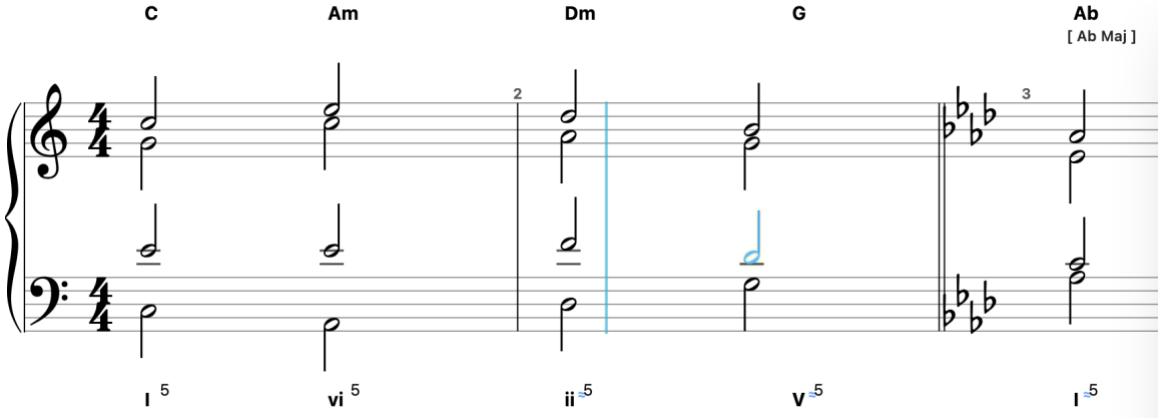
Default meter: 4/4. Available: 2/4, 2/2, 3/4, 3/8, 4/4, 5/4, 6/8, 7/8, 12/8 and any custom meter. Meter changes from the T panel. Metronome toggled with K.

The **BPM** field in the toolbar shows the tempo in force where the caret is: if there is a metronome mark («♩ = 120») earlier on, it shows that one; otherwise the starting tempo of the piece. During playback it follows the changes.
Typing a number into it, or using the up and down arrows, changes what you see: the metronome mark that governs that point, or the starting tempo if there is no mark before it. A mark written in another unit («𝅗𝅥 = 60») stays in its own unit.

The metronome (K) beats this tempo: with the piece stopped it follows the number in the field; while the piece plays it follows the metronome marks, rit. and accel., and waits during a fermata's pause. Real-time recording (REC, chapter 12) also starts from the number in the field, then follows the metronome marks, rit. and accel. it meets.

## 6.5 Key change within the piece

Besides the main key, the key signature can change at any point of the piece. In the signs palette, Structure group, pick the key and drag the ♯♭ button onto the measure it applies from: the new key signature appears, preceded by the naturals that cancel the previous one, and the following systems carry it at the start of the staff.




*The key change at bar three: the double barline marks the boundary and the analysis carries on in the new key.*

Notes are not moved. What changes is the key signature, and therefore which accidentals are printed: from there on you write in the new key, and clicking the A line in A flat major gives you an A flat, with nothing to add by hand.
It applies to the whole piece — choir and tracks alike — because the key belongs to the piece, not to the staff.
The analysis follows: from that point Roman numerals and figures read in the new key. The T panel is still there for when the two do not coincide, that is a modulation without a key change.
The change brings a double barline with it, which the analysis treats as a boundary: fifths and octaves across the change are no longer reported. Crossing a key change is not a voice-leading mistake.
To remove a change: right-click on the key signature, or «Remove key change» in the palette, which acts on the measure at the cursor.

### 6.6 Anacrusis (pickup)

A piece that starts with a pickup is declared from the Score menu → Anacrusis, or from the palette, Structure tab. You choose the length of the pickup — in 4/4 an eighth, one, two or three quarters — and the first bar becomes bar 0: shorter, and not counted. The first full bar is bar 1, as in printed editions.
The pickup starts empty and you fill it by writing. The metronome beats the downbeat on bar 1, not on the first note of the pickup, and the analysis reads the pickup as the weak beat it is. In MusicXML files the pickup is written as an implicit bar, the way MuseScore, Finale and Sibelius write it; a file that declares one reopens with its pickup.

#### The repeat compensates

When a repeat goes back to the beginning of the piece, the bar that closes it is shortened by the length of the pickup: going back, that bar and the pickup make one full bar, as it is written on paper. With first and second endings, the first-ending bar is shortened. A final bar without a repeat stays full: where nothing repeats there is nothing to compensate. The «The repeat compensates» tick turns this off.

#### Shorter bars in the middle of the piece

A section in the middle of the piece can also start with a pickup: then it is the bar before the double barline that holds only the beginning of the bar. In the palette, Structure tab, the «Bar length» row gives any bar a length shorter than the time signature: a click applies it to the bar at the cursor, or you drag it onto the bar you want; «full» makes it whole again. The bar keeps its number, and the notes of the following bars stay where they are. If the part that would be removed holds notes, the bar is not shortened and a message says why.

## 7. Harmonic analysis

### 7.1 How it works

Harmony Tutor analyses the notes in real time through an 8-block pipeline. The result is a harmonic label (Roman numeral) below the staff for each chord and/or modern chord symbols above the staff, toggleable from the toolbar selector.

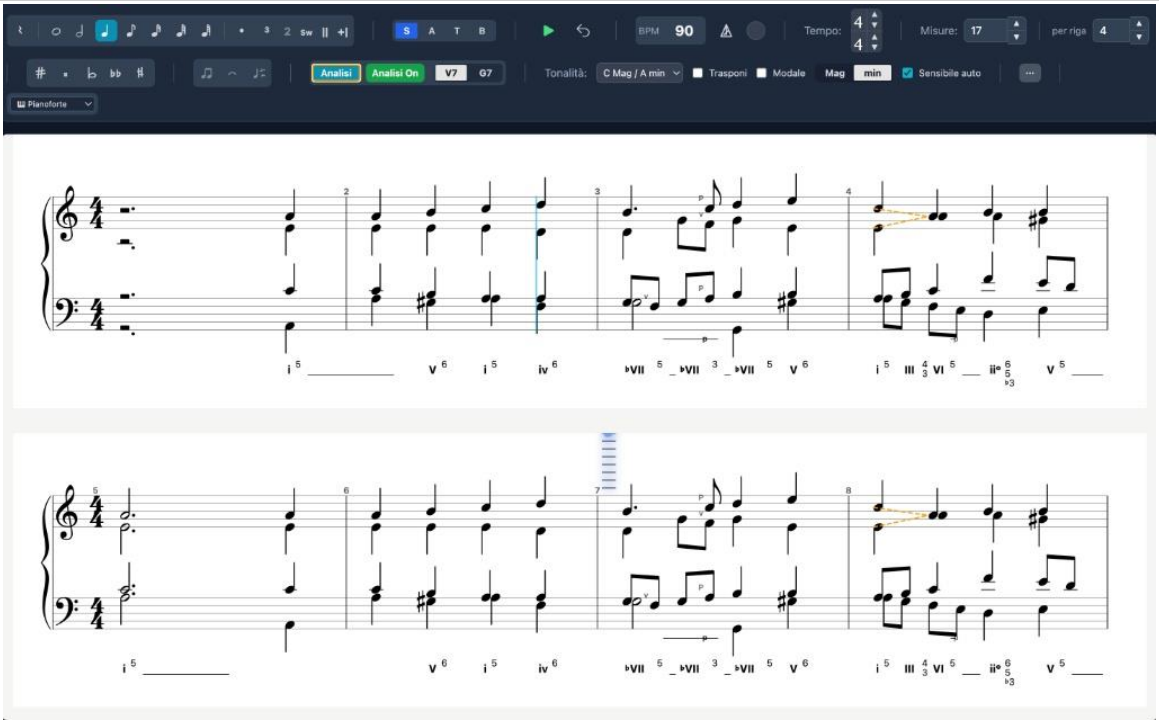



*Figure 10 — Score with active harmonic analysis*

## 7.2 The pipeline

The diagram below illustrates the full flow from note entry to label rendering.

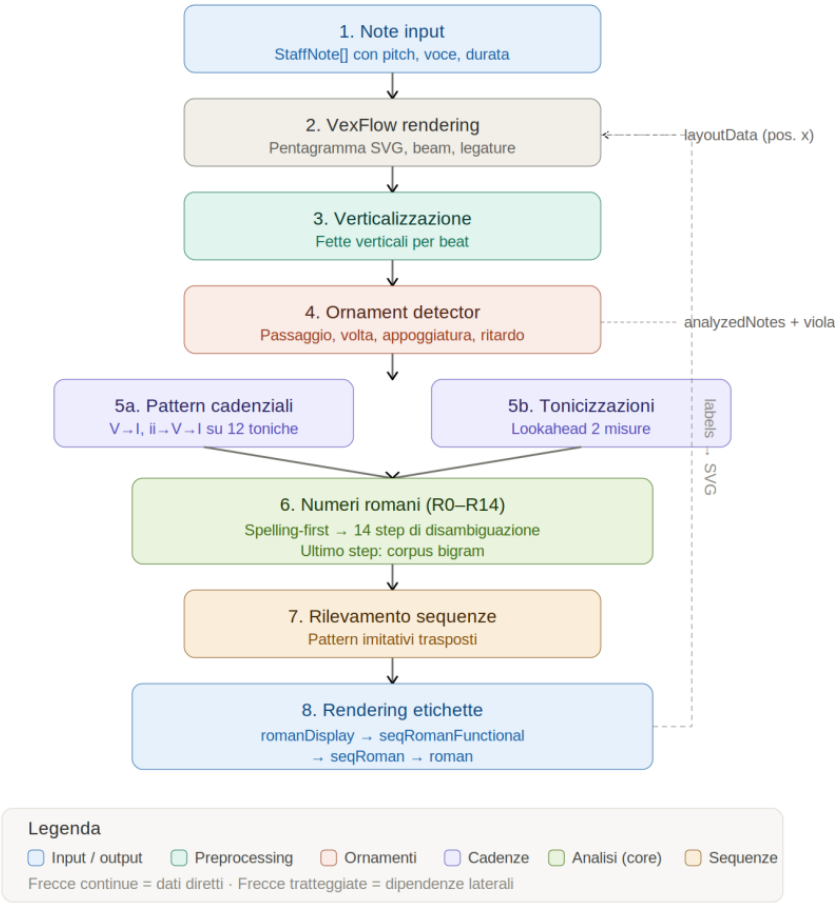



*Figure 11 — Harmonic-analysis pipeline: 8 blocks*

- **1. Verticalisation:** grouping of simultaneous notes.
- **2. Ornaments:** identification and separation of non-chord tones.
- **3. Chord identification:** spelling-first analysis, candidates ranked by plausibility.
- **4. Roman numeral (R0–R14):** 14 disambiguation steps; the last one uses the statistical corpus.
- **5. Cadential patterns:** recognition of cadences and tonicisations.
- **6. SATB rules:** over 40 voice-leading rules.
- **7. Sequences:** recurring melodic patterns (optional).
- **8. Rendering:** labels below the staff (and symbols above, if enabled).

## 7.3 Recognised chords

### Triads

| Type | Example (C) | Intervals |
|---|---|---|
| Major | C–E–G | 1–3–5 |
| Minor | C–E♭–G | 1–♭3–5 |
| Diminished | C–E♭–G♭ | 1–♭3–♭5 |
| Augmented | C–E–G♯ | 1–3–♯5 |

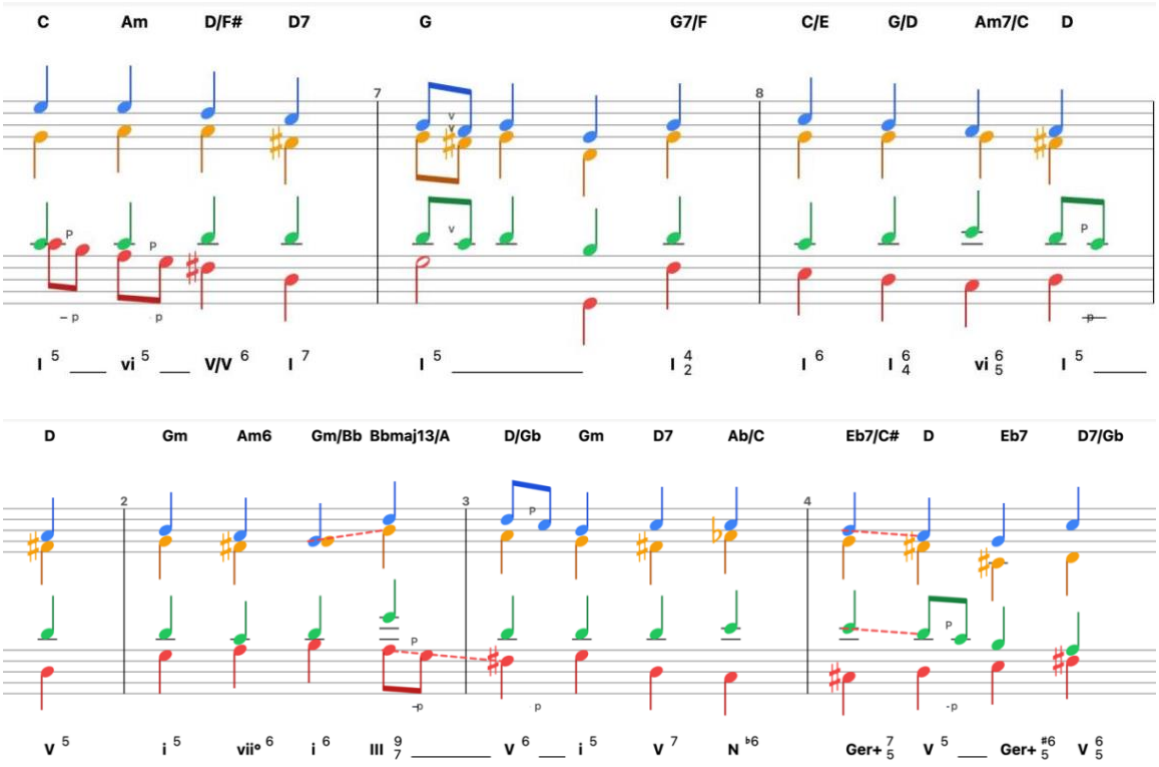
### Sevenths

| Type | Example (C) | Intervals |
|---|---|---|
| Dominant 7 | C–E–G–B♭ | 1–3–5–♭7 |
| Major 7 | C–E–G–B | 1–3–5–7 |
| Minor 7 | C–E♭–G–B♭ | 1–♭3–5–♭7 |

| Type | Example (C) | Intervals |
|---|---|---|
| Diminished 7 | C–E♭–G♭–B𝄫 | 1–♭3–♭5–𝄫7 |
| Half-diminished | C–E♭–G♭–B♭ | 1–♭3–♭5–♭7 |

### Special chords

- **Augmented sixths:** Italian (It6), French (Fr6), German (Ger6).
- **Neapolitan (♭II):** major triad on the lowered 2nd degree.
- **Dominant ninths:** V9, V7♭9, V7♯9.




*Figures 12-13 — Coloured voices with symbols and Roman numerals, ornaments (P, v)*

**Analysis of instrumental tracks (ACC).** The engine described in this chapter works on the SATB chorale. Accompaniment tracks have a dedicated harmonic analysis (chord symbol above the staff and Roman numeral below, with harmonic rhythm and manual overrides): see §11.8.

## 7.4 Alternative readings are chosen on the label

A chord often admits more than one reading: the same sonority can be a seventh degree or a rootless dominant, and at a modulation it may belong to two keys. Clicking a Roman numeral opens a compact menu below the label with the possible readings: you pick one and that is that. The choice is stored in the file.

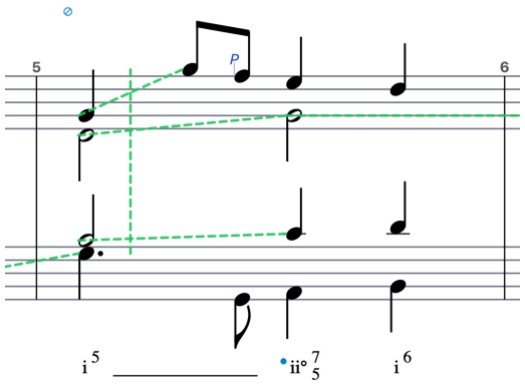
The measure properties panel (the T key) remains the place where modulations and tonicisations are declared.

A reading chosen in another key — for example «I⁶ in E♭» — does not apply to that chord alone: the chosen key carries on over the following chords as long as they stay in it, and stops at the first one that contradicts it. The latest choice always wins.

### Your choices show

Where you wrote or chose a label yourself, there is a light-blue dot next to the Roman numeral: hovering over it says so, and a click opens the menu to correct or remove it.

Where you removed a key the program had inferred, «∅» appears above the staff. Right-click the sign to bring the key back.




*Your choices on the staff: «∅» where you removed the key the program had inferred; the light-blue dot next to the label where you chose the key yourself.*

### 7.5 Labels in modulations

When the piece moves to another key, the chord that belongs to both — the pivot — carries two readings joined by «=»: its degree in the old key and in the new one. Moving from C major to G major, for example, the A minor chord that prepares the cadence reads «vi=ii». After the pivot the degrees are read in the new key: its tonic is «I».
A dominant that leads to a degree of the key, after which the music goes straight back home, is a secondary dominant, written «V/V», «V⁷/vi»… If instead the music rests on the arrival chord for a long time, more than half a bar, it is a cadence in the key of that degree: the dominant is called «V», and the arrival carries the double reading — «I=V», for example, if the degree is the fifth.
A chain of dominants — E⁷, A⁷, D⁷, G⁷, C — is read link by link: each chord is the dominant of the next.
In minor the natural seventh degree (F in G minor) is written «VII», without ♭: «vii°», lower case, is the chord built on the leading note.

## 8. Non-chord tones

Harmony Tutor automatically recognises non-chord tones and excludes them from the chord.

| Type | Marker | Feature |
|---|---|---|
| Passing tone | P | Stepwise between chord tones, weak beat |
| Neighbor tone | v | Steps away from and back to the real note |
| Incomplete neighbor tone | V / A | Reached by a leap, resolves by step in the opposite direction |
| Appoggiatura | A | Dissonance on a strong beat, resolves by step |
| Anticipation | N | Anticipates a note of the next chord |
| Escape tone | S | Leap + stepwise resolution |
| Suspension/Retardation | R | Held note, resolves downward |
| Changing tone | C | Ornament with a leap from the dissonance |

Suspensions are classified as 4–3, 7–6, 9–8, 2–1. Manual marking uses the Option (Mac) or Alt (Windows) key + the corresponding letter.
The incomplete neighbour note is reached by a leap and resolves by step in the opposite direction: on a weak beat it is marked V, like the neighbour note, on a strong beat A, like the appoggiatura.

### 8.1 Suspension or appoggiatura: the tie decides

The two figures are written the same way and are told apart by how the dissonance arrives. A suspension is prepared: the note was already sounding in the previous chord and stays put while the harmony changes underneath it — and that preparation is written with a tie. Without the tie the same note is an appoggiatura, a dissonance that lands on the accent unannounced.

The label says which: R for the suspension (ritardo), A for the appoggiatura. Remove the tie from a suspension and you will see the R turn into an A.

### 8.2 The ornaments the program learns from your corrections

When you mark by hand an ornament the program had not recognised, that correction does not stay locked inside the file: it is collected into a table which the analysis consults whenever it meets a similar case.
It is there to break ties, not to force a reading: it only steps in where two readings are equivalent and something has to decide. The mechanism is the same one the program has long used for harmonic progressions, where the profile extracted from the corpus guides the choice of degrees.

### 8.3 Letters on the staff and the panel

The passing notes, neighbour notes and escape tones the program recognises carry their letter on the staff (P, V, S) and do not appear in the panel, which is kept for errors, warnings, exceptions and observations.
A note moving on a subdivision of the beat, over a bass that holds the harmony, does not create a new chord: the label stays that of the chord that continues. The beat is the real beat of the metre — in 4/2 it is the half note, in compound metres the dotted quarter.

## 9. Rules and violations

The analysis panel flags violations with coloured badges. Every flag is expandable with a description and a tip.

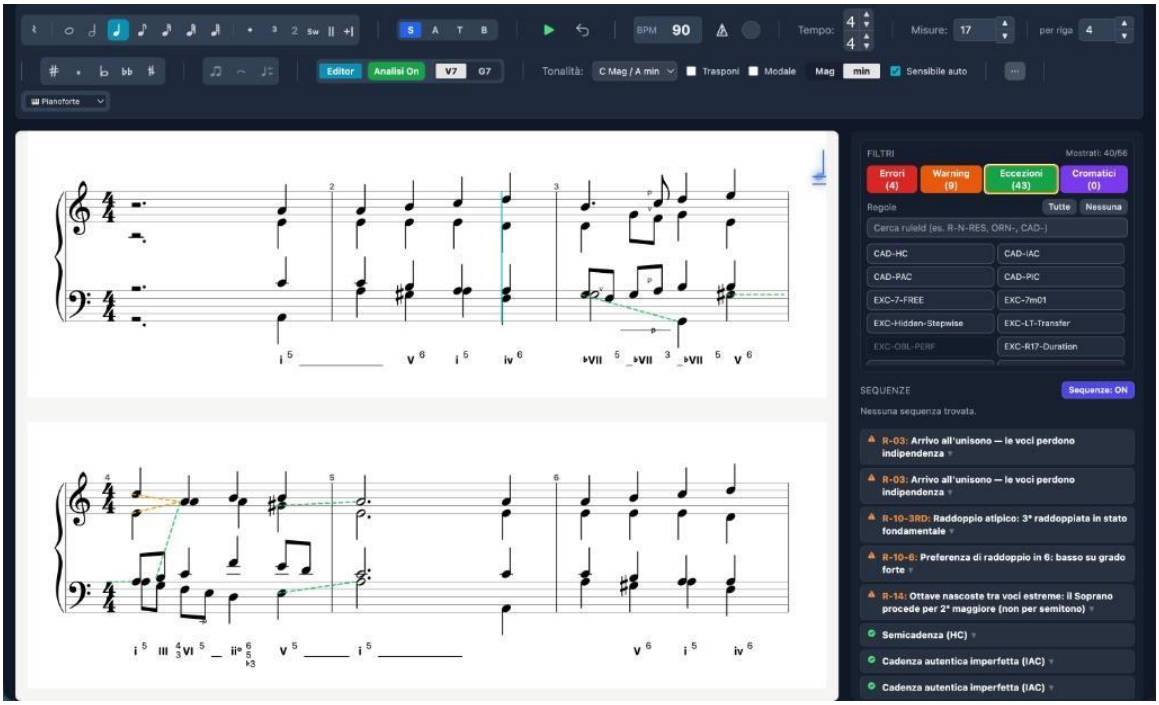



*Figure 13 — Analysis panel with filters, violations and cadences*

### 9.1 Severity levels

| Colour | Severity | Meaning |
|---|---|---|
| Red | Error | Serious violation of fundamental rules |
| Orange | Warning | Situation to avoid, not always forbidden |
| Green | Exception | Special case tolerated by tradition |
| Purple | Chromatic | Advanced chromatic harmony |
| Light blue | Observation | Correctly written but placed so that it does not perform its function |

## 9.2 Errors (red)

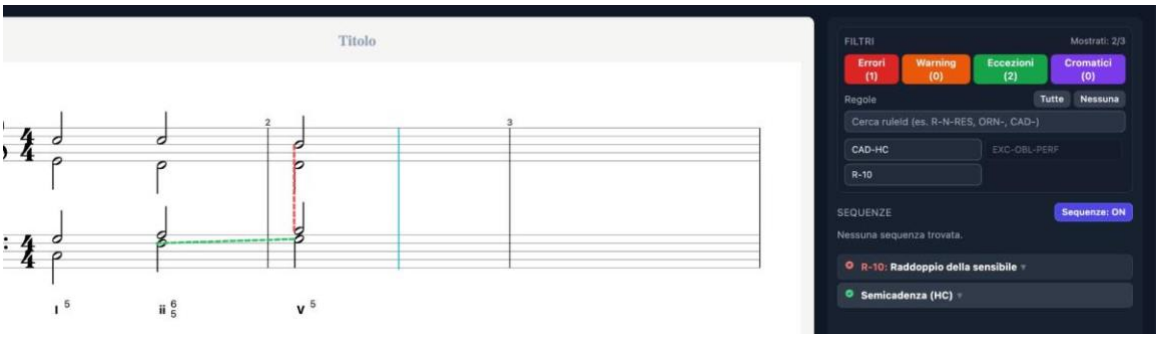



*Figure 14 — Example error R-10 (doubled leading tone)*

| Code | Rule |
|---|---|
| R-01 | Parallel octaves/unisons |
| R-02 | Parallel fifths |
| R-04 | Voice crossing |
| R-07 | Leading tone does not resolve |
| R-09 | Chromatic false relation |
| R-10 | Doubled leading tone |
| R-12 | Seventh does not resolve |

## 9.3 Warnings (orange)

| Code | Rule |
|---|---|
| R-05 | Hidden fifths/octaves |
| R-08 | Excessive spacing S–A / A–T |
| R-13 | Parallel motion in all voices |
| R-15 | Wide leaps in an inner voice |
| R-16 | Harmonic syncopation |
| R-17a/b/c | Problematic melodic leaps |
| R-RANGE | Note out of range |

## 9.4 Exceptions (green)

Special cases tolerated by tradition, flagged with a green badge in the analysis panel (e.g. hidden octaves/fifths between outer voices moving by step).

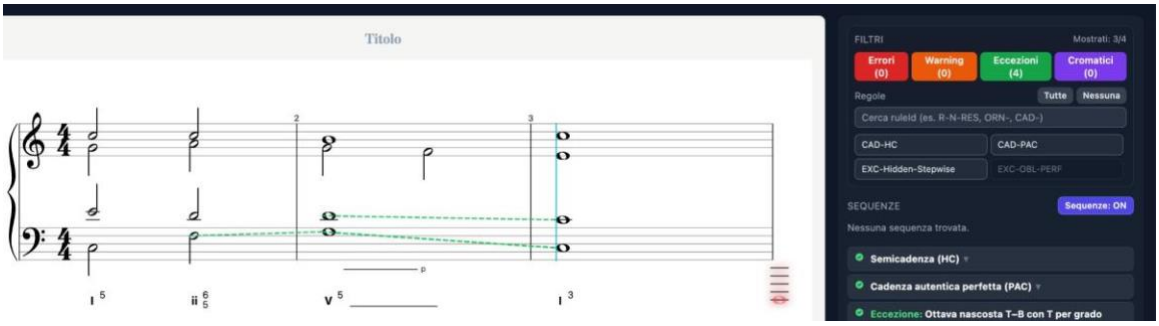



*Figure 15 — Exceptions: PAC cadence, HC half cadence, hidden-by-step*

### Allowed syncopation: two exceptions to the barline rule

The “barline rule” (R-16) discourages a chord that enters on a weak beat and simply continues onto the downbeat of the next measure: the metrical accent would fall on harmony already heard, and the downbeat would carry nothing new. There are two cases where the writing is sound, and the analysis recognises them as green exceptions.

The first is a change of INVERSION: if the chord returns on the downbeat over a different bass, the harmony is renewed and the accent finds a new event. It is the bass that decides this, not the upper parts.
The second is a RHYTHMIC PATTERN: if the same anticipation returns on the same beat for at least three consecutive measures, it is not an oversight but a deliberate figure — typical of ostinato accompaniment and compound metres. With only two measures the repetition may be accidental, and the rule stands.
Every rule that speaks of strong and weak beats reads the metre in force in that measure: in a piece that moves between 3/4 and 4/4 the third beat changes nature.

## 9.5 Recognised cadences

| Code | Type | Formula |
|---|---|---|
| CAD-PAC | Perfect authentic | V → I (root position) |
| CAD-IAC | Imperfect authentic | V → I (with inversion) |
| CAD-HC | Half cadence | Phrase ending on V |
| CAD-PLAG | Plagal | IV → I |
| **CAD-DEC** | **Deceptive cadence** | V → vi (maj.) / V → VI (min.) / V → ♭VI (chrom.) |

Among the clues the analysis weighs when deciding whether a passage changes key or stays put there is also the half cadence: a pause on the dominant which, rather than announcing a modulation, confirms the key you are already in.

## 9.6 The analysis panel

- **Filters:** Errors, Warnings, Exceptions, Chromatic checkboxes with counts.
- **Rules:** clickable chips + search field (e.g. “R-01”, “CAD”).
- **Click a Roman numeral:** opens the Explanation card with all the details and alternative readings (≈).
- **Disable rule:** can be silenced individually; persists across sessions.
- Copying a text: highlight with the mouse the title or the explanation of an entry, or the text of a window, and press Cmd+C (Ctrl+C on Windows), or right-click → «Copy». Useful, for example, to paste it into a report or an e-mail.

### From the warning to the place where you fix it

Analysis labels are not only there to be read: clicking a roman numeral or a chord symbol takes you to where that reading is changed, without hunting for the right command in the panel.
Alternative readings — when a chord can be understood in more than one way — appear where you write, next to the music, instead of sitting in a separate list.
A rule's advice can change with the situation, too. On a crossing between notes that are all STEMLESS, for instance, the advice no longer talks about part writing: it tells you to check which voice the flagged notes ended up in. On whole notes, a crossing is nearly always an entry mistake.

## 9.7 Cadential 6/4 → V⁶⁄₄

In the cadential progression I⁶⁄₄ → V → I, the I⁶⁄₄ is automatically relabelled as V⁶⁄₄, following the modern functional reading.

- Relabelling happens automatically when the engine detects the V⁶⁄₄ → V → I progression.
- Manual overrides (T panel) always take priority.

> ***Note:*** *With frequent modulations, check the tonal context before interpreting the V⁶⁄₄ label.*

## 9.8 Inconsistent spelling

When the notes as written do not form any chord — for instance D sharp, G and B flat, where D sharp to G is a diminished fourth rather than a third — the analysis reports Inconsistent spelling and points at the note to respell. The J key fixes it.

In that case the chord symbol shown is a reading of the sounds and does not match what is written: to make that explicit, it appears in parentheses. The Roman numeral survives the wrong spelling, because it is computed from the sounds; the figured bass does not, and shows distorted intervals until the note is corrected.

## 9.9 Collapse into a single chord (Opt+Shift+H)

This is for harmony spread over time rather than stacked: an arpeggio, for instance, produces confusing symbols note by note. Selecting the notes — even on different beats — and pressing Opt+Shift+H makes the analysis read them as one chord: the full label is pinned to the onset nearest the cursor. The other onsets lose their own label, so one chord no longer produces four identical symbols — except for those still sounding at that point, which are a harmony in their own right and keep theirs.

How to do it. 1) Move the cursor — the vertical line — to the point where you want the label to appear: it is the cursor, not the selection, that decides, because the command anchors the selected onset nearest to it. 2) Select the notes to be read together. 3) Press Opt+Shift+H. If the cursor is not placed, the label goes to the first onset of the selection.

With the analysis in SATB mode the command gathers the selected notes of the choir AND those of the accompaniment tracks. This is how a passage is disambiguated: select the choir notes together with the track note — typically a bass that clarifies an ambiguous chord — and the label is recomputed on the union. Selecting only the track notes, instead, derives the chord from those alone: it is the easiest mistake to make.

To undo it, repeat Opt+Shift+H on the same selection: the collapse is removed and the automatic analysis returns. Alternatively, use "remove override" in the T panel. Note that overrides are NOT undone by Cmd+Z, which acts on notes.

The label is not frozen. It remembers which notes it was made from and follows them: delete one of them and it is recomputed on the rest — take the B out of a C major seventh and it goes back to reading C — and when fewer than two are left it disappears, letting the automatic analysis return. Note that the lowest of the selected sounds decides the inversion, so annexing a note below the choir bass also changes the figure on the Roman numeral: the chord you declared has a different bass.

## 9.10 Observations (light blue)

The fifth category in the panel, alongside errors, warnings, exceptions and chromatic findings. It holds what is correctly written but placed so that it does not perform the function it claims: not a voice-leading error, and not merely a matter of style either.
The founding case is the cadential 6/4 placed the wrong way round. In the compound cadence the 6/4 falls on the strong beat and the dominant on the weak one: it is not the succession of chords that makes the cadence, it is the metrical relation between the two. If the 6/4 sits in a weaker position than its resolution, or lasts less, the figure is well built — the sixth falls to the fifth, the fourth to the third — but the two terms are reversed and it does not work as a cadence.
The check first works out which species of 6/4 it is looking at, because their metrical requirements are opposite. The passing 6/4 (the bass reaches it and leaves it by step) and the neighbour 6/4 (the bass stays put, and the following chord is the previous one returning) want the weak beat, and are not flagged. It also stays silent when the 6/4 is preparing a suspension — that is, when one of its dissonances goes on sounding into the chord of resolution: there the weak beat is the right place, because a prepared dissonance is prepared on the weak beat.

## 9.11 The augmented sixth

In an augmented sixth chord two notes form the interval that defines it: the augmented sixth — A♭–F♯ in C — or, inverted, the diminished third F♯–A♭. The note a semitone above the dominant (♭6, the A♭) tends to fall, the one a semitone below (♯4, the F♯) to rise: they converge on the octave of the dominant. The

program recognises the sixth from the spelling of the two notes, whatever the chord is called, and follows its resolution:

- no signal: the two notes open onto the octave of the dominant, also through the cadential six-four or a passing chord;
- observation: the two notes do what they should, but the arrival chord is not the dominant;
- warning: one of the two does not resolve, or the resolution passes to another voice;
- error: one of the two goes against its tendency — the ♯4 falls or the ♭6 rises. If the ♯4 falls by a chromatic semitone (F♯ → F), the advice suggests writing it as a seventh (G♭): there it works as a seventh, not as a leading note.

An ornamented resolution counts as a resolution: the note that arrives through an appoggiatura or a suspension (D♯ – F♯ – E), or that moves away and comes back before resolving. A sixth made by a passing note, or by a short ornamental note, is not a chord and is not judged. On the staff the dashed line starts from the note that does not go where it should.

## 9.12 Doublings, crossings and spacing: when they are checked

The checks on the chord — doublings (of the leading note, of the seventh, in the six-four…), crossings between voices, spacing between parts — are made at every change of harmony, on any beat, and when on a strong beat a voice changes position while staying on the chord. What counts are the notes that make the harmony: a passing note does not create a doubling, and instead of a suspension its note of resolution counts. Crossings and doublings are reported only between notes that really sound together.

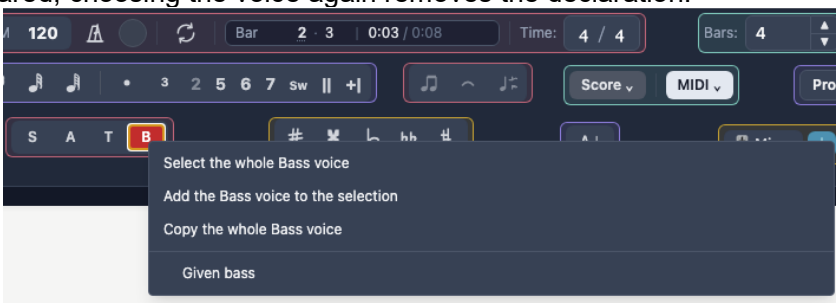
The range of the voices is checked on every note, passing notes included. Whether the chord is complete (a missing third) is checked on strong beats.

## 9.13 Exercises with a given voice (given bass, given soprano)

In harmony exercises one voice is often given by the assignment — the bass, sometimes the soprano — and the student writes the others. If you tell the program, mistakes lying entirely in the given voice are not reported: they are not the student's.

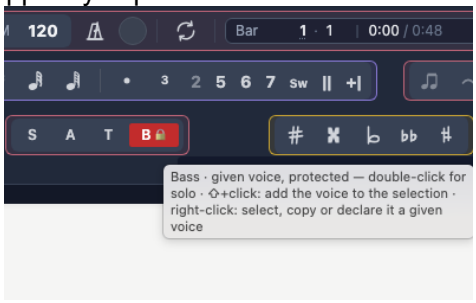
Step 1. In the toolbar, right-click the voice button: S, A, T or B.

Step 2. In the menu choose «Given bass» (or «Given soprano», «Given alto», «Given tenor»). Several voices can be declared; choosing the voice again removes the declaration.

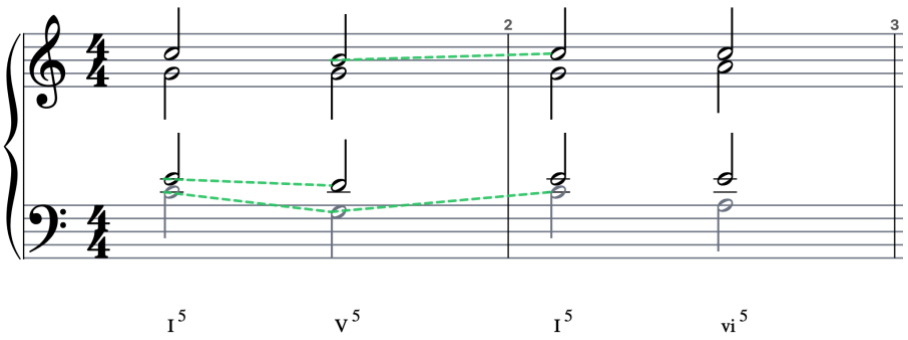



*The right-click menu of the «B» button: at the bottom, «Given bass».*

The notes of the given voice turn grey and its button shows a padlock. They are protected: a gesture that would change them by mistake — moving, deleting, changing their duration — stops with a message. To correct the assignment, choose «Unlock the bass for editing» (or the voice you declared) in the same menu: the unlock lasts while the app stays open. The declaration is saved in the piece.




*The given bass button shows the padlock; its tooltip says the voice is protected.*




*The notes of the given bass in grey; the voices written by the student in black.*

In the analysis, a mistake involving the given voice and a written voice — fifths between bass and tenor, for example — stays reported, with the advice to change the voice written by the student: «The bass is given: change the tenor».

## 9.14 Notes on the piece

Every piece has a place for writing about it: the teacher's explanation for the students and the notes of whoever studies it. The text is saved in the piece's file and travels with it.

### Opening the panel

Use the «Notes» button in the toolbar, next to «Analysis», or the View › «Notes on the piece» menu (Cmd+Alt+4, Ctrl+Alt+4 on Windows). The panel sits in the right-hand column and alternates with the findings list: the «Analysis» button brings the findings back, and a second click on «Notes» closes the column so the score takes the whole width.

### The two parts

- «Teacher's analysis» — the explanation of the piece, written by the teacher for the students. A piece that contains one opens the panel by itself.
- «Reminders» — free notes of whoever studies the piece: observations, doubts, an AI's answer (see below).

### Writing

The «Write» button (or «Edit», if there is already some text) opens the writing box; «Done», or the Esc key, closes it. While you write, the keys go to the text, not to the score.

Above the box, the symbol bar inserts at the cursor the signs that are not on the keyboard: ♭ ♯ ♮ ° ø $^{6}$ $^{7}$ $^{6}_{4}$ $^{6}_{5}$ $^{4}_{3}$ $^{4}_{2}$ → –.

Bars mentioned in the text become links. Writing «bar 3», «b. 3», «bars 9–12» or «m. 5», the reference shows in light blue, underlined, when the text is read: a click takes the cursor to that bar.

If the teacher has locked the piece (chapter 15), the teacher's analysis can be read but not changed; the reminders belong to whoever uses the file.

### Prepare for your AI

If you subscribe to an artificial intelligence — ChatGPT, Claude, Gemini… — you can ask it to comment on the piece starting from Harmony Tutor's analysis. There is nothing to connect or set up: you copy and paste.

Step 1. Press «Prepare for your AI», at the top of the notes panel, or choose File › «Prepare for your AI…».

Step 2. The program copies the piece to the clipboard — title, key, time signature, the notes of the four voices at every onset, the chord symbol, degree and figures of every chord, the findings — together with a ready-made analysis request. A message in the panel confirms the copy.

Step 3. Open your AI's chat, paste (Cmd+V, Ctrl+V on Windows) and send.

Step 4. If the answer is useful, copy it and paste it into «Reminders»: it stays saved in the piece.

The request asks the AI to check Harmony Tutor's analysis, to say where it disagrees and to organise its answer in eight parts: overview, functional analysis, chords outside the key, cadences, style and repertoire, voice leading, the debatable points of the analysis, three questions or exercises for the student. An AI can be wrong even when it sounds sure, though: always check its answer.

The copied text is in the app's language. If the teacher has disabled exporting for the piece, nothing is copied, and the degrees, chord symbols or findings the teacher has hidden are left out.

# 10. Audio and MIDI playback

## 10.1 Internal playback

Internal playback uses the Web Audio API. Instruments use good-quality sampled timbres (orchestral set, double bass and basses) loaded locally; timbres without local samples fall back to the General MIDI set (FluidR3). The default timbre is Piano. The full palette of selectable timbres (keyboards, strings, basses, woodwinds, brass and tuned percussion) is listed in 11.6 Available instruments.

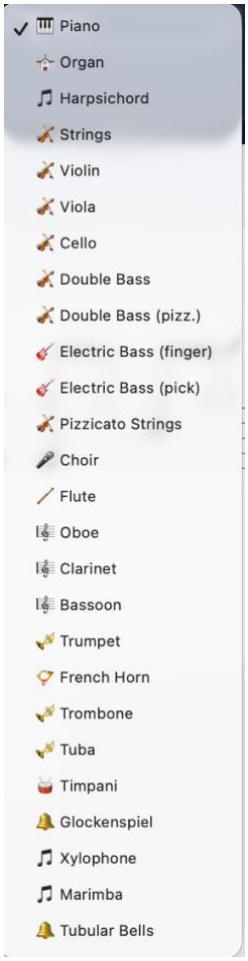



*Figure 16 — Available timbres*

### Dynamics (velocity)

Each note has a dynamics (velocity) value that affects playback and export. Velocity is captured from MIDI import and live recording, and is used for more expressive playback, for real-time monitoring and for external MIDI output. It is also preserved faithfully in MIDI export.

The velocity of every note can be seen and edited in the velocity lane (see 11.13).

## 10.2 Controls

| Shortcut | Action |
|---|---|
| Space | Play / Stop |
| Enter | Return to start |

| Shortcut | Action |
|---|---|
| O | Loop the selection (O again turns it off) |
| K | Metronome |
| + / – | Tempo |
| Double-click S/A/T/B | Solo voice |

### The loop (O key)

To hear a passage again, select the notes or measures to repeat and press O, or the loop button — the two arrows in a circle, next to the metronome in the toolbar: an amber band marks the stretch, and playback repeats it from the attack of the first selected note to the end of the last, even when it starts mid-measure. There is no gap between one pass and the next.
Pressing O again, or the button, turns the loop off; if you turn it off while playing, the piece ends with the current pass. Changing piece turns the loop off by itself, and audio export ignores it. The loop also works towards an external MIDI instrument: on stop every note is turned off, from every pass.

### The position and jumping to a bar

In the toolbar, next to the loop, a box tells you where the caret is: «Bar 24 · 3 | 1:12 / 3:40» means bar 24, third beat, one minute twelve seconds from the start of the piece, out of three minutes forty. Beats are counted in the unit of the meter: crotchets in 4/4, quavers in 6/8. During playback the numbers move on. The length takes into account how the piece is played: repeats and endings, metronome marks, accelerando and rallentando. Fermatas are not counted.
To go to a bar, click its number, type the bar and press Enter (Esc cancels): the caret moves to the start of that bar and the page brings it into view. If the piece is playing, it starts again from there.

## 10.3 External MIDI

Harmony Tutor supports MIDI keyboards on input and sending MIDI out to an external instrument or DAW (on Mac, via the IAC Driver). When an external MIDI output is selected:

- by default the internal sound is silenced and you listen through the external instrument or DAW; each mixer channel can, however, also or only play inside the app (see «Where each channel plays»);
- the controls that come from the player — the sustain pedal (CC64) and the other pedals, the modulation wheel, expression — arriving from the keyboard are forwarded, to avoid stuck notes; volume and pan are sent by the mixer, and are never sent back;
- on stop, any notes still active are explicitly turned off (note-off flush).

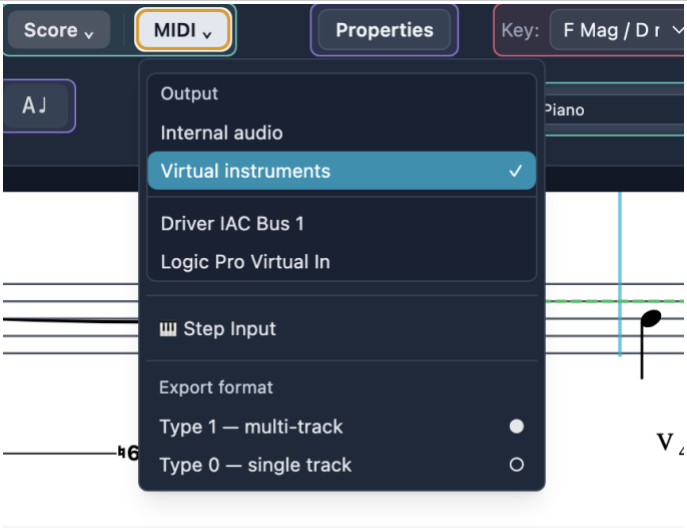
External playback is velocity-sensitive. Configuration is done from the Preferences panel (MIDI tab) or the side menu.

### The MIDI menu in the toolbar: where the piece plays

The «MIDI» button in the toolbar opens, under the heading «Output», the list of places where the piece can play:

- «Internal audio»: Harmony Tutor's own sounds. It is the default; the ✓ tick shows which choice is active.
- «Virtual instruments»: the plug-in instruments loaded into the parts (see 11.15). On the Mac it appears when Harmony Tutor Plugin Host is running; on Windows it is always there.
- Below the line, the external MIDI ports — for example «IAC Driver Bus 1» towards a DAW — or «No MIDI device». The list is read again every time the menu opens: an interface connected later appears straight away.

At the bottom of the same menu are «Step Input» and «Export format».




*The «MIDI» menu: the output (here «Virtual instruments»), external ports, Step Input and the export format.*

### Where each channel plays: MIDI, Internal, Both

With an external MIDI output selected, every mixer strip — choir voices, tracks and added instruments (see 11.11) — shows a menu with three choices under the MIDI channel:

9. Plays: MIDI — only towards the external instrument. It is the default.
10. Plays: Internal — only inside the app, with the program's sounds; nothing goes out. Useful to keep a sound that the external setup does not have.
11. Plays: Both — goes out to the external instrument and is also heard inside the app.

The menu turns green when it is not on «MIDI», and the choice is saved with the piece. Without an external output the menu does not appear: everything plays inside. Audio export always contains every channel, whatever the choice.

If the same bus (for example «IAC Driver Bus 1») carries the notes to the DAW and is also an input, the program recognises the messages that come back to it: it does not send them out again or write them as notes, so there are no endless loops. If instead a DAW track sends MIDI back on the same bus, the loop has to be broken in the DAW, giving it a second bus for its output.

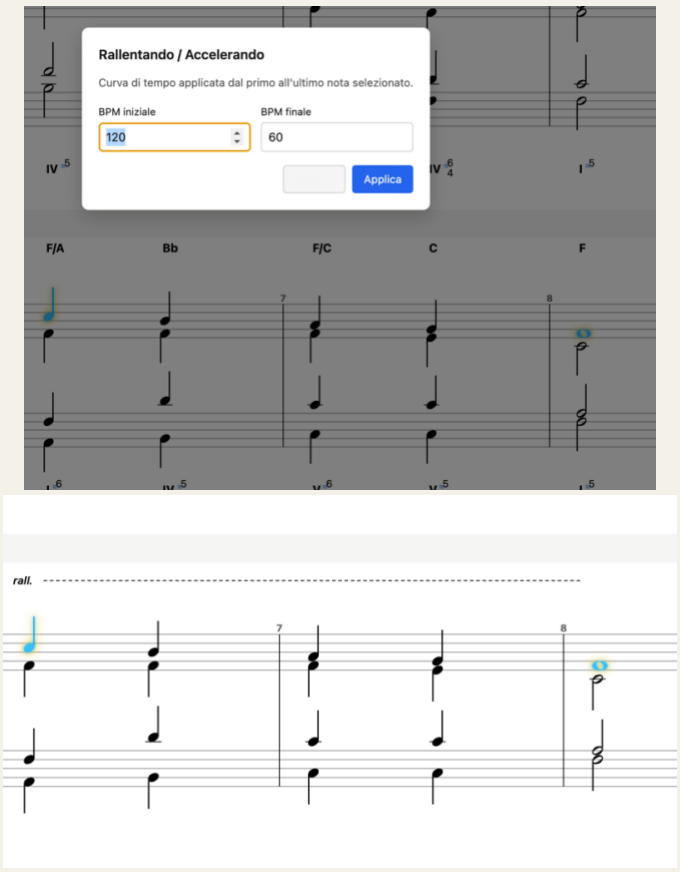
## 10.4 Tempo curve — Rallentando / Accelerando

The TempoCurve function creates gradual tempo changes over a range of notes. Playback interpolates the BPM linearly between the start and end value. The curve is shown as an overlay on the staff.

### How to insert a tempo curve

1. Select the start and end note of the range (Shift+Click for multi-selection).
2. TempoCurve menu, or Shift+Option/Alt+R for a rallentando.
3. Set the start BPM and end BPM.
4. Confirm: the graphic overlay appears on the staff.

> ***Note:*** *The rallentando/accelerando also applies correctly to the notes of ACC tracks and to loaded files.*




*TempoCurve — overlay on the staff with start and end BPM*

### Moving the start and the end

When the mouse passes over the ends of the sign, small arrows appear. Dragging the end of the dashed line lengthens or shortens the curve; dragging the left edge of the word «rall.» or «accel.» moves its start. The end snaps to the nearest note, and the curve can carry on into the next line. A click on the word edits the values; right-click removes the sign.

## 10.5 Restart the audio engine

If sound stops coming out — headphones unplugged, output device changed, audio system not ready when the app started — «Restart audio engine», in the Score menu of the toolbar, rebuilds the engine without closing the program. Your score is untouched.
It is needed because from the inside there is no way to tell that sound is not reaching the speakers: the system reports whether the engine is running, not whether sound comes out. A typical symptom is seeing the mixer meters move in silence.

## 10.6 A chord from the MIDI keyboard enters on every part

In step input — the Step input button in the MIDI menu of the toolbar — a chord played on the keyboard enters on all the choir voices at once. The notes are distributed from the highest to the lowest across the active voices, according to the number of parts set in Score → Parts (4, 3 or 2), and the playhead advances only once.
A chord played by hand does not arrive as a single message: the notes arrive separately, a few milliseconds apart. The program collects them for an eighth of a second before deciding whether it is looking at a chord or a single note. The wait applies to step input only: real-time recording is untouched.

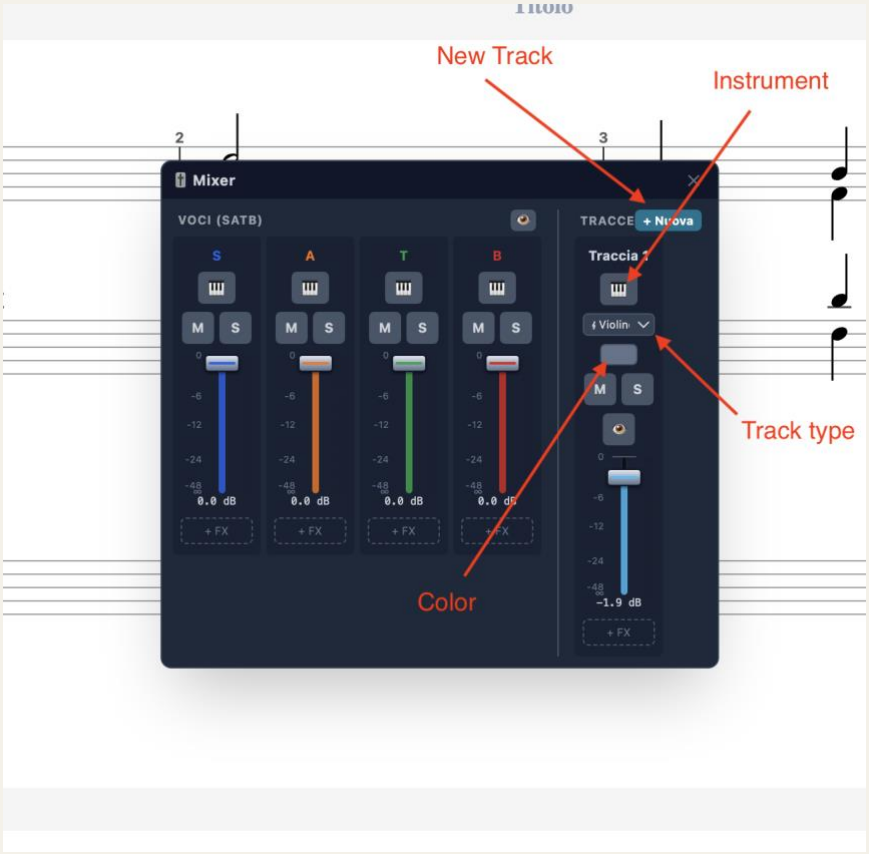
On accompaniment tracks the number of choir parts does not count: the chord goes in with all its notes. On a grand staff with voices the notes are distributed across the voices — the highest to 1, the lowest to 4, the others to 2 or 3 —; on a plain grand staff they all go in together.

### Which port it listens to

With Step input on, the MIDI menu shows the «Listen to» list: choose the port of your keyboard there. Notes arriving from other ports — for instance from another music program that is open and uses the system MIDI bus — are ignored and do not enter the score by themselves. «All ports» listens to all of them. The choice is remembered.

## 11. Accompaniment tracks (ACC)

Accompaniment (ACC) tracks are independent of the SATB voices. They are ideal for figured bass, rhythmic patterns or arrangements, without interfering with the harmonic analysis of the chorale.




*ACC tracks panel — add track, name, MIDI instrument*

### 11.1 Adding and managing tracks

1. Open the tracks panel from the Mixer or the side menu.
2. Click the “+” button and choose the track type (see 11.4).
3. Assign a name and a MIDI instrument to the track.

ACC notes are visually distinct from SATB notes on the staff, including via the track colour. Right-clicking a Mixer strip deletes the track.

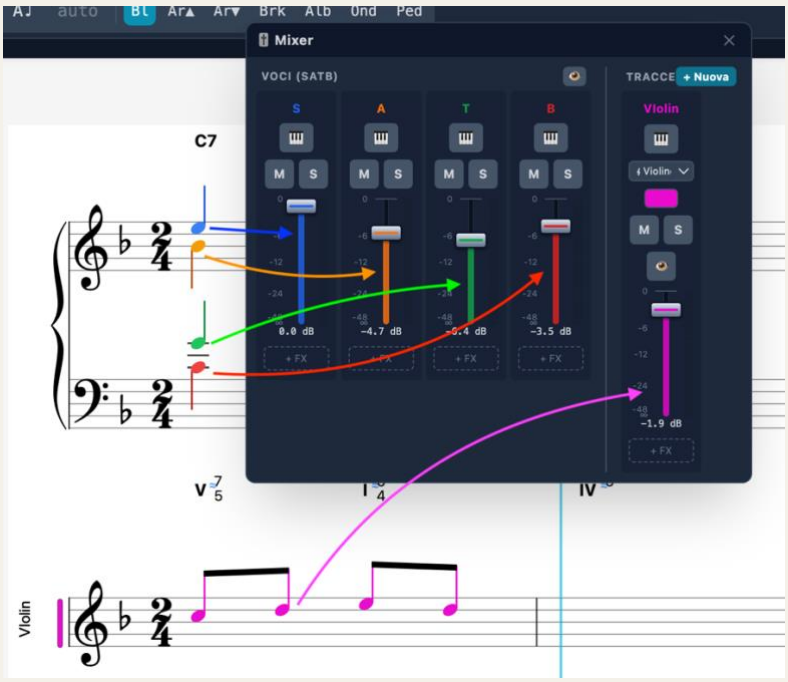
### Orchestral section brackets

Two different brackets can appear to the left of the staves, answering two different questions. The black one gathers instruments of the same family — strings with strings, woodwinds with woodwinds — and is derived from the instrument chosen for the track. The blue one, further out, gathers the staves the

analysis reads together. So an analysis group can span several families, and two instruments of the same family can be analysed separately.

## 11.2 Mixer

The Mixer is a floating, draggable panel that gathers control of the four SATB voices and all ACC tracks in a single window. It replaces the old sidebar mixer and opens from the Mixer button in the toolbar. The panel is split into two sections: VOICES (SATB) on the left and TRACKS on the right.




*Mixer panel — VOICES (SATB) and TRACKS sections*

*Figure — Mixer: VOICES (SATB) and TRACKS sections*

### VOICES section (SATB)

One strip per voice: Soprano, Alto, Tenor, Bass. At the top of the section, a visibility button hides or shows the whole SATB block in the editor.

| Control | Description |
|---|---|
| Instrument | MIDI instrument assigned to the voice. |
| M (Mute) | Mute / unmute the voice. |
| S (Solo) | Isolate the voice in playback. |
| Volume fader | Voice gain (0–100%), coloured with the voice colour. Adjustable without stopping playback. |
| MIDI channel | Outgoing MIDI channel for the external MIDI output: Auto (default) or a fixed 1–16. |

### TRACKS section (accompaniment)

One strip per ACC track. The “+” button (with its menu) adds a track; right-clicking a strip deletes it.

| Control | Description |
|---|---|
| Track name | Editable label, shown in the track colour. |
| Staff / clef | Grand staff (treble+bass with a brace) or a single staff (treble, bass, soprano, alto, tenor). Bass and treble are also available as octave-down (8vb) transposing clefs (double/electric bass and guitar): they sound one octave lower than written, while the notation is unchanged. |

| Control | Description |
|---|---|
| Colour | Colour picker: vertical band to the left of the staff, tints the track's notes and the fader. SATB voices keep their fixed colours. |
| M (Mute) | Mute / unmute the track. |
| S (Solo) | Isolate the track in playback. |
| Visibility (eye) | Show / hide the track in the editor without removing it. |
| Volume fader | Track gain (0–100%). |
| MIDI channel | Outgoing MIDI channel for the external MIDI output: Auto (default) or a fixed 1–16 (Auto: voices 1–4, ACC tracks from 5 up, drums always 10). |

***Note:*** *Mute, Solo and Volume act in real time during playback (no need to stop and restart).*

***Note:*** *The Mixer settings (instruments, volumes, mute, solo, colour and the track staff/clef configuration) are saved in the project file and restored on reopening. Older files without this data start from the defaults (piano, full volume, no mute).*

### Adding tracks — the “+” menu

In the TRACKS header, a single “+” button opens a small menu: “New track” (a pitched accompaniment track) or “Drums” (a percussion track). This keeps the fader area uncluttered. The new track appears as a strip on the right.

### Master faders (SATB, ACC and global)

Besides the per-channel faders, the Mixer has three master faders: SATB (controls the four voices together), ACC (controls all accompaniment tracks together) and MIX (the global output of the whole mixer). Routing is hierarchical — each channel feeds its group master, and both group masters feed the global master — so the SATB master scales only the voices, the ACC master only the tracks, and the MIX master everything. Each master has its own level meter, and the three values are saved with the project.

### MIDI output channel

Every voice and track strip has a MIDI-channel selector for the external MIDI output (toward a DAW such as Logic): Auto (default) or a fixed channel 1–16. With Auto, the four voices use channels 1–4 and the accompaniment tracks are assigned from channel 5 upward; a drum track always goes to channel 10.

### Changing the instrument of several voices/tracks at once

The instrument selector in the toolbar normally changes the active voice or track. If you first select notes belonging to several voices (or several ACC tracks), choosing an instrument applies it to all the voices/tracks present in the selection at once — handy for assigning, say, the same string sound to two inner voices in a single step.

### Per-channel FX: EQ, Compressor, Pan and Reverb

Below the pan, every strip has its INSERTS: a «+ FX» menu to add an effect and a small rectangle for each one inserted. You only see what is actually there — an effect you do not use takes no space — and clicking the rectangle opens that effect's window. The × that appears on hover removes it. The effects

available on every channel are the three-band Equalizer, the Compressor and the Reverb, each with its own plug-in style window. Pan stays on the strip, because it is not an effect but a position.
Next to each inserted effect there is □: it switches the effect off without removing it, settings included, so you can hear the difference; the rectangle turns grey and a click switches it back on. The × removes it.

### Reverb: a few shared spaces

Reverb is not an effect that each channel owns: there are A FEW SHARED SPACES — two out of the box, «Reverb A» and «Reverb B» — and every channel chooses which one to send to, and how much. This is not about saving resources, it is a musical choice: putting all the voices in the same space is what makes them sound like one ensemble. Four voices in four different halls sound like four recordings glued together, and it is a flaw you notice at once.
The window opens from the «Rev» insert and is divided in three. At the top, what concerns ONLY the channel you opened it from: how much it sends and into which space. In the middle the space itself, with a drawing of how it is built and its knobs — and the window says plainly that it is shared, and with how many channels, because moving it there moves it for all of them. At the bottom, an explanation of what you are touching.
Mix is the channel's knob, and it is the only one you need for normal use. At 0% you hear the channel only; at 100% you hear it ONLY through the space. It does not add reverb on top of the sound: it moves the channel from one side to the other, taking away direct sound as it puts in tail.

12. Decay is the length of the tail; Pre-delay is the silence before the space answers, that is how far away the walls are; Early is the first reflections, and it is the knob that makes you hear A ROOM instead of a generic fog; Damping darkens the tail; Shape decides whether it dies away at once or lingers; Level is how much of that space reaches the master.
13. Room, Hall and Plate are not three fixed reverbs but three STARTING POINTS: you load one and then you move it. The name of the space («Reverb A») stays whatever you call it.
14. The master has its own «Rev» insert too: it adds space to the whole ensemble in one go, without going channel by channel.




Figure — The reverb window: the channel's Mix, the shared space with its knobs and the drawing of the tail.

### Group EQ and Compressor (SATB and ACC)

Besides the inserts on individual channels and on the global master, the two group buses — SATB and ACC — each have their own three-band EQ and Compressor, opened from the EQ/Comp buttons below

their master faders. They act on the whole block: the SATB group EQ/Comp shapes all the voices together, the ACC group one all the tracks, before the signal reaches the global master. This is handy to 'glue' and balance the choir and the accompaniment separately. The settings are saved with the project.

### Sound bank per voice and track: Orchestral or GM

Next to the instrument selector in the toolbar, a second menu chooses the sound bank of the active voice or track: Orchestral (the local orchestral samples, the default) or GM (the General MIDI soundfont). The two timbres have a different character — the orchestral one has a more realistic attack but can sound a little 'harder', GM is more uniform and softer — and they can be mixed freely, choosing a different one for each voice or track (for example GM on the inner parts and Orchestral on the melody). The choice is saved with the project.

### Multiple channel selection and group actions

At the top of every strip there is a selection dot: click it to select the channel, and the strip is highlighted with a coloured border. Shift-click a second channel to also select every channel between the two. When at least one channel is selected, an action bar appears at the top of the mixer with the group actions: Mute and Solo (toggle all selected channels together), Delete (removes the selected ACC tracks — SATB voices cannot be deleted) and Deselect. In addition, dragging the fader of a selected channel moves all selected faders together by the same amount, preserving the group's relative balance.

### Reordering and copying track settings

ACC tracks can be reordered from the right-click menu on the strip (Move left / Move right): the order changes both in the mixer and in the score. Also from the right-click menu, Copy settings and Paste settings transfer the clef and the whole effects chain (EQ, Compressor, Pan, Reverb) from one track to another, without touching the notes or the volume.
Mixer channels can be selected, and the Delete key removes the selected ones.
The equaliser window shows the spectrum of the sound, before and after the correction: you can see the effect of what you are adjusting instead of having to imagine it.

## 11.3 Accompaniment patterns

On an ACC track a chord need not be written as a block: it can be written with an accompaniment FIGURE — an arpeggio, an Alberti bass, a waltz, a ballad — in piano notation, with two hands. The figure is chosen in the palette, Arpeggiator section, from a drop-down menu; below the menu a line says how the chosen figure is written, so you know what will appear on the staff before you write it.
The figures come in four groups:

15. Chords — Block chords; Repeated chords in eighths.
16. Basses — Bass and chord (oom-pah); Waltz (bass-chord-chord); Alberti bass; Pop ballad.
17. Arpeggios — ascending, descending, wave; broken chord in the left hand with a block right hand, and its reverse (broken chord in the right hand, left hand held); wide arpeggio (1-5-8-10); arpeggio over a held bass; two-hand arpeggio; ascending arpeggio and broken chord in triplets.
18. Styles — Ballad, Pop rock, Arpeggio 1, Arpeggio 2, Arpeggio 3.

### How a figure is written

The figure takes from the chord only WHICH notes it has and which one is in the bass: the bass is the lowest written note, so an inversion stays an inversion. It then lays them out in a register suited to the piano — the bass low, with the octave where the figure calls for it, the right hand in close position around middle C. So even a chord entered from its symbol, which the program voices as for a choir, comes out the way a pianist would write it.
It is written on the grand staff with voices (see 11.4): the right hand in voices 1-2, the left in voices 3-4, with durations split on the beats and ties where they are needed. On plain grand-staff tracks, where each staff has a single voice, the figures that need two overlapping layers in the same hand stay visible but greyed out, and the tooltip says why.

### Voicing, subdivision, metre

19. Voicing — the button above the menu turns the right hand: each press changes the note on top (for C major: C-E-G, then E-G-C, then G-C-E, and back to the start). The bass does not move, and the counter shows where you are in the cycle.
20. Subdivision — for the simple figures (arpeggios, Alberti, broken chords) you choose the note value they are written in: quarters, eighths, sixteenths or triplets. In compound metres (6/8, 9/8,

12/8) eighths and sixteenths remain: there the beat's three eighths already are the figure. In the style figures the rhythm is part of the figure, and the selector is off.

21. Metre — a figure meant for one metre is greyed out in the others, with the reason in the tooltip: the Waltz only in 3/4, the Ballad in 4/4. What counts is the metre of the measure where the chord falls; the figure menu never changes the time signature of the piece. A triplet figure is written with the bracket and the 3 in simple metres, as plain eighths in compound ones.

### How long a figure lasts

When you enter a chord from its symbol with a figure active, the chord lasts until the end of the measure, whatever duration is chosen in the toolbar: the common case is one chord per measure, and you type the symbols one after another. A chord entered in the middle of another shortens it: the first is rewritten, with its figure and voicing, up to the second one's attack.
For a different duration, select the chord and choose the value in the toolbar (or with keys 1-7, or the dot): on a chord written by a figure, the value is the duration of the WHOLE figure, not of each of its notes. A one-measure arpeggio in eighths, given a half note, becomes half an arpeggio, still in eighths. When lengthened, the figure stops at the next chord's attack. With «Block chords» the toolbar duration works as always.

### Changing the figure of a chord already written

Select the chord — one note is enough — and choose another figure, or another subdivision: the chord is rewritten at once. Several chords can be selected and changed together, each on its own. Going from one figure to another the chord stays the one you wrote: the notes a figure adds (the octave of the bass, the tenth) or leaves out neither swell it nor use it up.
The «with pedal» switch, below the menu, makes every figure write its own pedal sign, lifting and pressing again at every change of harmony (see 20.8). With no ACC track in the piece, the section says an accompaniment track is needed for the figures.

## 11.4 Accompaniment track types

When creating a track with “+ New”, you can choose between several types, depending on the writing complexity needed:

| Track type | Description |
|---|---|
| Single staff (Treble, Bass, Soprano, Alto, Tenor) | A single staff in the chosen clef. Suitable for melodic lines, basses, single-voice parts. |
| Grand staff | Double staff (treble + bass) with a brace, for two-staff writing. |
| Grand staff with voices | Double staff with the full voice management of the SATB (see below). |

### Grand staff with voices

This is the most articulated track type and the only one that allows true polyphonic piano writing. It reuses the same engraving engine as the SATB, applied to a single renamable track. It has four independent voices, distributed across the two staves:

| Voice | Staff | Stems |
|---|---|---|
| Voice 1 | Treble (right hand) | up |
| Voice 2 | Treble (right hand) | down |
| Voice 3 | Bass (left hand) | up |
| Voice 4 | Bass (left hand) | down |

Because the voices are independent, each can have its own rhythm (polyrhythm) and contain chords: this is the substantial difference from the simple Grand staff, which does not handle rhythmically independent voices.

> ***Note:*** *In practice, a “Grand staff with voices” track is equivalent to a SATB, but as a standalone, renamable accompaniment track: ideal for piano parts or multi-voice writing with rhythmic independence.*

On “voiced grand staff” tracks the voice a note will belong to is chosen by WHERE you click — as in the chorale — and the ghost note shows it in the colour of that voice before you write. A single voice can carry a chord: notes stacked on the same attack stay in the same part, under one stem.

Keyboard tracks (piano and the like) are created already «with voices». To move notes already written into another voice, select them and press S, A, T or B: the note goes to that voice, and to the staff it belongs to (S and A above, T and B below).

### What each type is for

The three types are not levels of difficulty: they serve different purposes, and it pays to choose them for the job at hand.

22. **Single staff** — for parts that play a single line: a violin, a flute, a bass. From the clef menu (see 11.10) you choose the instrument directly, which gives the staff its name, clef and sound.
23. **Grand staff** — the piano as it is played. Each note goes by itself to the right staff according to its pitch: from middle C upwards on the upper staff, the rest below, with a single stem per chord. It is the right type when voices do not exist or do not matter: recording from a MIDI keyboard, importing a MIDI file, writing accompaniment figures such as arpeggios and repeated chords. A chord played in step input goes in whole, with all its notes.
24. **Grand staff with voices** — the piano written in independent parts, like a keyboard chorale or a page of Bach: each voice has its own rhythm and its own stem. The voice is chosen by where you click; a chord played in step input is distributed across the voices, the highest note to voice 1, the lowest to voice 4 and the others to voice 2 or 3.

## 11.5 Copy and paste between tracks

Copy/paste works between any tracks: between different ACC tracks and between SATB and ACC, in both directions. The flow is: copy the selection, choose the destination, paste.

- **Choosing the destination:** a single click on an ACC track's staff (or the SATB staff) sets that track as the active destination and positions the cursor, without inserting anything.
- **Automatic conversion:** pasted notes are adapted to the destination — on an ACC track they take the track's clef (or are split between treble and bass if it is a grand staff); in SATB they take the selected voice and clef.
- **Robust copy:** ACC-track selections and triplets are also copied correctly.

> ***Note:*** *Manual per-note markings (e.g. structural note, ornaments) are preserved during copy/paste between tracks.*

After every paste the cursor moves just past the pasted fragment, to the next whole beat: pressing Cmd/Ctrl+V several times repeats the same fragment — a measure of drums, an arpeggio — one after another. The fragment's length includes any rests that were copied.

### Copying an entire voice (button and shortcut)

To move a whole SATB voice onto an accompaniment track in one operation — instead of copying it line by line — copy the entire voice: right-click the S, A, T or B button in the toolbar and choose “Copy entire voice”, or select the voice and press Shift+Cmd/Ctrl+C. Then paste onto the destination ACC track as usual (Cmd/Ctrl+V, then click the track’s staff). This is the fastest way to send each of the four voices to a different instrument for a wider orchestration.

### Usage tip — from chorale to orchestration

One of the most effective uses of copy/paste between tracks is to use the SATB as the harmonic foundation for an arrangement. The typical flow: write and correct the four-part chorale using the real-time analysis, until the harmony is valid and the part-writing is flawless; then copy the individual voices onto the accompaniment tracks, assigning each one an orchestral destination (for example the four voices to a string section). This way the parts enter the orchestral texture already correct in terms of harmony and voice-leading: the checking work done on the chorale is not lost, but becomes the solid base on which to build the orchestration.

### Copying a whole track

Just as a whole choir voice can be copied, so can a whole accompaniment track: all its notes go to the clipboard and can be pasted wherever they are needed, even into another piece.

## 11.6 Available instruments

For every SATB voice and every accompaniment track you can choose the timbre from the instrument button in the Mixer. The sound palette has been expanded with sampled orchestral timbres, grouped by family:

- **Keyboards: Piano, Organ, Harpsichord.**
- **Strings: Strings (section), Violin, Viola, Cello, Double Bass, Pizzicato strings, Choir.**
- **Basses: Double Bass (pizzicato), Electric Bass (finger), Electric Bass (pick).**
- **Woodwinds: Flute, Oboe, Clarinet, Bassoon.**
- **Brass: Trumpet, Horn, Trombone, Tuba.**
- **Tuned percussion: Timpani, Glockenspiel, Xylophone, Marimba, Tubular bells.**

The chosen instrument applies both to internal playback and to external MIDI output. The basses (pizzicato double bass and electric basses) are natural-decay (one-shot) sounds, sampled across the instrument's typical low register; outside that range a note may not sound.

## 11.7 Drum track (percussion)

In addition to pitched accompaniment tracks, you can add a dedicated percussion track, on MIDI channel 10, to write drum-kit parts using standard notation.

### Adding a drum track

- **In the Mixer, TRACKS section, press the button to add a percussion (drum) track.**
- **On creation the floating Percussion module opens automatically, from which you insert the hits.**
- **Alternatively, you can include a drum track right when creating a new project, from the initial-tracks dialog.**

### Notation

The drum track uses standard percussion notation: a percussion clef and × noteheads for cymbals. Each kit piece (kick, snare, hi-hat, toms, cymbals…) has its own fixed position on the staff.

### The two kits

Two kits are available, selectable per track: Orchestral (default) and Rock. They change the sounds, the piece map and the overall timbre.

### Entering hits

- **Percussion module: toggled with the button in the toolbar (it appears only when a drum track exists). It shows the kit pieces; click a piece to insert it at the cursor position.**
- **With the mouse: click directly on the percussion staff at the desired point; the hit snaps automatically to the standard line for that piece.**

On playback the track plays the selected kit; on MIDI export it is sent on channel 10. If swing is active, the drum eighth notes also follow the triplet feel.

## 11.8 Harmonic analysis of ACC tracks

Besides the SATB chorale (Chapter 7), Harmony Tutor can harmonically analyse an instrumental piece written on an accompaniment (ACC) track: useful for studying the harmonic structure of an imported or

freely composed piece. For each chord it shows the **chord symbol** above the staff and the **Roman numeral** below.
**Activation.** A global switch toggles the analysis subject between **SATB** and **ACC** (the two modes are mutually exclusive). In ACC mode you choose which track to analyse with the dedicated selector.
**Harmonic rhythm.** The engine automatically segments the flow into chords, detecting where the harmony changes even with irregular durations; a long note is taken into account in every beat where it sounds. Segmentation favours the points of greatest harmonic clarity.
**Manual overrides.** The automatic analysis can be corrected by hand:
• **Ornament marking** (Opt+O on a note): excludes that note from the chord reading — useful for passing or neighbour tones.
• Collapse to chord (Opt+Shift+H on a selection): declares that the selected notes form a single chord and writes the label where the cursor is. The same shortcut on the same selection undoes it. See §9.9.
ACC-track overrides are saved per track together with the project.

### Combined analysis (parts on separate staves)

When a piece is spread across several ACC staves — typically the parts of a MIDI file imported as separate staves (see 13.1) — the analysis reads them as a whole. Tracks belonging to the same group are combined by onset and analysed together: the chord symbol and Roman numeral appear above the group's top staff and reflect the complete harmony, not the single part.
Analysis group. Multi-track import automatically assigns the same group to all parts imported together. To add or remove a track from the group, click the clef on its staff and use «Include in combined analysis» at the bottom of the menu. An unrelated track is not included until you add it explicitly; the group is saved with the project.
Setting the analysis to an ACC track. The group forms around the currently analysed ACC track, so first point the analysis at an accompaniment track: with analysis on, in the toolbar pick ACC in the SATB / ACC switch and choose the track to analyse from the drop-down that appears next to it. Then, from the clef menu of the other tracks, use «Include in combined analysis» to add them to the group.
Layout. Staves in the same group are joined on the left by a bracket and connected by the barlines, like a score system, and are shown closer together for an immediate overview.

### Effects heard while auditioning

Volume, mute, solo, pan, EQ, compressor and reverb are heard when clicking or entering a note as well, not only during playback: auditioning goes through the same mixer channels. Adjusting an effect can be judged immediately, without starting playback.

### Bringing back a hidden choir staff

Creating a project without SATB does not delete the choir staff but hides it: the four voices exist, empty. In the mixer, at the top of the "Voices (SATB)" column, the dot switches the staff on and off — the tooltip reads "SATB staff hidden — click to show". From there the choir comes back and can be written on.

## 11.9 Transposing instruments

Many wind instruments do not sound the note they read: a trumpet in B flat reading a C sounds a B flat. Harmony Tutor keeps the two apart — what sounds and what is written — and shows one or the other on request.
The file always holds the real sound. When a track is given a transposing instrument, its part is rewritten as the player reads it: the notes on that staff change, and so does that staff's key signature, while analysis, playback and export go on working from the true sound.
The «Concert pitch» checkbox switches between the two views: on, it shows what you hear; off, what each player reads on the stand. It affects the display only and moves nothing in the piece.
When «Concert pitch» is ticked, in the clef menu, under the choice of transposing instrument, a line reminds you why the part is shown as it sounds and where to untick it: in the toolbar, next to the key, or in the View menu.
A transposing instrument can also be given to a choir staff, from the clef menu, but only in the four-staff layout: on the grand staff two voices share the same staff and could not have two different key signatures. To orchestrate the choir with transposing instruments, switch to the four staves.
When the piece changes key, the key signature of the transposed staves changes with it, at every change.

## 11.10 Clef, part and instrument

These are three different things, and confusing them is why certain operations look impossible when in fact they are simple.

25. The PART — soprano, alto, tenor, bass — is the identity of the staff. It is the part that says which notes live on that staff, and it never changes: it is also what harmonic analysis looks at when it talks about parallel fifths or voice leading.
26. The CLEF is a property of that staff, not its identity. You can change it whenever you like: the notes stay where they are and still belong to the same part. Only the way they are written changes.
27. The INSTRUMENT is the sound the part is played with, and it touches neither the notation nor the analysis.

The practical consequence is the one that matters: you can write for string quartet starting from the choir. Put the four staves in C clefs, then give each staff the clef it needs — violin, violin, viola, bass — and write. Harmonic analysis keeps working exactly as before, because as far as it is concerned the four parts are still the same four: it never noticed a thing.
If you assign a transposing instrument, what is said in 11.9 applies: the file always keeps the real sound, and the written notation is derived from it.

### The clef menu

Clicking the clef at the start of a staff opens a menu that is the same for choir staves in the four-staff layout and for tracks, with the same items in the same order:

28. **Clef** — the tick shows the current one.
29. **Part** — the instrument or voice the staff stands for: violin, viola, clarinet, soprano and so on. Choosing it gives the staff its name, clef and sound in one go; on the choir it also sets the range that the analysis checks. Tracks also offer «Piano» and «Piano with voices», the two forms of the grand staff (see 11.4). Here «part» means the role of the staff: the identity described above — which choir voice lives on that staff — does not change.
30. **Select the whole part** and **Copy the whole part** (on tracks: «the whole track»).
31. **Transposing instrument** (see 11.9). On choir staves it appears only in the four-staff layout.

On the grand staff and in treble-clef-only view, the choir clef opens the choice of Staves instead: grand staff, old clefs (four staves) or treble clef only — the same as in the View menu. In the four-staff layout the same choice sits at the top of the clef menu, so you go back with the same gesture.
On tracks the menu continues with ensemble analysis and the orchestra section.
On tracks, at the bottom of the menu, «Hide the track» removes it from the score — you show it again from the mixer — and «Delete the track…» deletes it with all its notes, after a confirmation. Cmd+Z (Ctrl+Z on Windows) undoes it.

## 11.11 Several instruments on the same staff (doublings)

In an orchestral score two instruments often play the same line: violin and oboe in unison, for example. Instead of copying the part onto a second staff, you add the oboe to the violin's staff: the staff stays one, and is called «Violin and Oboe».

### Adding and removing an instrument

In the mixer, next to every choir voice and every track (except drums), there is a small «+». Pressing it adds, just to the right, a strip tinted with the staff's colour: that is the added instrument. It starts as an oboe, and you change it with the instrument selector as on any strip. Right-click the strip for «Remove this instrument». Deleting a track also removes its doublings.
The strip has everything the others have: volume, pan, mute and solo, the level LEDs — at the same height as the staff's, so they can be compared —, the EQ, compressor and reverb inserts, the MIDI channel and the choice of where it plays (see 10.3). Solo on the oboe lets you hear the oboe even when the violin is silent.

### How it sounds

The added instrument plays the staff's notes with the same dynamics, articulations and pedal. It is not a perfect copy, though: each of its notes starts a few milliseconds earlier or later and drifts by a few cents of tuning, as happens between two real players. The offset depends on the note's position, so the same piece always sounds the same and the exported file matches what you hear.

### The staff name

The staff name is built automatically: «Violin and Oboe», and with three instruments «Violin, Oboe and Flute». In the margin the names are stacked, one column per instrument, with the added ones to the left,

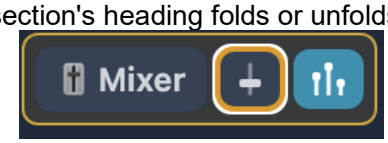
towards the edge. A name that does not fit the staff's height switches to the short form (Vln., Ob., Vc.…). The name you gave the track stays as it is: removing the doubling brings it back by itself, and renaming the staff «Violin I» makes it «Violin I and Oboe». For choir voices the name appears with separate staves.

### Towards a DAW and in files

Towards an external instrument each doubling has its own MIDI channel. «Auto» takes the first free one, after those of the voices and tracks and skipping 10, and the menu says which — «MIDI: Auto (6)» —, so you know which channel to put the oboe on in the DAW. If channels run out the menu turns red: the instrument plays only inside the app.
In the MIDI file the oboe is a track of its own; in MusicXML the staff is called «Violin and Oboe» and declares both instruments; in audio export by part the oboe gets its own file (see 13.2 and 13.3). Re-importing such a MusicXML brings the staff back with its added strip.

## 11.12 The channel strip

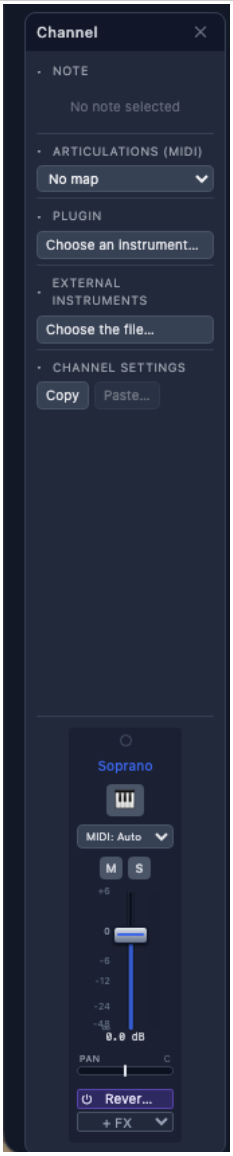
The fader icon in the Mixer group of the toolbar opens a single channel strip on the left side of the window: the one of the track or voice of the selected note (with nothing selected, the voice chosen in the toolbar). It is the same strip as in the mixer — instrument, fader, pan, mute and solo, effects — and every change shows in both. The score narrows to make room for it.
The strip is divided into sections — «Note», «Articulations (MIDI)», «Plugin», «External instruments», «Channel settings» —: a click on a section's heading folds or unfolds it, and the choice is remembered.




*The Mixer group of the toolbar: the mixer button, the channel strip icon and the lane icon (here switched on).*

### The Note section

At the top of the strip, in the «Note» section, is the selected note: its name — useful with the old clefs — and its velocity. Pitch and value are changed from the toolbar and with the keys, as usual; with several notes selected, their number appears next to the name.
With several notes selected, the velocity shows a small graph, one bar per note, and two modes: ± shifts them all by the same amount and keeps the differences, = sets them all to the same value. The notes are written when you let go of the slider. The ↺ button returns hand-edited notes to the dynamic markings.




*The channel strip: at the top the Note section (here with no note selected), Articulations, Plugin, External instruments and Channel settings; below, the mixer strip.*

## 11.13 The velocity and expression lane

The three-bar icon in the Mixer group opens the lane at the bottom of the page. While it is open the score switches to the continuous view, so every bar sits under its note; closing it brings back the previous view.

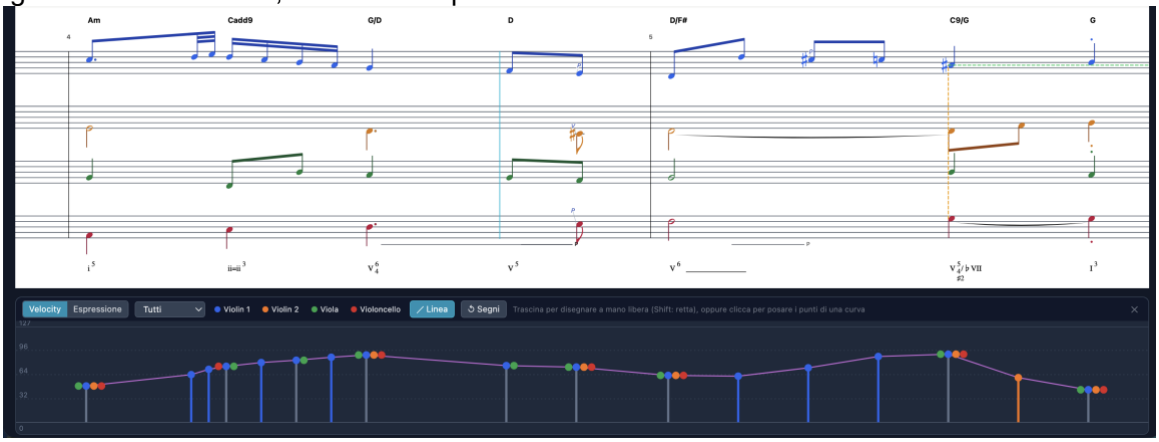
### Which part is shown

The lane shows the part of the last note you clicked, or the one chosen in the menu at its top. The menu also offers «All» and the orchestra sections present in the piece (Strings, Woodwinds, Brass…): there the notes of several instruments are shown together, each in its instrument's colour, and the legend next to the menu switches instruments on and off. A click shows or hides an instrument, Option-click (Alt-click on Windows) leaves only that one. An instrument that is switched off is neither shown nor edited.

### Velocity

Every note is a dot on its bar, as high as the velocity it really plays at: for notes that follow the dynamic markings (hollow dot) the one given by p, f and hairpins, for the others (filled dot) their own. Notes sounding together share one bar, one dot per instrument. Hovering shows the instrument name and the value.
Accent and marcato add a boost on the attack: in the lane it shows as a dashed line above the dot. With no dynamic markings, a newly written note plays at velocity 80.
Drag a dot up or down; if the note is selected, all selected notes move by the same amount. With the Line button on, you draw freehand over the bars and the velocities follow the stroke; with Shift you draw a straight line; clicking places the points of a smooth curve, applied with a double-click or Enter (Esc cancels).

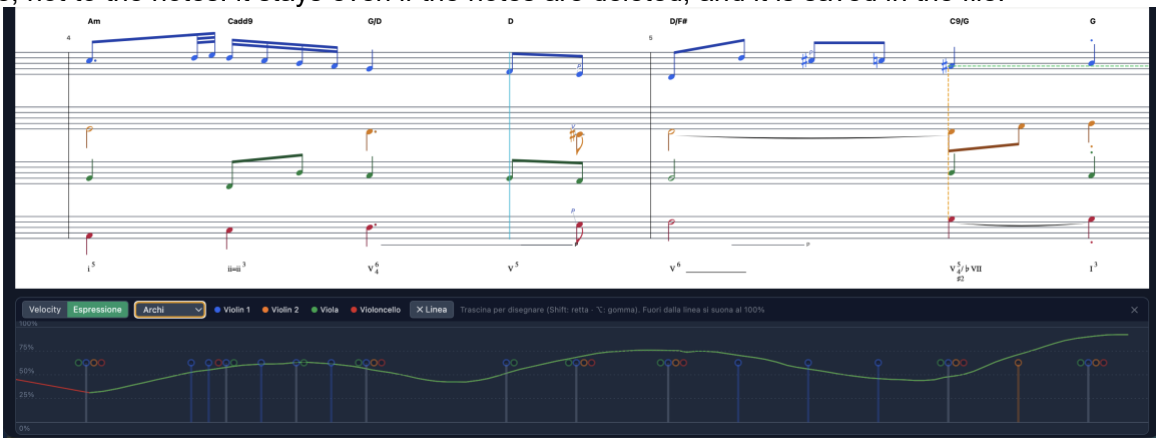
A velocity set by hand overrides the dynamic markings for that note. The orange line shows the level of the markings, the fuchsia one joins the hand-edited notes. The ↺ Dynamics button returns them to the markings: the selected ones, or the whole part shown.




*The lane in «All» under the continuous score: one dot per instrument on the bar of each attack; the fuchsia line joins the hand-edited notes.*

### Expression

The lane's second mode, in green, is expression: a line that stays, like the «expression» volume (CC11) of sequencers. It works below the mixer fader, which sets the level of the part, while expression draws the breathing of the phrase, even inside a held note or a chain of tied notes.
Draw it freehand; with Shift you draw a straight line, with Option (Alt) you erase the stretch you pass over. Drawing over a stretch replaces it. Outside the line the part plays at 100%; 50% lowers it by about 18 dB. The ✕ Line button removes the part's expression.
Strings, winds and voices follow the line inside the note too; piano and plucked sounds take the value at the attack; drums have none. Towards external MIDI instruments expression goes out as CC11. It is tied to time, not to the notes: it stays even if the notes are deleted, and it is saved in the file.




*The expression of the Strings section: the green line draws the breathing of the phrase; the faded bars show where the notes fall.*

## 11.14 External sounds: libraries and articulations

Harmony Tutor can have the score played by instruments outside the app, far more realistic than the built-in sounds: for example Spitfire strings opened in Kontakt. This section explains how to connect them and, above all, how to make the external instrument play legato, staccato, pizzicato and so on according to what is written in the score.

### The words you need

- Articulation: the way a note is played. For strings, for example: legato (notes joined in one bow stroke), staccato (short, separated notes), pizzicato (the string plucked with a finger), tremolo (the bow moving back and forth very fast).
- Library: a collection of sounds recorded from real instruments, for example «Spitfire Studio Strings». To play, a library must be opened with a dedicated program.
- Kontakt: the most widely used program for opening libraries. Other libraries open with a program from their maker (a «player»), or inside a DAW, that is a program for recording and mixing such as Logic, Cubase or Reaper.

- MIDI channel: Harmony Tutor sends the notes of each part on a channel, numbered 1 to 16; in Kontakt each instrument listens to one channel. For Harmony Tutor's cello to play Kontakt's cello, both must have the same channel number.
- Keyswitch: a key, usually among the lowest on the keyboard, that does not sound but makes the instrument change articulation. In Kontakt keyswitches appear as slots at the bottom of the instrument panel: each slot has the articulation's name and, above it, the key's name (for example E-1).
- Map: in Harmony Tutor, the list that says which keyswitch to press for each articulation of a library. Each part of the piece can have its own map.

### What you need before starting

- The program that plays must be open, with the instrument loaded (for example Kontakt with «Stradivari Cello»).
- In Harmony Tutor the MIDI output to that program must be chosen: the «MIDI» menu in the top bar (see 10.3).
- The part must play on the instrument's channel: in the mixer or in the channel strip, the «MIDI» menu under the instrument's name (see 10.3).

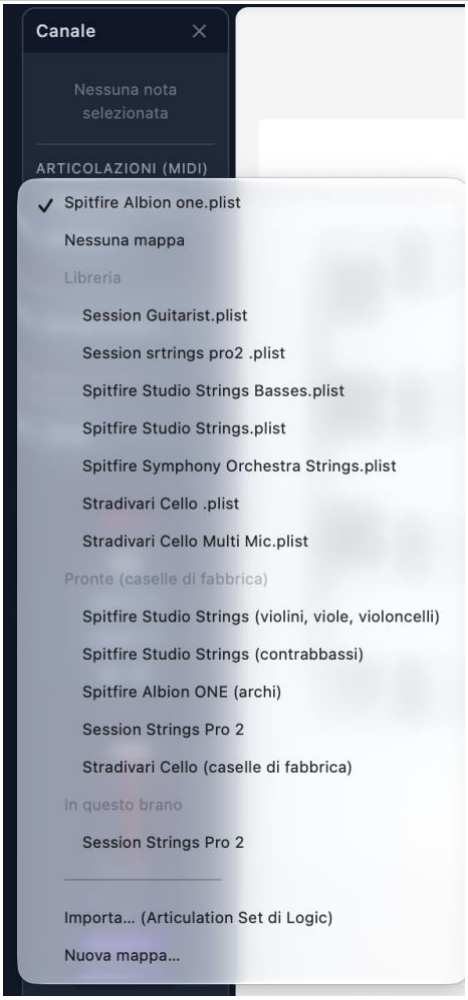
Once this is done the part already plays with the external instrument. The following steps make the articulation change.

### Step by step: giving a part a map

Step 1. Select a note of the part, for example the cello, and open the channel strip with the fader icon in the Mixer group of the bar (see 11.12). A column opens on the left side of the window, with the part's name at the top.
Step 2. In the column find the «Articulations (MIDI)» section. Under the heading there is a drop-down menu: if the part has no map yet, it reads «No map».
Step 3. Click the menu. The maps you can choose are divided into groups, each with a heading that cannot be clicked:

- «Ready-made (factory slots)»: the maps already included in Harmony Tutor for some popular libraries (see below).
- «Library»: the maps you imported or created; they work in all your pieces.
- «In this piece»: the maps already given to other parts of this piece.

A group appears only when it contains something: at first, for instance, there is no «Library» group. At the bottom of the menu there are two commands instead, which do not choose a map but bring in a new one: «Import… (Logic Articulation Set)» and «New map…» (see below).




*The map menu open: at the top the chosen map (ticked) and «No map»; then the groups «Library», «Ready-made (factory slots)» and «In this piece», with grey headings; at the bottom the «Import…» and «New map…» commands.*

Step 4. Choose the map for your library. Its name appears in the menu and the «Edit…» button appears below it.
Step 5. Check that it works. Press «Edit…»: a window opens with the list of articulations. Press the ▶ button next to one of them and look at the Kontakt panel: the slot with the same name must light up. If another slot lights up, or none, the map needs correcting (see «Correcting a map»).
From now on, when you play the piece, a note with a staccato dot plays the staccato, notes under a slur play legato, and so on. The map is saved in the piece: reopening it, it is still there, even on another computer.

### The ready-made maps

Maps are included for Spitfire Studio Strings (violins, violas, cellos and basses), Spitfire Albion ONE (strings), Session Strings Pro 2, Stradivari Cello and BBC Symphony Orchestra Discover (strings, basses, woodwinds and brass, harp and celeste, timpani and percussion, tuned percussion). They work with the instrument as the library opens it the first time, that is with the keyswitch slots in their factory layout. If you changed the slots in Kontakt (added, removed or moved articulations), the ready-made map no longer matches and must be corrected.
The BBC Symphony Orchestra Discover maps already have «Dynamics also as CC1» ticked. Those for harp and celeste, timpani and percussion and tuned percussion are «by instrument»: each slot is a different instrument (for example Marimba or Xylophone), chosen in the palette by its name; with no text written the first one plays.

### Importing Logic Articulation Sets

Logic Pro users may already have prepared maps for their libraries: in Logic they are called Articulation Sets. Harmony Tutor reads them as they are.
Step 1. In the map menu choose «Import… (Logic Articulation Set)». The file dialog opens.
Step 2. Logic's files are in a folder that is usually hidden. Press Shift, Command and G together (⇧⌘G): a field appears where you can type a path. Paste ~/Music/Audio Music Apps/Articulation Settings and press Return.

Step 3. Click your library's file; to take several at once, select them while holding Command (⌘). Then press «Open».
Every chosen file goes into the Library. The first one is also given to the part you started from, and its «Edit…» window opens so you can check it as in Step 5.

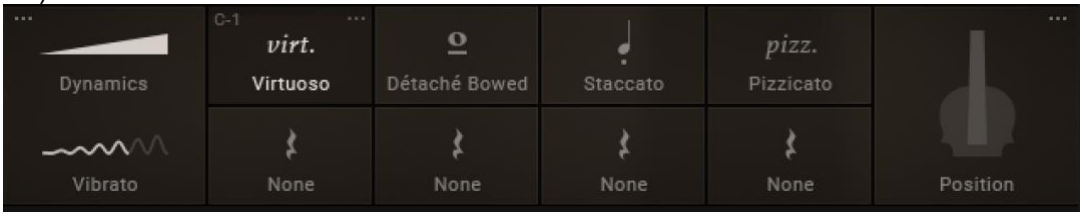
### Correcting a map, or creating one

A map is corrected when its keys do not match the instrument's slots; it is created from scratch with «New map…» when nothing is ready-made for your library. Either way you work in the «Edit…» window. At the top of the window is the map's name, which you can change: it is the name you will find it under in the Library. Below, in the «Articulations and keys» part, each row is an articulation: on the left the name, then the key that controls it (for example E-1), the − and + buttons to move it by one key, «Learn», ▶ to try it and ✕ to remove it. «+ Add articulation» adds a row.



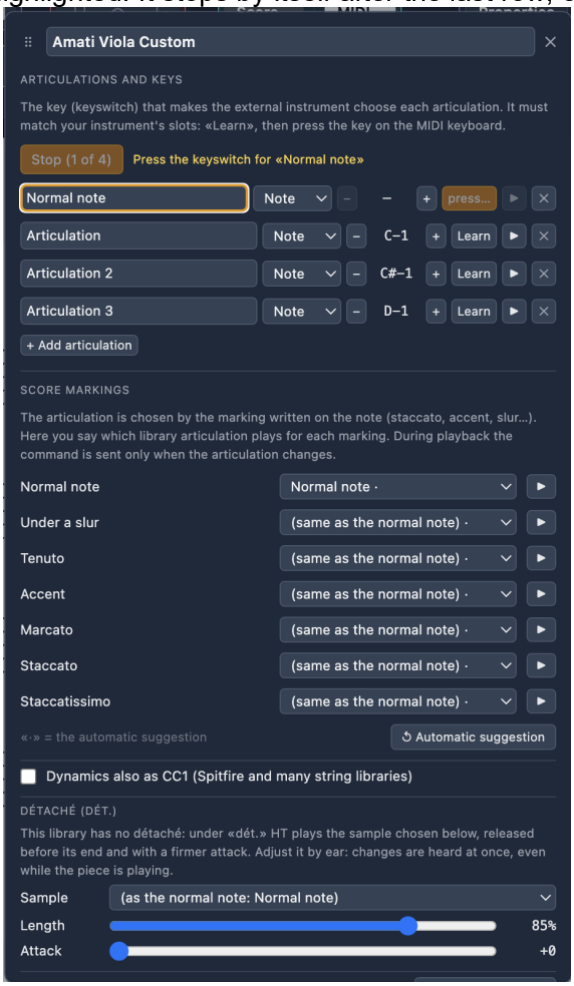

*The «Edit…» window of the Albion ONE map. At the top the articulations, each with its key (from C-2 upwards), the − and + buttons, «Learn», ▶ and ✕; below, which articulation plays for each score marking.*

The easiest way to set the right key is «Learn»; you need a MIDI keyboard connected.
Step 1. In Kontakt look at the articulation's slot and read the key name written above it (for example E-1 for Staccato).



*The keyswitch slots in an instrument panel in Kontakt (here Stradivari Cello): each slot is an articulation; the key's name appears above the slot (here C-1 above Virtuoso). Kontakt is a trademark of Native Instruments.*

Step 2. In Harmony Tutor press «Learn» on that articulation's row: the button turns orange and reads «press…».
Step 3. Press that key on the MIDI keyboard. The key's name appears in the row and the button returns to normal; the key you pressed is not written into the score.
Step 4. Press ▶ on the same row: the right slot must light up in Kontakt.
To set the keys of every row one after another there is «Learn in sequence», next to the «Articulations and keys» heading. Press it, then press the keyswitches on the MIDI keyboard in the order of the rows: next to the button it says which articulation it is waiting for, for example «Press the keyswitch for «Staccato»», and the row is highlighted. It stops by itself after the last row, or when you press «Stop».




*The «Edit…» window of a map, with «Learn in sequence» waiting for the first keyswitch; below, the score markings, dynamics as CC1 and détaché.*

«Learn» listens to every connected MIDI keyboard, and also to the drawn keyboard of a plug-in instrument inside Harmony Tutor (see 11.15), if the plug-in sends out the notes you play there. The learned key does not sound and is not written in the score.
Some libraries use controllers (CC) or program changes (PC) instead of keys: the small menu next to the name lets you choose «Note», «CC» or «PC» and type the number.
In the «Score markings» part you choose which articulation plays for each marking: the note without markings («Normal note»), notes under a slur, staccato, staccatissimo, accent, marcato and tenuto.
Items with the «·» dot are the automatic suggestion, which Harmony Tutor derives from the names; «↺ Automatic suggestion» puts it back. A marking left on «(same as the normal note)» plays like the note without markings.
If the library has both a Staccato and a Spiccato, the automatic suggestion gives the Staccato to the dot and the Spiccato to the wedge (staccatissimo).
The «Dynamics also as CC1» box should be ticked for libraries that set the volume with the modulation wheel instead of with how hard the key is struck: Spitfire and many string libraries do this. If with one of these p and f all sound the same, this is the box to tick.
When you close the window the map is also saved to the Library, and from then on it is available in every piece. The «Remove from library» button takes it off the list; pieces that use it keep it.

If the library has no détaché, the window also has a «Détaché (dét.)» section: it says what Harmony Tutor plays under the «dét.» text (see below). «Sample» chooses the articulation to use — the Staccato, for example, gives the bow attack the Sustain lacks —, «Length» how much earlier to release the note, «Attack» how much to strengthen its start. It is heard at once, even while the piece is playing, and is saved with the map.
The window is moved by dragging the ⠿ handle at the top.

### Techniques written above the staff

Some articulations in the score are not a sign on the note but a word above the staff, valid from that point on: pizz. (pizzicato), arco (back to playing with the bow), con sord. (with the mute), trem. (tremolo), sul pont., col legno. They are written from the palette.
Step 1. Select the notes where the technique should start. You can select them on several staves at once, for example across the whole string section.
Step 2. Open the palette with the «Palette» button in the bar and open the purple group «Articulations — external libraries». It shows the articulations of the map of the staff chosen in the channel strip, already with the name written in the score; in brackets, the name the library uses.
Step 3. Click the technique. The text appears above the first selected note of every staff; with no notes selected, at the cursor on the channel strip's staff.
A technique applies from that point on, on that staff, until you write another one, and while it applies it rules, even over the note markings. To go back to the markings write «ord.», which Harmony Tutor turns into «arco» after a pizz. and into «senza sord.» after a con sord. by itself. Legato, stacc., spicc. and marc. are written the same way when they apply to a whole passage: for a single note use the sign on the note.
«dét.» (détaché, separate bow strokes) is always in the group, except with «by instrument» maps. If the library has a détaché, that plays; if not, Harmony Tutor plays the normal note and releases it a little before its end, as set in «Edit…». The written note does not change, and the internal sounds ignore it.
To remove a text, right-click it. To remove many at once, select the notes that include them and press «Remove articulations from the selection» in the same palette group; the number in brackets says how many it will remove.

### External instruments: finding the instrument as you left it

The map says which key to press, but it is the instrument that decides which articulation sits on that key. In many libraries the keyswitch slots can be changed, and the layout is saved in the file of the program that plays: a Kontakt multi (an .nkm file, holding one or more instruments with their channels), a Kontakt snapshot (an .nksn file, the settings of a single instrument) or a DAW project. Reopening the «factory» instrument, your slots are gone and the map does not match.
So you do not have to think about it, Harmony Tutor remembers in the piece which file to open. To save a multi in Kontakt:
Step 1. In Kontakt load the piece's instruments, each on its own MIDI channel, and arrange the slots as you need them.
Step 2. In the «FILE» menu at the top of the Kontakt window choose «Save multi as…». In the save dialog choose the «Patch only» option and save, for example next to the piece. The other options («Patch + samples», «Monolith») also copy every sound of the library, many gigabytes: you do not need them.
Then in Harmony Tutor:
Step 3. In the channel strip, under «External instruments», press «Choose the file…» and point to that file.
Step 4. The button becomes «▶» followed by the file name. From then on, when you reopen the piece, just press it: the file opens in its program and instruments, slots and channels come back as you saved them. The ✕ next to it removes the file from the piece.
Kontakt multis, instruments (.nki) and snapshots are accepted, and projects from Logic, Cubase, Studio One, Reaper, Ableton Live, Bitwig, Pro Tools and Dorico. If the file is on an external drive, the drive must be connected.

### Copying settings from one channel to another

When several parts use the same library there is no need to repeat everything for each one.
Step 1. Open the channel strip of the part already set up and, in the «Channel settings» section, press «Copy». «Copied from:» appears below with the part's name.
Step 2. Open the channel strip of another part and press «Paste…». A window opens with checkboxes: tick what you want to bring over (output, MIDI channel, articulation map, instrument, effects, volume and pan); next to each item you see the value that will be pasted. Press «Paste».

The checkboxes stay as you left them while the app is open, so for the other parts just repeat Step 2. The MIDI channel should normally not be copied: each instrument in Kontakt listens to its own.

### If the instrument does not respond

- Nothing plays: check the MIDI output in the «MIDI» menu of the bar, and that the part's channel is the one the instrument listens on in Kontakt (the «MIDI Ch» field in the instrument header).
- It plays, but the articulation never changes: does the part have a map? In the channel strip, under «Articulations (MIDI)», it must not read «No map».
- The wrong articulation changes, or none: the instrument's slots are not those of the map. With ▶ in «Edit…» see which slot lights up and correct the key with «Learn», or reopen the instrument saved with your slots (see «External instruments»).
- Soloing a part silences the instrument: in Kontakt the instrument listens to all channels («MIDI Ch: Omni») and also receives the zero volume sent to the parts left out of the solo. Give each instrument its own channel.

## 11.15 Plug-in instruments inside Harmony Tutor

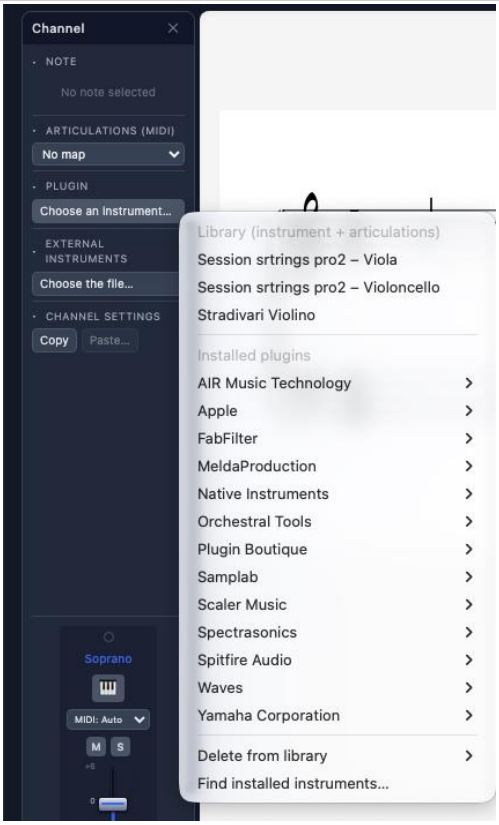
Besides sending notes to an external program (see 11.14), Harmony Tutor can host a plug-in instrument — Kontakt, a library's player — for each part. The sound enters the mixer like that of the built-in instruments, with volume, pan and effects, and the plug-in is saved with the piece.

### The words you need

- Plug-in: a program that does not open on its own but inside another one, which hosts it. Library instruments (Kontakt, the players from Spitfire and other makers) also exist as plug-ins.
- AU and VST3: the two plug-in formats. On the Mac Harmony Tutor uses AU (Audio Units), on Windows VST3.
- Harmony Tutor Plugin Host: the helper program, installed together with Harmony Tutor, that hosts the plug-ins. Harmony Tutor starts and controls it by itself; it has no icon of its own in the Dock.

### Step by step: giving a part a plug-in

Step 1. Select a note of the part and open the channel strip with the fader icon in the Mixer group of the toolbar (see 11.12). In the column find the «Plugin» section.
Step 2. Press «Choose an instrument…». If Harmony Tutor Plugin Host is not running yet, the button reads «Starting…» for a moment, then the menu opens by itself.
Step 3. The first time the list of instruments is empty: open the menu and choose «Find installed instruments…». Under the button «Searching… 12 of 140» appears: the search tries the instruments one at a time and can take a few minutes. The list is remembered and is ready straight away the next times; search again when you install a new instrument.
Step 4. Open the «Choose an instrument…» menu. Under the «Installed plugins» heading the instruments are grouped by maker, in submenus, each with its format in brackets, for example «Kontakt 8 (AU)». Choose the instrument: while it loads, «Loading «Kontakt 8»…» replaces the menu — with an instrument and its samples it can take some tens of seconds — then its window opens and the part plays it on its MIDI channel.



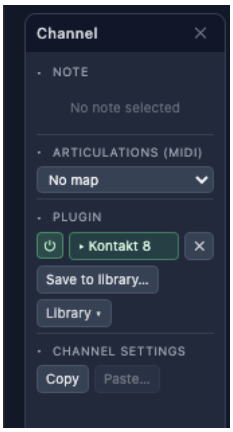

*The «Choose an instrument…» menu: at the top the instruments saved in the library, then the installed plugins, by maker; at the bottom «Delete from library» and the search for instruments.*

Step 5. In the plug-in window prepare the sound as in its own program: in Kontakt, for example, load the library's instrument. From then on the part plays with the plug-in.
When a part has a plug-in, the MIDI output switches by itself to «Virtual instruments» (see 10.3); parts without a plug-in keep playing with the built-in sounds.

### The plug-in row

With the plug-in loaded, the section shows three buttons:

- ⏻ switches the plug-in off without removing it: the part plays with Harmony Tutor's built-in sound, for comparison; a click switches it back on. To silence the part use mute.
- ▸ followed by the name opens and closes the plug-in window.
- ✕ removes the plug-in from the part at once. To change instrument, remove the plug-in and choose another.




*The «Plugin» section with Kontakt loaded: ⏻, the name that opens its window, and ✕; below, «Save to library…» and «Library ▾».*

One plug-in window is shown at a time: opening another hides the first. The window stays above Harmony Tutor and does not take the keyboard: the space bar still starts and stops the piece. With a window open, clicking a note of another part that has its own plug-in opens that part's window. Closing the window only hides it: the instrument keeps playing.

### Where the sound goes

The plug-in's sound enters the mixer channel of its part: it goes through the fader, mute and solo, the effects, pan and reverb, and comes out in the audio export (see 13.3). The strip's «Plays» menu applies here too: with «Internal» the part plays only the built-in sound, with «Both» both.
Harmony Tutor sends each plug-in only its own part's notes and receives its first stereo output. In Kontakt, therefore, set the instrument to output 1 («st. 1») and to the part's MIDI channel, or to «Omni».

### Playing the part from the MIDI keyboard

With the MIDI output on «Virtual instruments» the MIDI keyboard plays the part you are writing: the chosen voice of the choir or, when you work in the accompaniment or record, the active track. If the part has a plug-in, the notes and the pedal go to it. If it has none, if the plug-in is switched off (□) or if the strip says «Internal», the built-in sound plays; with «Both», both.
The plug-in's sound arrives a little later than the built-in instruments, about a tenth of a second, because it comes back into Harmony Tutor to go through the mixer. When the piece plays you do not notice it, because Harmony Tutor sends the notes to the plug-in early; playing from the keyboard you do. To record (REC, chapter 12) without that delay, switch the plug-in off with □ while you play: the part answers at once with the built-in sound, and the recording is the same. Then switch it back on to listen.

### The instrument library

An instrument prepared with care — in Kontakt, the instrument with its keyswitch slots arranged — can be kept in the instrument library, which works in every piece.
Step 1. In the «Plugin» section press «Save to library…». A field appears with a suggested name (the map's or the plug-in's name, followed by the part's): correct it if you like and press «OK»; Esc cancels. The plug-in, its whole state and the part's articulation map (see 11.14) are saved together. A name already in the library is replaced.
Step 2. In another part, or another piece, open «Choose an instrument…»: at the top of the menu, under the heading «Library (instrument + articulations)», are the saved instruments. Choosing one gives the part the instrument and its map; the window does not open by itself.
To remove an instrument from the library choose «Delete from library ▸» and its name, at the bottom of the «Choose an instrument…» menu or from the section's «Library ▾» button. Parts and pieces that use it keep it.

### What the piece keeps

The piece remembers, part by part, which plug-in plays and in which state. When you reopen it, Harmony Tutor starts Harmony Tutor Plugin Host by itself and reloads the instruments: with large libraries the sounds may take a few seconds to arrive. The plug-in's state is updated while its window is open and when you close it: before saving the piece, close the plug-in window.
With plug-ins the «External instruments» section (see 11.14) does not appear: it would open a second Kontakt, outside Harmony Tutor.

### If something goes wrong

- The plug-in does not load: a message says «Harmony Tutor Plugin Host could not load…» and why. Search for the instruments again, or open the instrument in its own program to check that it works.
- The button stays on «Starting…»: after 20 seconds a message says so. Usually the system has opened a window asking for a permission, left behind Harmony Tutor: answer it, then press «Choose an instrument…» again. If there is no window, quit Harmony Tutor with Cmd+Q and reopen it.
- «Loading…» does not finish: here too a permission window may be waiting — for example the request to read the libraries' external disk, described below — or the external disk is still waking up.
- The instrument does not find the library's sounds (Kontakt shows «Browse» instead of the library): on the Mac, if the library is on an external disk, Harmony Tutor Plugin Host must be allowed to read it. When the Mac asks whether it may access files on a removable volume, answer «Allow»; if you refused, in System Settings → Privacy & Security → Files and Folders switch on removable volumes for Harmony Tutor Plugin Host, then quit Harmony Tutor with Cmd+Q and reopen it.
- The part does not play with the plug-in: check that the output is «Virtual instruments» (the «MIDI» menu in the toolbar), that the plug-in is not switched off (□) and that in Kontakt the instrument listens to the part's channel or to «Omni».
- On Windows instruments are searched in the VST3 folder (C:\Program Files\Common Files\VST3): an instrument installed elsewhere does not appear.

# 12. Real-time recording (REC)

Captures MIDI notes from the connected keyboard in real time, with an automatic count-in and post-recording quantisation.




*REC button — active state (red) and count-in*

## 12.1 Recording flow

1. Select the destination voice or ACC track.
2. Arm recording with **Shift+R** (requires a visible ACC track).
3. Press **Space** to start: the automatic count-in (default 1 measure) runs and recording begins at beat 1.
4. Play on the connected MIDI keyboard.
5. Press Stop (Space) to finish.
6. Post-recording quantisation is applied automatically.

How it rounds. Each note goes to the simplest nearby point: first the beat, then the eighth, then the chosen grid (see 12.2), with a tolerance of one fifth of the subdivision. So a note played a little before or after the beat goes on the beat, and a real off-beat sixteenth stays a sixteenth. A note lasts until the next one if less than a beat separates them; a longer gap stays a rest. The «Q» button redoes the rounding with the chosen grid, starting again from what you played.

> Note: on ACC tracks too, the R key writes a rest at the cursor (see 4.4a); recording is armed with Shift+R.

## 12.2 REC settings

| Parameter | Description |
|---|---|
| Count-in | Number of count-in measures (default 1). |
| Quantisation grid | Minimum rhythmic value to which notes are snapped. |
| Overlapping-note trimming | Overlapping notes in the same voice are shortened automatically. |

> ***Note:*** *Make sure the MIDI keyboard is configured in Preferences → MIDI.*

# 13. Import and export

## 13.1 Import

| Format | Extension | Notes |
|---|---|---|
| Native | .htp / .json | Complete format |
| MIDI | .mid | See below |
| MusicXML | .xml / .musicxml / .mxl | From MuseScore, Dorico, Finale, Sibelius; compressed too (.mxl) |
| MuseScore | .mscz / .mscx | From File › Import MuseScore...: needs MuseScore installed (see below) |

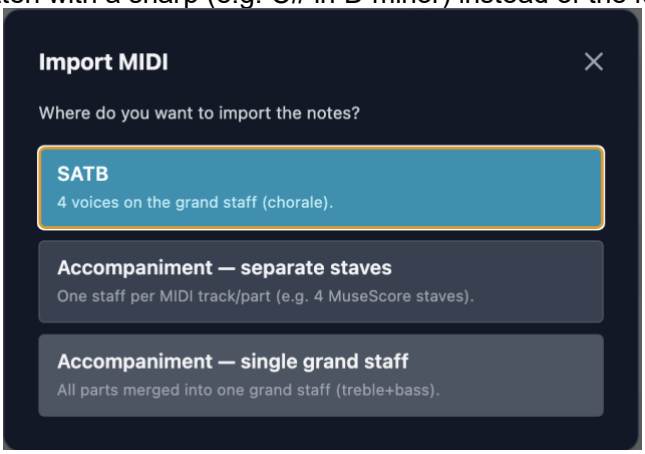
### Advanced MIDI import

MIDI import is no longer limited to a simple heuristic voice assignment. When importing a MIDI file (e.g. a piano part) into an accompaniment track:

- **Automatic voice separation:** on a “Grand staff with voices” track, the part is rendered up to two voices per staff (fixed stems: voice 1 up, voice 2 down), without pre-splitting the tracks in a DAW.
- **Automatic clef assignment:** notes below middle C (MIDI 60) go to the bass staff, those from middle C upward to the treble staff.
- **Triplet-aware quantisation,** ties across the barline (a held note survives into the next measure) and correct measure counting, including for ACC-only imports.

### Multi-track import and destination

When importing a MIDI file, a window asks where to place it, with three choices: SATB (four voices on the grand staff), Accompaniment — separate staves (one staff per MIDI track) and Accompaniment — single grand staff (all parts merged into two staves).
Separate staves. A multi-track MIDI file — for example a MuseScore score on four staves — is imported as one ACC track per part, each on a single staff, with the clef chosen from its range (treble or bass) and the name taken from the MIDI file. This is the right format for combined analysis (see 11.8); the single grand staff is still handy for a piano written on two staves.
Spelling in minor keys. MIDI does not store note spelling (C# and Db are the same key): with a minor key set, the leading tone is written with a sharp (e.g. C# in D minor) instead of the key-signature flat.




### The key of MIDI files

A MIDI file contains no accidentals: it contains note numbers. The spelling of every C♯ or D♭ is decided by the key signature, and the format provides a message for declaring it — but many files do not write it, and those exporting from a sequencer often leave the default, which is C major.
When the declaration is missing or is the default, the program now infers the key signature from the notes in the file and shows it in the import dialog before going ahead, so it can be corrected in advance instead of respelling the accidentals afterwards. Key changes written in the file are preserved.

### MusicXML with one part per voice

Importing as choir a MusicXML written with one part per voice, each on a single staff — a string quartet, a chorale on four staves —, the piece comes in on separate staves, as in the file: each voice takes the clef (including the treble clef with the 8 below), the name and the instrument of its part. Clefs, names and instruments of the piece opened before do not linger.
A three-part file comes in as soprano, alto and bass (voices 1, 2 and 4), with three-part writing (see 4.10).
A staff that declares several instruments in the file — «Violin and Oboe» — comes back as it was: a staff with the first instrument and, in the mixer, the strip of the second (see 11.11). Compressed .mxl files, the default format of MuseScore and Dorico, open like the others.

### MuseScore files

Files saved by MuseScore (.mscz and .mscx) open from File › «Import MuseScore...». Harmony Tutor has them converted to MusicXML by the MuseScore program installed on the computer, then imports them like any MusicXML, with the same window that asks where to put the parts. The conversion takes a few seconds.

So MuseScore must be installed, where its installer puts it: the Applications folder on the Mac, Program Files on Windows.
If MuseScore is missing, or cannot convert the file — this can happen with a file saved by a newer MuseScore than the one installed, or with a damaged file — a window says so. Then you do it by hand: open the file in MuseScore, on any computer, choose File › Export… and the MusicXML format, and open the new file with File › «Import MusicXML...».
A chorale written in MuseScore on two staves, with two voices per staff, comes in with its four voices.

## 13.2 Export

| Format | Extension | Notes |
|---|---|---|
| Native | .htp | Complete, reopenable |
| MusicXML | .musicxml | MuseScore, Finale, Sibelius |
| MIDI format 0 | .mid | All voices on a single track |
| MIDI format 1 | .mid | Each voice on a separate track |
| PDF | .pdf | For printing |
| PNG | .png | Screenshot |

### Export music

The File → Export music menu item (⇧⌘E) opens a single dialog with all export forms: MIDI (the piece as you hear it, for Logic and other sequencers), MusicXML (with the visible analysis — Roman numerals above and real figures below —, a universal format for MuseScore, Finale, Sibelius, Dorico…) and, in the «For blind users» section, the Spoken and Token modes described below. PDF and PNG export remain in their own menu items.

### The MIDI file for a sequencer

The MIDI file is for carrying on the work in a sequencer — orchestrating in Logic, Cubase or Reaper, having a library play the piece. That is why it carries the piece as you hear it in Harmony Tutor, not just the written notes:

- Tempo. The toolbar tempo, or the metronome mark of the first bar if there is one, then every metronome mark, every rit. and accel. A fermata becomes a slowdown on the note under it: it lasts as long as in playback, and the bars stay in place.
- Bars. A pickup and irregular bars have their real length — an eighth-note pickup is a 1/8 bar — so the sequencer's barlines match the score's, together with changes of time and key signature. The sequencer counts bars from 1: with a pickup, every number is one higher.
- Repeats and first and second endings are written out in full, because in a sequencer repeat signs are only drawings. After a repeat, therefore, the sequencer's bar numbers no longer match the score's.
- Dynamics. Each note has the velocity of the markings. On string, wind and choir parts the dynamics also come out as controller CC1, the wheel with which orchestral instruments (Logic's, Spitfire's and many libraries') change loudness and tone: so a hairpin grows on a held note too. On piano, organ, guitars and synths the wheel does something else — on Logic's piano it moves the pitch — so there the dynamics stay in the velocity. If the part has an articulation map, its «Dynamics also as CC1» box decides. The Expression lane comes out as CC11.
- Articulations. Staccato and staccatissimo shorten the note, accent and marcato strengthen it, as in playback. If the part has an articulation map (11.14), its keyswitches come out too.
- Tied notes are a single held note; octave signs and swing are included.

In Logic, when you drag the file into a project that is already open, it asks whether to import the tempo information: answer yes, otherwise the project's tempo stays.
To take the score into a notation program (MuseScore, Sibelius, Dorico) the right format is MusicXML: MIDI contains no markings, and whoever opens it sees only notes and durations — a staccato would come back as an eighth note followed by a rest.

### Printing, PDF and images

Printing and PDF fit the page to the sheet: if a system is wider or taller than the space available, everything is reduced just enough for each system to fit whole. On a score with many parts the reduction can be considerable — that is a matter of room on the paper, not a fault: fifteen staves on an A4 cannot come out any larger.

Image export produces several numbered files when the piece is long, and always cuts between one system and the next: no staff is ever split in half.
The paper format, portrait or landscape, is chosen in the layout settings. For a score with many parts portrait almost always works better, because height is the constraint.

### What goes into the file: the three toggles and the staves

From version 1.6 the export follows what you see. The three analysis toggles — Roman numerals, chord symbols, figured bass — no longer govern the page alone: they also decide the contents of the files. What is off is not exported.
The same applies to the staves. A hidden staff — the choir or an accompaniment track — stays out of the file. This is how you export a single track, or two out of three: switch off what you do not need and export.
If no staff is left on, the export includes them all. A valid file with no music would be the worst outcome, because it looks like a successful export.
The commands live in the toolbar and in the View menu, with their shortcuts (§18.4). Before saving, the Export music dialog spells out what is about to be written — “Will be exported — Staves: Choir, Guitar. Analysis: Roman numerals, figured bass.” The summary is tied to the Export button, so a screen reader reads it together with the button.
MIDI is the exception for the analysis: it carries the notes, not the labels, so the three toggles do not affect it. The choice of staves does apply to it as well.

### Chord symbols in the files

Chord symbols are exported as real chords, not as text: a <harmony> element in MusicXML and <Harmony> in MuseScore's native format. Whoever opens the file sees them as chords — they transpose together with the notes, they can be edited, and they are reprinted according to the conventions of the program reading them.
A symbol the application cannot classify is not lost: it is exported anyway, with its exact spelling. Better a chord declared “other” but spelled correctly than a missing symbol.

### Accessible export of the analysis (MuseScore / VoiceOver)

Harmony Tutor can export the harmonic analysis in a form readable by a screen reader (VoiceOver), designed for blind students and musicians. From the **File** → **Export analysis…** menu (Cmd+Shift+X) a dialog opens with three modes:
• MusicXML — universal format (.musicxml): Roman numerals above the staff and figured-bass figures below. This is the format to open the file in any notation program (MuseScore, Finale, Sibelius, Dorico).
• **Spoken** — **MuseScore** format (.mscx): the analysis is written **in words** (e.g. “dominant seventh”) in the figured bass, in a small font. Navigating the bass notes in MuseScore, VoiceOver reads the phrase aloud. It requires no installation: the file is self-contained. (Verbalisation is currently in Italian.)
• **Token (for a VoiceOver dictionary)** — MuseScore format (.mscx) with compact labels in the figured bass: scale degree (V7, IV43) or actual chord symbol (Bb7, F#65). It requires a VoiceOver pronunciation dictionary that expands the labels into words.
**About the formats.** The **Spoken** and **Token** modes produce **.mscx** files, which only **MuseScore 4** can open — that is where the screen reader reads the analysis while navigating the notes. For a file that opens anywhere, use the **Standard** (MusicXML) mode.
**Typical use (blind users).** Export as **Spoken**, open the .mscx in MuseScore 4 with VoiceOver on and navigate the bass notes: for each chord you hear the scale degree with any inversion.
Chord symbols in words. In Spoken mode the symbols are said in full: Bb/D becomes “B flat bass D”, Gm(add9)/A “G minor with added ninth bass A”. The major mode is not named, because in a chord symbol it is implied: C is said “C”, Cm “C minor”.
Symbol and function together. If both Roman numerals and chord symbols are on, the file says both, separated by a comma: “B flat bass D, dominant seventh in first inversion”. The symbol says which chord it is, the Roman numeral says which function it has — and that is the comparison one makes while studying.

### What the file contains (import dialog)

Before asking where to put the notes, the dialog lists what it found in the file: the name of each part, its number of notes, one or two staves and the clef. The "separate staves" option states how many staves will be created.

The instrument is taken from the file — from Program Change in MIDI, from the part list in MusicXML — instead of assigning a piano to everything. Percussion is recognised from channel 10 or from the track name and becomes a drum track, with the kit (orchestral or rock) chosen from the pieces the file actually uses. Next to each part a label states how it will be treated, "percussion" or "pitched": tap it to correct it. Percussion stays on its own track even when choosing "single grand staff", and with percussion only the SATB destination is not available.

Instrumental pieces. An imported piece (a guitar, a piano, the parts of a chorale) comes in as an accompaniment track, not as the choir. The accessible export follows it: it writes the notes the analysis is about, and if four parts are analysed together it exports all four.

Transposing clefs. The tenor of a chorale is written in treble clef with an 8 below: on import and export the program recognises and declares it, so the part stays where its author wrote it.

### The choir on separate staves and doublings

If the choir is written on separate staves, in MusicXML each voice comes out as a part of its own, with its name, clef and instrument: whoever opens the file finds the score as it is, not a piano grand staff. A voice's dynamics go only into its part, and the analysis under the lowest part. On the grand staff the choir comes out as before, a single part on two staves.

A staff with added instruments comes out with the combined name («Violin and Oboe») and declares every instrument; some programs, MuseScore among them, use only the first one for playback. In MIDI each added instrument is a track of its own, on its channel.

## 13.3 Audio export

From the File menu, «Export audio», the piece becomes an audio file you can play anywhere, send to someone, or take into an editing program.

Without plug-in instruments it is not a recording. The program does not listen to itself while it plays: it works the piece out again from scratch, faster than real time. The difference is audible, and it is the reason it is done this way — the file does not depend on what the computer happened to be doing at the time, and it cannot contain either hesitations or glitches.

What comes out is what you hear on playback: the mixer with its volumes and panning, the written dynamics, the articulations, the reverb. A voice muted in the mixer comes out muted, and if you have soloed a voice only that one comes out. The metronome never does.

There are four formats. WAV at 24 bit is lossless and leaves headroom for anyone who will work on the sound afterwards; WAV at 16 bit is CD quality and is a third smaller; MP3 at 320 is compressed but in practice indistinguishable, and is the one to send by mail; MP3 at 192 is the lightest. To give an idea: one minute of stereo is about 17 MB in WAV 24 bit and about 2.3 MB in MP3 320.

The «One file per voice» tick changes the result: instead of a single file you get as many as there are voices and tracks playing, each in its own, with its own reverb. They are all the same length, so importing them together into an editing program lines them up by themselves. You choose where to put them once: the files appear there, named after the piece followed by the name of the part.

Instruments added to a staff (see 11.11) come out in a file of their own, for example «Oboe (Violin)», and the staff's file contains the staff alone: this way, in an editing program, the two can be balanced separately.

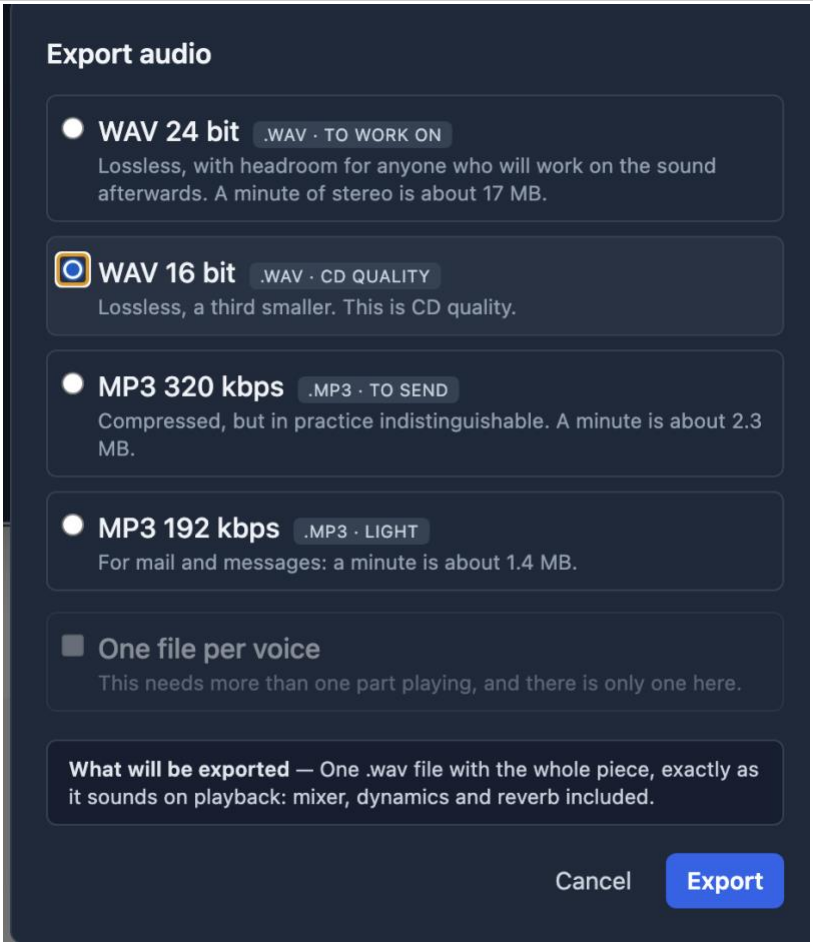



Figure — The audio export window: the format, the tick for separate voices and the summary of what will be written.

### With plug-in instruments: in real time

If the piece uses plug-in instruments (see 11.15) and the MIDI output is «Virtual instruments», the plug-ins' audio cannot be computed in advance: the piece is played from the first bar to the end and recorded while it plays, so the export takes as long as the piece. You hear it while it records; metronome and loop stay out.
A window shows the elapsed and total time, with «Cancel», which leaves no half-written file: do not touch anything until the end. The computer does not go to sleep until it has finished. With plug-ins only the full mix comes out: the «One file per voice» tick does not apply, and a message says so.
If the plug-ins' sound was interrupted during the recording, a message at the end says how many times: you might hear small gaps. Close the other applications and try again.

## 14. Automatic chorale generator

Generates the four SATB voices from a progression of Roman numerals. Chorale menu → “Insert progression”.

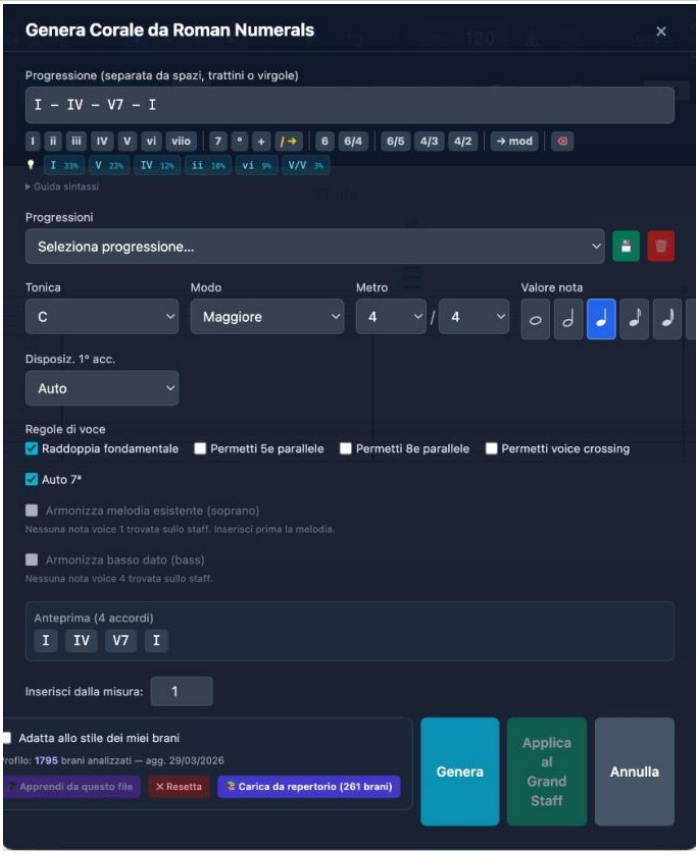



*Figure 17 — Chorale generator*

## 14.1 Interface

- **Progression field:** enter chords separated by spaces or dashes.
- **Quick buttons:** I, ii, iii, IV, V, vi, vii°, 7, °, +, 6, 6/4, 6/5, 4/3, 4/2, →mod.
- **Corpus suggestions:** coloured chips with the most likely progressions.

## 14.2 Settings

- **Tonic, Mode, Meter, Harmony duration: realisation settings.**
- **Voice rules:** double the root, allow parallel 5ths/8ves, voice crossing, auto 7th.
- **Constraints: a given soprano, a given bass or a given inner voice (alto or tenor) as a fixed constraint.**

## 14.3 Syntax

| Syntax | Meaning |
|---|---|
| I ii IV V vi | Degrees (upper-case=Major, lower-case=minor) |
| I6 I6/4 | Inversions |
| V7 V6/5 V4/3 V4/2 | Seventh and inversions |
| vii° iiø7 | Diminished / half-diminished |
| V/V V7/IV | Secondary dominants |
| It6 Fr6 Ger6 | Augmented sixths |
| →G: →f#: | Modulation |
| \| | Barline (new measure) |

## 14.4 Corpus and learning

The generator uses a statistical system that learns from the pieces entered and analysed. Through bigram and trigram statistics, the software learns the horizontal behaviour of the voices and the most frequent harmonic progressions, guiding the choices of inversion, voice-leading and doubling.

- **Learn from this file:** adds the current exercise to the corpus.
- **Load from repertoire:** imports the pre-computed profile from the corpus.
- **Reset:** restores the default stylistic profile.

### 14.5 How it chooses the harmony

The generator does not choose one chord at a time. It looks at the whole phrase and searches for the best path taken as a whole: a succession may contain a less likely step and still be the right choice, if what comes after justifies it. This is how one harmonises by hand, and it is why the result can change at a point far from the note you edited.
The probabilities come from the repertoire, not from a hand-written table: how frequent each step is, which harmony is found under which note of the melody, and how the choice shifts according to the chord before and the chord after.
The generator also knows where in the bar it is writing. Half the rules of harmony speak of strong and weak beats: the seventh, the six-four, the doublings and the cadences are judged from that too.
Before writing, every solution passes through the real checker — the same one that flags violations on the page, not a copy of it. What the program would flag as an error, the generator does not write.

### 14.6 Phrases and cadences

A fermata marks the end of a phrase. Where there is one, the generator places a cadence on it and builds it properly, with the dominant and the closing chord in root position. Without fermatas the piece is divided into phrases of plausible length, and closes are judged also by where the phrase sits in the piece: an inner phrase does not close like the last one.
The six-four is not an inversion like the others: it is three different chords written the same way. The generator writes the cadential one only where it stands up — on the strong beat, with the dominant resolving it on the weak one and not longer than it — and never opens a piece with a six-four.

### 14.7 Harmonising from a given voice

The given voice can be the melody, the bass, or an inner voice — alto or tenor. In the panel you switch off the voices the generator must not write: those stay exactly as written on the staff and take part in the choice of harmony, instead of being contradicted by the chord chosen for the others.
The bass remains the most decisive constraint, because besides the note it also fixes the inversion. An inner voice is worth as much as the melody: a note belongs to three different chords whoever sings it.
If the chosen harmony does not contain one of the given notes, the panel says so instead of deleting it silently.

### 14.8 Harmony duration, voicings, stages

“Harmony duration” sets how long each chord lasts, and applies both to hand-written degrees and to the harmonisation of a melody. The duration of a single chord can also be changed with the note-value icons beside the field. These used to be two separate commands with two different names in two corners of the panel — “Note value” and “Harmonic rhythm” — doing the same thing.
“Voicing” cycles through the possible voicings of the same chord, and the button says where it is in the cycle. Select some voices first and the cycle moves only those.
The generator’s three stages — choosing the degrees from the repertoire, the voice leading, and the best path across the phrase — can be switched off one by one from the panel: useful for seeing what each of them does, and for comparing results.

## 15. Teacher mode — Analysis Lock

Lets the teacher prepare exercises by locking the analysis engine and selectively hiding solutions or visual warnings. The lock is saved in the .htp project file: the student can open the file, view the score and play the audio, but the interface hides the elements chosen by the teacher. It cannot be bypassed without the password.

**[ SCREENSHOT TO BE INSERTED ]**
*Analysis Lock modal — password field and masking options*

*Figure — Analysis Lock panel*

## 15.1 Activating the lock

32. Help menu → “Analysis Lock”.
33. Set a teacher password.
34. Select the elements to hide.
35. Click “Apply lock”.

## 15.2 Maskable elements

| Option | Effect when locked |
|---|---|
| Coloured lines and errors | Hides voice-leading flags and disables the violations panel. |
| Roman-numeral labels | Hides the Roman numerals below the staff. |
| Chord symbols | Removes the symbols above the staff. |
| Recognised ornaments | Hides the ornament markers (P, v, A, etc.). |
| Alternative readings | Removes the ≈ indicator. |
| Disable export and note copy | Blocks export (MIDI, MusicXML, PDF, PNG), note copy/paste and «Prepare for your AI». |

## 15.3 Unlocking

Click the padlock icon and enter the teacher password. To prevent accidental bypassing, there is no keyboard shortcut for this function.

> ***Note:*** *There is no password-recovery mechanism. Keep it in a safe place.*

## 15.4 Method texts

The explanations in the analysis panel — the title of each finding, its explanation and its suggestion — are written in general terms. Teachers can rewrite them in the words of the method or textbook they use in class, and hand them to their students in a file.
Method texts change the words, not the analysis: the findings stay the same, only the way they are explained changes.

### Writing the texts

Step 1. Open Preferences › Analysis and click «Method texts»: the section opens.
Step 2. Write the method's name (for example «Dubois — Year 3») and the author: the students will see them under every rewritten text.
Step 3. Find the finding to change: the list is grouped by rule families, and the «Filter by name or code…» field finds it at once. A click on the rule opens its fields: «Title», «Explanation», «Suggestion».
Step 4. Write only what you want to change: an empty field keeps the app's text, which shows in it in grey. «Start from the app's text» copies the current explanation and suggestion into the fields, to correct rather than rewrite; «Back to the app's text» deletes what you wrote for that rule.
Some findings have a title that changes with the case: melodic leaps, for example, carry the name of the interval. For these the editor shows every form of the title, gathered in the «Titles» group; where the app adds the end of the title — the notes, the voices, the key — you rewrite only the start.

### Handing them to the students

«Export…» saves the texts in a .htmethod file, to send to the students like any attachment.
The student opens it with a double-click, or from Preferences › Analysis › «Method texts» › «Import…». Harmony Tutor asks whether to use them, saying whose they are and how many findings they rewrite, and the student confirms. The texts apply to every piece.

In the student's analysis panel, under every rewritten text, «📝 Method text:» appears with the method's name and author, to tell it apart from the app's texts. «Remove the method texts», in the same section of the preferences, brings back the app's texts.

# 16. About the application

Help menu → “About Harmony Tutor…” shows: the installed version, copyright and the website URL.

The PDF manual travels inside the application: Help menu → “User manual (PDF)…” opens it in the language of the interface. There is nothing to download separately, and it is always the one matching the installed version. In the same menu, “Shortcuts…” opens the full list of keyboard commands.

> ***Note:*** *On Windows this is the only way to check the installed version number.*

# 17. Preferences and settings

The Preferences panel is organised into six tabs. Settings are saved automatically.

## 17.1 Editor tab

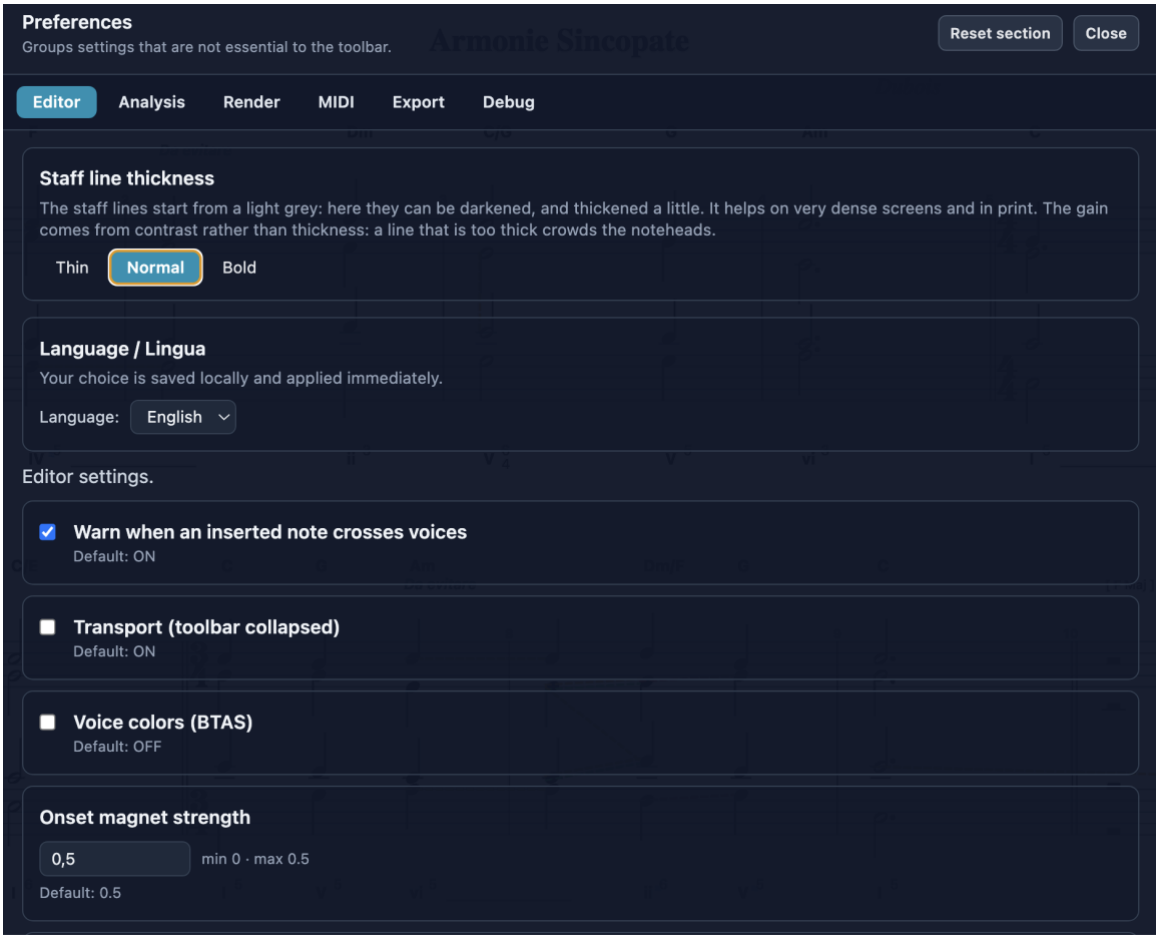



*Figure 18 — Preferences: Editor tab*

- **Voice colours (BTAS):** distinct colouring per voice.
- **Staff layout:** Grand Staff, Old SATB clefs, Treble-only, Close score, Open score.
- **Hide toolbar / Measure numbers / Autosave:** interface options.
- **Select current voice only:** rectangle selection filtered by voice.
- **Transport:** keeps the transport controls visible with the toolbar hidden.

Magnet strength on attacks — how strongly the note you are placing is drawn towards attacks already written in that measure (0 to 0.5 of the reference value). At zero only the grid applies; raising it makes stacking a chord easier and stepping slightly off one harder.
Warn when an entered note crosses the voices — shows the warning described in 4.12, with the option to swap the two voices. Anyone writing counterpoint with deliberate crossings turns it off.
Staff line thickness — the staff lines can be given more body, for printing or for very dense screens.

## 17.2 Analysis tab

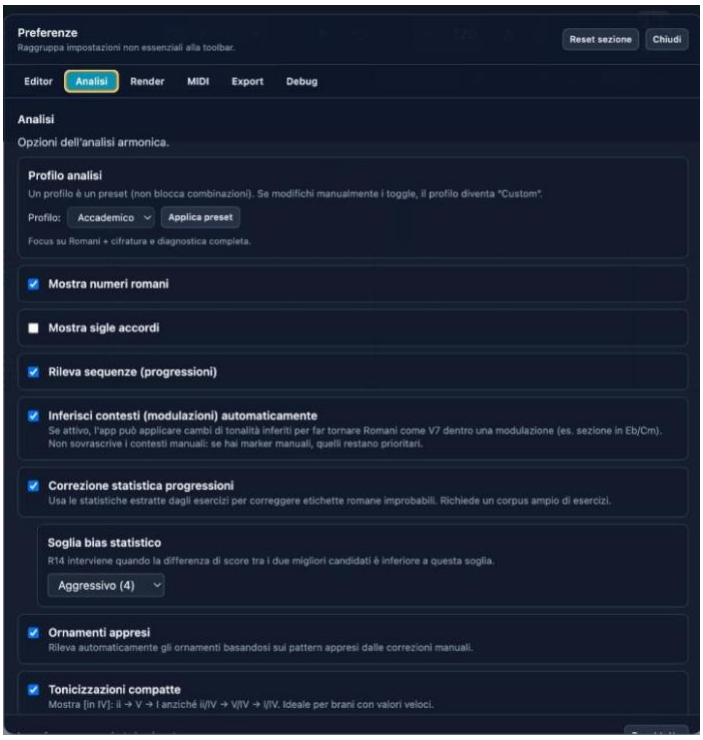



*Figure 19 — Preferences: Analysis tab*

- **Analysis profile:** Academic preset (default). Customisable.
- **Show Roman numerals / chord symbols:** display toggles.
- **Detect sequences:** recognition of imitated harmonic sequences.
- **Infer contexts:** automatic modulations (manual markers stay priority).
- **Statistical correction:** uses the corpus to correct unlikely labels.
- **Statistical-bias threshold:** Conservative, Moderate, Aggressive.
- **Learned ornaments:** uses patterns from manual corrections.
- **Compact tonicisations / Cadential patterns:** compact format and automatic cadence recognition.
- Method texts: the texts of the findings rewritten by the teacher, to hand to the students (see 15.4).

The settings of this tab apply to every piece: opening a piece does not change them.

## 17.3 Render / MIDI / Export / Debug tabs

Render: graphic rendering settings. MIDI: input/output devices. Export: format options (including MIDI format 0 and 1). Debug: advanced diagnostics.

> ***Note:*** *The "Reset all" button restores all settings. "Reset section" only the current tab.*

## 17.4 View commands live in the menus

The commands you switch on and off while looking at the score live in the menus, where they have a shortcut, and not among the preferences: a preference is configured once, these are commands.
The View menu now holds bar numbers, voice colours, the toolbar (⌥T) and the floating command bar, the bar customisation, the Staves submenu with staff layout and grouping by orchestral family, the Roman-school numbering and sequences among the analysis layers, and the title to include in exports.

In the Analysis tab of the preferences the four settings that are actually touched stay in view: analysis profile, automatic context inference, cadential-pattern recognition and harmony hints from the accompaniment. The others sit behind the “advanced” switch. The texts of the findings — title, explanation and suggestion — are rewritten in «Method texts», in the same tab (see 15.4).

# 18. Keyboard shortcuts

## 18.1 Global

| Shortcut | Action |
|---|---|
| Cmd/Ctrl + N | New project |
| Cmd/Ctrl + O | Open |
| Cmd/Ctrl + S | Save |
| Cmd/Ctrl + Shift + S | Save as |
| Cmd/Ctrl + I | Import MIDI |
| Cmd/Ctrl + Shift + E | Export MIDI |
| Cmd/Ctrl + Z | Undo |
| Cmd/Ctrl + Shift + Z | Redo |
| Cmd/Ctrl + P | Print |

## 18.2 Editor

| Shortcut | Action |
|---|---|
| Space | Play / Stop (starts recording if armed, with count-in) |
| Enter | Return to start |
| O | Loop the selection (O again turns it off) |
| S | Score as page / ribbon |
| V | Cycle voice |
| 1–7 | Duration |
| Shift + 1–7 | Adds a tied value to the selected note |
| Shift + Opt/Alt + 1–7 | Tied value, pushing the following notes forward |
| # / b / n | Sharp / Flat / Natural |
| . | Augmentation dot |
| R | Rest at the cursor |
| Control + R | General rest (every part) |
| **Shift + R** | Arm / disarm / stop recording (with a visible ACC track) |
| L | Tie |
| K | Metronome |
| T | Measure-edit panel |
| **Option/Alt + F** | Fermata |
| **Opt + H** | Mark the selected note(s) as structural (enters the analysis) |
| **Opt + Shift + H** | Collapse the selection (≥2 notes, even an arpeggio) into a single chord |
| **Shift + Opt/Alt + R** | Tempo curve (rallentando/accelerando), also on ACC selections and loaded files |
| Option/Alt + ↑/↓ | Move note to treble/bass staff |
| Option M | Insert close/open score from the playhead onward |
| Alt + L | Cycle layout |
| Alt + H | Toggle labels |

| Shortcut | Action |
|---|---|
| Cmd/Ctrl + / – | Zoom the score, keeping the selected note still |
| Cmd/Ctrl + 0 | Zoom to 100% |
| Backspace | Delete selection |
| ↑/↓ | ±1 semitone |
| Alt + ↑/↓ | ±1 octave |
| Shift + Cmd/Ctrl + C | Copy the entire selected SATB voice (then paste onto an ACC track) |

> ***Note:*** *On different keyboard layouts, the # key may arrive as à, and the . key may produce > with Shift: the app accepts both.*

### 18.3 Ornament marking (Option/Alt + letter)

| Shortcut | Ornament |
|---|---|
| Option+H | Structural |
| Option+P | Passing tone |
| Option+V | Neighbor tone |
| Option+A | Appoggiatura |
| Option+N | Anticipation |
| Option+S | Escape tone |
| Option+R | Suspension/Retardation |
| Option+O | Ornamental |
| Option+C | Changing tone |

### 18.4 Analysis shown and staves (View menu)

| Shortcut | Action |
|---|---|
| Cmd/Ctrl + Alt + R | Roman numerals |
| Cmd/Ctrl + Alt + S | Chord symbols |
| Cmd/Ctrl + Alt + F | Figured bass |
| Cmd/Ctrl + Alt + C | Choir (SATB): show or hide |
| Cmd/Ctrl + Alt + 3 | Violations list |
| Cmd/Ctrl + Alt + 4 | Notes on the piece |

Accompaniment tracks are switched on and off from View → “Staves shown and exported”, each by its own name. These commands also choose the contents of the exported files (§13.2).

## 19. Melodic transformations

Harmony Tutor provides tools to study and compose with the transformations of a melodic motif: **transposition** (T), **inversion** (I), **retrograde** (R) and **retrograde-inversion** (RI) — the symmetries at the heart of imitation and thematic development. Transformations can be **tonal** (within the key) or **real** (at exact intervals).

### 19.1 Motif detector

The detector automatically finds, throughout the piece, occurrences of the same motif transformed — transpositions, inversions, retrogrades and retrograde-inversions — both **within a single voice** and **across voices** (cross-voice), in tonal or real form.
**Activation.** The “Melodic motifs” toggle. When active, the notes belonging to a recognised transformation are **colour-coded by type**, and a **bracket** marks the reference model motif.
**Note.** In very dense textures the matches may be hard to see; that is why the detector is off by default.

## 19.2 Transformation tools

They let you **generate** a new phrase by transforming a selection of notes:
- **Transposition** — moves the selection by a chosen interval (tonal or real).
- **Inversion** — flips the melodic contour around an axis (per voice).
- **Retrograde** — reverses the temporal order of the notes (onset-based).
- **Retrograde-inversion** — combines inversion and retrograde.

**Tonal** transformations stay within the key; **real** ones keep the exact intervals (a warning flags when a real inversion leaves the scale). The result is **inserted after** the selection, extending the measures automatically if needed.

## 19.3 View and layout

Under Score ▸ View you can choose between the page, where lines wrap, and the continuous ribbon, which puts every measure on a single staff that scrolls horizontally. The layout follows the zoom: zooming out fits more measures per line instead of leaving empty space on the right.

The S key switches between page and ribbon and back, without opening the menu.

In the continuous ribbon the mouse wheel scrolls the score horizontally (down to move forward), and Shift + wheel vertically; the trackpad works as usual.

Choosing the ribbon, two ways of following playback appear in the same menu: «Score scrolls», where the playback line stays still and the music scrolls under it, and «Line moves (by pages)», where the score stays still and the line crosses it: at the edge the view jumps one screen ahead, turning the page at a barline. The choice belongs to the computer, not to the piece.

Content-aware spacing, in the same menu, gives each measure the width its content needs: a measure of eighth notes takes more room than one of whole notes, instead of cramming them. Analysis labels follow the layout in every case.

### System breaks fixed by hand

Measures per line is a good ceiling to start from, but it does not know where a phrase ends. ⌥Enter ends the line at the measure where the caret is; the same is available from the right-click menu on a note ("break the line after this measure").

On very dense music the line may close before the number you set: if the notes would not fit without crowding, the program wraps early. The number is a ceiling, not an obligation.

A break fixed by hand overrules everything else: neither the measures-per-line ceiling nor the available width can move it. You can see it — a ⏎ mark at the end of the fixed line — and you click it to remove it. Without a visible mark, layout would become a list of invisible decisions with no way back.

### Measures per line, view and paper format stay with the file

The three layout settings — the measures-per-line ceiling, the on-screen view (systems or continuous ribbon) and the paper format (A4 portrait or landscape, which concerns printing and PDF) — are saved in the project and restored when you reopen it. They are choices about the PIECE, not about the working session: a close-spaced chorale and a keyboard piece are not read on the same page.

Files saved with earlier versions do not carry them: they open with the starting values, without inheriting the layout of the file opened before.

### Back to 100%: double-click on empty space

Zoom resets with a double-click on an empty part of the page — where nothing is drawn. Anything under the pointer (a note, a tie, a sign, a label, the staff lines, the clef) blocks it. ⌘0 and the toolbar command are still there.

### Zoom keeps the point you are looking at still

Zooming with the trackpad (pinch) or with Ctrl + wheel keeps the music under the pointer still: the very passage you are looking at gets larger, even when the lines are laid out again.

From the keyboard: Cmd + and Cmd − (Ctrl on Windows) zoom the score in and out keeping the selected note still — with nothing selected, the playback position, otherwise the centre of the window; Cmd 0 returns to 100%. The same commands are in the View menu.

# 20. Score signs

Signs are placed from the signs palette, opened with the *f* button in the toolbar. It is a floating window:

drag it by its title, and it stays open while you work.

## 20.1 Using the palette

Signs are collected in three collapsible groups, by what the sign is about: Dynamics (intensity), Articulations and expression (how sounds are attacked and joined), Structure (the layout of the page). One group opens at a time, and stays open while you drag.
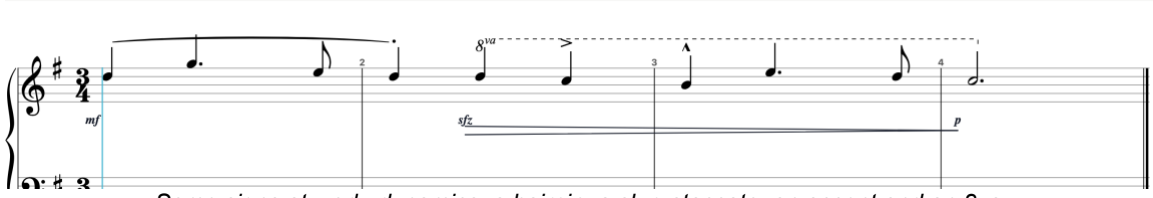



*The palette docked to the left edge: the search box at the top, and several groups open at once.*

Drag: grab the sign and drop it where you need it on the score.
Or: select a note and click the sign. A hairpin placed this way covers the chosen note; select two, the first and the last, and it runs from one to the other. A slur needs two notes.
Right-click a sign to remove it, wherever it is: on the note, on the curve, on the text.
The palette is docked to the left edge and its body scrolls: several groups can stay open at once, and a search box at the top takes you straight to the sign you need, instead of opening and closing sections by trial.
It also holds durations, accompaniment patterns and melodic transformations; the “Notes” group gathers everything that concerns the note itself (accidentals, dot, tuplets, articulations, octave signs).
Transposition, mode and the automatic leading note sit next to the key signature; the time signature is written with its two numbers.

Signs are picked up and put down: hairpins are drawn by dragging them onto the music instead of being written as text, and accidentals are proper icons rather than text characters.
Besides dragging them, signs can be placed with a single click: pick the sign from the palette and it is applied where the cursor is.

## 20.2 Dynamics

Levels from ppp to fff, accents (sf, sfz, rf), fp, and crescendo/diminuendo hairpins. They are not drawings: they drive playback volume. Each sign belongs to the staff it sits on — a p under the upper staff concerns soprano and alto, not the bass — whether you drag it or place it by clicking a note. To give it to every part in one go, turn on «to all parts» in the palette, and the sign appears under every staff.




*Some signs at work: dynamics, a hairpin, a slur, staccato, an accent and an 8va.*

Hairpins are stretched by dragging their ends. On a held string note a hairpin swells the sound inside the note; a piano, once the key is struck, cannot grow.
A dynamic marking written inside a hairpin becomes its goal: the hairpin grows (or fades) up to that marking, and from there on the marking applies.
With no signs at all, the notes keep the dynamics they came with (from a MIDI import or a recording); as soon as a sign is written, the sign takes over. The exception is a note whose velocity was set by hand, in the velocity lane or in the channel strip: there the note decides (see 11.13).

## 20.3 Articulations

Staccato, staccatissimo, accent, marcato, tenuto. They combine — accent and staccato on the same note is ordinary writing — and are placed on a note or on a whole chord.
Dragged onto a chord they apply to the whole chord, because the sign is drawn once above the group and could not say «the soprano only».
Selecting specific notes and clicking applies to exactly those: that is how you staccato the soprano alone.
They sound: staccato shortens the note and its tail, the accent reinforces the volume. Tenuto is a written sign and does not change the sound, because full length is already the norm.

## 20.4 Slurs

Not to be confused with the tie (key L), which joins two equal sounds into one. A slur connects different notes and is a sign of phrasing.
Place it by selecting two notes and clicking the sign, or by dragging it onto a note: it then reaches the next note of the same voice.
Lengthen or shorten it by dragging the ends of the curve, where a handle appears on hover.
If the passage breaks across a line, the curve stops at the edge of the staff and resumes on the next system.
In the choir the curve sits on the stem side — above for soprano and tenor, below for alto and bass — so one voice's slur does not land on the other.

## 20.5 Octave signs (8va and 8vb)

For passages that would otherwise need too many ledger lines: they are written an octave lower (or higher) and the sign says to play them shifted.
Notes are not moved: you write where it should be read, and the sign changes how it sounds.
To tidy a passage already written high: select it, move it down an octave with ⇧+down arrow, then add the sign.
The sign applies to the voice it is written in: an 8va on the soprano does not raise the bass sounding at the same time. It also covers the note under the end of the bracket.

- The sign also applies to the notes going to external instruments and plug-ins. On an accompaniment track it applies to that track only.

## 20.6 What survives in files

All of these signs are written to MusicXML and read back when the file is reopened: dynamics, articulations, slurs, octave signs, key changes, repeats, endings and double bar lines. MIDI carries dynamics (as note velocity) and octave signs (as sounding pitch), which is all that format can say.

### 20.7 Moving signs on the page

A metronome mark landing on a chord symbol, or two texts overlapping, can be moved out of the way by dragging: the same gesture carries the sign to another point in TIME and to another HEIGHT. Where you release decides the measure, the vertical shift decides the height, and the offset stays with the sign in the file as well.
Engraving has no rule for every possible collision between signs: whoever writes moves them just as much as is needed.
A free text (“dolce”, “CVII”) is corrected with a double-click on it, without deleting and retyping.

### 20.8 The sustain pedal

On piano tracks the sustain pedal is written in the traditional notation: «Ped.» where it goes down, «✱» where it comes up, below the track's bass staff. Each track has its own pedal; single staves and drum tracks have none.

36. Select the notes and press «Ped» in the palette's Arpeggiator section: the sign covers the span of the selected notes, rests included. Press again and the sign goes away.
37. Drag its ends to move it: «Ped.» moves where the pedal goes down, «✱» where it comes up. Right-click the sign to remove it.
38. With the «with pedal» switch on, accompaniment figures write their own pedal: at every change of harmony, ✱ and straight back to Ped.

The pedal drives the sound, as on a piano: it keeps the dampers raised, so UNDER THE PEDAL A REST DOES NOT STOP THE SOUND — notes already played keep ringing until the ✱. A rest under «Ped.» says the hand lifts, not that the sound stops. For silence on a rest, put the ✱ before the rest or on it.
The pedal stays in the files: the exported MIDI carries the pedal controller (CC64), which a sequencer shows and lets you edit; MusicXML carries the «Ped.» and «✱» signs. Pieces written with earlier versions, where the pedal lengthened single notes, sound as before.

### 20.9 First and second endings

Endings (voltas) are the different closings of a repeat: the first ending is played the first time through and goes back, the second takes its place the next time. They are placed from the palette, «Measures» group, with the «Volta» button:

39. with a click, the ending goes on the bar where the caret is;
40. dragging it, it goes on the bar where you drop it;
41. with notes selected, it covers all their bars: that is how you make an ending of two or more bars.

The first one you place carries «1.»; the one straight after becomes «2.», and so on. An ending that goes back needs the repeat bar at its end: if it is missing, the program adds it and tells you so. Placing the ending again on the same bars removes it.
The number is changed with a **double-click** on it: «1.–3.» means that bracket is played three times before moving on to «4.». Enter confirms, Esc cancels.
On the page the bracket sits above the top staff: the first ending is closed on the right, the last stays open. If an ending runs over a line break, it carries on into the next line.
In the analysis, part-writing is interrupted between the end of the first ending and the start of the second, as at a double bar line: the two bars are close on paper but are never played one after the other, so fifths or octaves between them are not reported. The key, on the other hand, carries on.
Endings are written to MusicXML together with repeats and are read back when the file is reopened.

## 21. Frequently asked questions

**What is the difference between the free version and Harmony Tutor Pro?**
The free version includes all features for 10 days. After the trial period, the advanced features (harmonic analysis, chorale generator, export) are disabled. Buying the Pro licence unlocks everything with no time limit.

**Can I install the software on a new computer?**
Yes. Each licence allows 2 activations. If you change computer, you can deactivate the licence on the old device and reactivate it on the new one.

**The analysis shows the wrong chord. What can I do?**
Place the playhead at the desired point, press T and enter an override in the panel. Incomplete chords or tonicisations can create ambiguity.

**What is CAD-DEC?**
The deceptive cadence: V resolves to vi/VI/♭VI instead of I. It is flagged with a green marker in the panel.

**What does V⁶⁄₄ mean instead of I⁶⁄₄?**
In the cadential progression, the I⁶⁄₄ is functionally relabelled as V⁶⁄₄. The override is available from the T panel.

**How do I insert a rallentando?**
Select the range of notes → Shift+Option/Alt+R → set the start and end BPM.

**How does the Fermata work?**
Select the note → Option/Alt+F. The duration is doubled in playback (default 2×). Repeat to remove it.

**How do I add an ACC track?**
Tracks panel → "+ New" → choose the type, then assign a name and a MIDI instrument.

**Does REC recording work with ACC tracks?**
Yes. Select the destination ACC track, arm with Shift+R and start with Space.

**Can I import files from MuseScore, Finale or Sibelius?**
Yes. MuseScore files (.mscz, .mscx) open from File → Import MuseScore..., if MuseScore is installed on the computer (see 13.1). From Finale and Sibelius, export to MusicXML and in Harmony Tutor choose File → Import MusicXML. MIDI import is also supported.

**What is the difference between MIDI format 0 and format 1?**
Format 0: all voices on a single track. Format 1: each voice on a separate track (recommended for editing in a DAW).

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## Third-party credits and licences

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